

Techniques in Combinatorics

Version 4.0

29-2-2024

Contents

Introduction	2
1 Counting	3
1.1 Basic Counting Tools	3
1.2 Recurrence Relations	10
1.3 Bijections	16
2 Extremal Combinatorics and Existence Problems	22
2.1 Induction	23
2.2 Pigeonhole Principle	29
2.3 Colouring	32
2.4 Double Counting	37
2.5 Extremal Principle	43
2.6 Majorization Inequality	46
3 Games and Operations	51
3.1 Algorithm	51
3.2 Invariance	60
4 Graph Theory	68
4.1 Concepts in Graph Theory	68
4.2 Applications of Graph Theory	91
5 Combinatorial Geometry	100
5.1 Combinatorial Ideas	100
5.2 Point Set Problems	106
5.3 Geometric Ideas	111
6 Generating Functions	118
7 Probabilistic Method	131
Links	142
References	148

Introduction

To many beginners, combinatorics is the easiest topic because one does not need any deep knowledge to work on combinatorial problems. Usually, the arguments are intuitive and obvious, and some are indeed common sense. However, due to the same reason, combinatorics is also the hardest topic to many people. This is because learning deep knowledge seems to be useless, and it is hard to train the intuition.

It is also hard to judge the difficulty of a combinatorial problem. If one looks into the IMO problems, there are occasionally some C3 or C4 problems for which only a small proportion of the contestants can get complete solutions. One reason is that a combinatorial problem usually only requires a key observation or a key idea. This makes the solution short. But sometimes the idea may be hard to think of, and there is no alternative method which may work.

To solve combinatorial problems, one first needs to learn a lot of tools which are often useful in this area. To solve a particular problem, perhaps only one or two of these tools are useful. But this also means that one may not be able to solve the problem if one has not seen the related tricks before. Therefore, it is necessary to learn all these tools and go through some standard examples. Afterwards, to solve a particular combinatorial problem, one can consider all the tools one by one and see which tool could be useful in this problem. It is like the situation that there are many roads in front of us, but in the end only one or two roads will lead to the goal. We have to explore every road one by one, but we will never know how far we need to go to judge whether it is the correct road or not. The only thing that could help us is the experience.

For this reason, we are going to learn various techniques which are commonly used in combinatorics. Although these techniques will be introduced under separated topics, one should bear in mind that all methods are useful in every topic.

In section 1, we introduce different counting methods, which are essential in many areas of combinatorics. In section 2, we will go through some standard methods, such as induction, the pigeonhole principle, colouring, etc. In section 3, we use algorithm and invariance to solve some problems involving games and operations. Section 4 is about graph theory, while section 5 is about combinatorial geometry. All techniques introduced in previous sections are useful for both topics, and there are some additional facts and tricks that one should learn in these topics. Lastly, we have two short sections about generating functions and probabilistic method.

1 Counting

Counting is the most elementary topic in combinatorics. Although there is seldom a purely counting problem in the contests, one may need to count different simple things when working on some problems. Therefore, it is necessary for us to learn how to count things.

There are various counting methods, each of which has its own advantage. We will go through different counting methods in this section.

1.1 Basic Counting Tools

First of all, we review the basic counting tools.

Definition 1.1. The *method of exhaustion*(窮舉法/枚舉法) refers to the counting method which lists out all possible cases.

This is the most fundamental way of counting. A clear disadvantage of listing everything out is that it is easy to make mistakes. It is very likely that some cases are missed out during the counting. However, sometimes the exhaustion method cannot be avoided in part of the steps. In order to count more accurately, one should follow a certain order when listing out the cases. In particular, the *lexicographic order* (or *dictionary order*(字典排列法)) is useful. The name is self-explanatory. For example, if we use an ordered tuple $(a_1, a_2, \dots, a_n)$ of numbers to identify a specific case, then we can list out all the cases in ascending order of their first entries a_1 . If some of them have the same first entry, then we order them using their second entries and so on. This allows us to count in a systematic way, so that it is easier to check for the mistake.

Theorem 1.1. (*Addition principle/Sum rule*(加法原理)) Let $S_1, S_2, \dots, S_n$ be pairwise disjoint finite sets. Then we have

$$|S_1 \cup S_2 \cup \dots \cup S_n| = |S_1| + |S_2| + \dots + |S_n|.$$

Recall that for any set S , the notation $|S|$ denotes the number of elements in S . Therefore, the addition principle states that the number of elements in at least one of the sets $S_1, S_2, \dots, S_n$ is equal to the sum of the numbers of elements of these sets, given that they are pairwise disjoint. This is rather obvious. For the application in counting, it simply means that we can count by partitioning all possibilities into a few cases. Each case has its own feature which allows us to count quickly by using some

other methods. Note that we should ensure that each situation falls into exactly one case in order to apply the addition principle (which is the disjoint condition).

Theorem 1.2. ([Multiplication principle/Product rule](#)(乘法原理)) Let $S_1, S_2, \dots, S_n$ be any finite sets. Then we have

$$|S_1 \times S_2 \times \cdots \times S_n| = |S_1| |S_2| \cdots |S_n|.$$

The Cartesian product $S_1 \times S_2 \times \cdots \times S_n$ is the set of all elements $(a_1, a_2, \dots, a_n)$ such that $a_j \in S_j$ for each $j = 1, 2, \dots, n$. For the application in counting, suppose we would like to make some choices in a few steps so that the number of choices in each step is independent of the other steps. Then we can simply take the product of the numbers of choices in all steps to obtain the total.

Problem 1.1. Let n be a positive integer. How many n -digit numbers are there such that all digits of n are nonzero and have the same parity?

Analysis. We demonstrate the simplest usage of the addition principle and the multiplication principle. For this simple problem, it is obvious that we should divide into two cases, according to the parity of the digits.

Solution. If all digits are odd, then there are 5 choices for each of the n digits, namely, 1, 3, 5, 7, 9. Hence, there are 5^n such numbers.

Similarly, if all digits are even, then each digit can be one of 2, 4, 6, 8. There are 4^n such numbers.

In total, we have $5^n + 4^n$ numbers satisfying the condition. □

Remarks. The step $5 \times 5 \times \cdots \times 5$ is an application of the multiplication principle, while the step $5^n + 4^n$ is an application of the addition principle.

Problem 1.2. Let n be a positive integer. How many n -digit numbers are there whose sums of digits are multiples of 5?

Analysis. Suppose we choose the digits one by one. Since the condition involves the sum of digits, it seems that the choice for each digit depends on the other digits. It is true that the choices are dependent, but the number of choices is still a fixed number. Therefore, we can apply the multiplication principle as usual.

Solution. When $n = 1$, there are two possibilities, 0 and 5. When $n > 1$, we choose the first $n - 1$ digits arbitrarily. The first digit is nonzero, and hence there are 9 choices. For each of the remaining $n - 2$ digits, there are 10 choices.

After the first $n - 1$ digits are chosen, let S be the sum of these digits. Then we have to choose a digit k for the unit digit such that $S + k$ is a multiple of 5. This means

k is one of $0, 1, 2, \dots, 9$ such that $k \equiv -S \pmod{5}$. Note that there are exactly 2 choices for k no matter what S is.

Therefore, there are $9 \times 10^{n-2} \times 2 = 18 \times 10^{n-2}$ numbers satisfying the condition for $n > 1$. $\square$

Remarks. The multiplication principle is useful as it helps us count the things quickly. Therefore, a common strategy is to consider different cases, each of which can be counted using the multiplication principle directly. Afterwards, we can add up the results using the addition principle.

Definition 1.2. (*Permutation*(排列)) Let n and r be nonnegative integers satisfying $0 \leq r \leq n$. Then the number of ways to choose r objects from n distinct objects in order is denoted by P_r^n .

Definition 1.3. (*Combination*(組合)) Let n and r be nonnegative integers satisfying $0 \leq r \leq n$. Then the number of ways to choose r objects from n distinct objects without specifying the order is denoted by $\binom{n}{r}$ or C_r^n .

In CMO and CGMO, it is common to use P_n^r (or A_n^r) to denote the number of permutations and use C_n^r to denote the number of combinations (with n and r being swapped). To avoid misunderstanding, we will use the notation $\binom{n}{r}$ in this note. This symbol is also known as the *binomial function*(二項式函數) or the *binomial coefficient*(二項式係數), due to the following theorem.

Theorem 1.3. (*Binomial theorem*(二項式定理)) Let n be a nonnegative integer. Then

$$(x + y)^n = \sum_{k=0}^n \binom{n}{k} x^{n-k} y^k = \binom{n}{0} x^n + \binom{n}{1} x^{n-1} y + \binom{n}{2} x^{n-2} y^2 + \dots + \binom{n}{n} y^n.$$

Proof. When we expand the left-hand side, each term corresponds to the product of n factors, each of which is either x or y . Therefore, each term must be of the form $x^{n-k} y^k$ for some $k = 0, 1, \dots, n$.

On the other hand, to obtain a term $x^{n-k} y^k$, we need to choose exactly k copies of y from n brackets (and the remaining $n - k$ copies of x from the other brackets). There are $\binom{n}{k}$ ways to do so by definition. (Although the brackets look the same, they are treated as different objects when doing the expansion.) This means the coefficient of $x^{n-k} y^k$ is $\binom{n}{k}$. Summing over k from 0 to n gives the desired identity. $\square$

Many counting problems can be solved using P_r^n and $\binom{n}{r}$ (indeed, the latter is more useful). The next two propositions give the useful formulae related to these numbers.

Proposition 1.1. Let n and r be nonnegative integers satisfying $0 \leq r \leq n$. Then the following hold.

$$(a) \quad P_r^n = n(n-1)\cdots(n-r+1) = \frac{n!}{(n-r)!}$$

$$(b) \quad P_n^n = n!$$

$$(c) \quad P_1^n = n$$

$$(d) \quad P_0^n = 1$$

Proof. To choose r objects from n objects in order, we can choose them one by one. There are n choices for the first one, $n-1$ choices for the second one, $\dots$, and $n-r+1$ choices for the last one (the r th one). By the multiplication principle, we have $P_r^n = n(n-1)\cdots(n-r+1)$. The second part of (a) follows from the definition of factorial.

The remaining parts can be deduced by substituting $r = n, 1, 0$ respectively in this formula. Note that the product $n(n-1)\cdots(n-r+1)$ is empty when we put $r = 0$. By convention, an empty product equals 1. Similarly, we define $0! = 1$ by convention. $\square$

Proposition 1.2. Let n, r, k be nonnegative integers satisfying $0 \leq r \leq n$. Then the following hold.

$$(a) \quad \binom{n}{r} = \frac{n(n-1)\cdots(n-r+1)}{r!} = \frac{n!}{r!(n-r)!}$$

$$(b) \quad \binom{n}{r} = \binom{n}{n-r}$$

$$(c) \quad \binom{n}{n} = \binom{n}{0} = 1$$

$$(d) \quad \binom{n}{1} = n$$

$$(e) \quad (\text{Pascal's identity (帕斯卡恒等式)}) \quad \binom{n+1}{r} = \binom{n}{r} + \binom{n}{r-1} \text{ for } r \geq 1$$

$$(f) \quad \binom{n}{n} + \binom{n+1}{n} + \cdots + \binom{n+k}{n} = \binom{n+k+1}{n+1}$$

$$(g) \quad \binom{n}{0} + \binom{n}{1} + \cdots + \binom{n}{n} = 2^n$$

Proof.

- (a) Suppose we want to choose r objects from n distinct objects in order. This can be done in two steps. Firstly, we choose r objects without specifying the order. There are $\binom{n}{r}$ ways to do so. Then we rearrange these r chosen objects to give the order, where there are P_r^r ways to do so. Hence, the total number of ways is $\binom{n}{r} \times P_r^r$. Note that this is equal to P_r^n by definition. This implies

$$\binom{n}{r} = \frac{P_r^n}{P_r^r} = \frac{n(n-1)\cdots(n-r+1)}{r!}$$

by [proposition 1.1](#).

- (b) The simplest way to prove this is as follows. If we want to choose r objects from n distinct objects, then we simply need to choose $n-r$ objects from these n objects and remove them. The remaining r objects will become our choice. This easily establishes the formula.

- (c) Putting $r = n$ in [\(a\)](#), we obtain $\binom{n}{n} = \frac{n!}{n!} = 1$. By convention, $\binom{n}{0} = 1$ as there is one way to choose nothing. This also agrees with [\(a\)](#) when $r = 0$.

- (d) Putting $r = 1$ in [\(a\)](#), we obtain $\binom{n}{1} = \frac{n}{1!} = n$.

- (e) Pascal's identity can be proved in a combinatorial way. Note that $\binom{n+1}{r}$ counts the number of ways to choose r objects from $n+1$ objects. We first call one of these $n+1$ objects special. If we do not choose the special object, then we have to choose r objects from the remaining n objects. There are $\binom{n}{r}$ ways to do so. If we instead choose the special object, then we only need to choose $r-1$ objects from the remaining n objects. There are $\binom{n}{r-1}$ ways to do so. Thus, the total number is the right-hand side of the identity.

- (f) Using Pascal's identity repeatedly, we obtain

$$\begin{aligned} \binom{n+k+1}{n+1} &= \binom{n+k}{n} + \binom{n+k}{n+1} \\ &= \binom{n+k}{n} + \binom{n+k-1}{n} + \binom{n+k-1}{n+1} \\ &= \binom{n+k}{n} + \binom{n+k-1}{n} + \binom{n+k-2}{n} + \binom{n+k-2}{n+1} \\ &= \dots \end{aligned}$$

$$\begin{aligned}
&= \binom{n+k}{n} + \binom{n+k-1}{n} + \binom{n+k-2}{n} + \cdots + \binom{n+1}{n} + \binom{n+1}{n+1} \\
&= \binom{n+k}{n} + \binom{n+k-1}{n} + \binom{n+k-2}{n} + \cdots + \binom{n+1}{n} + \binom{n}{n} \\
&\text{since } \binom{n+1}{n+1} = 1 = \binom{n}{n}.
\end{aligned}$$

(g) This follows from the binomial theorem by substituting $x = y = 1$.

□

Problem 1.3. (IMO Prelim 2012 Problem 14) Given a, b, c, d (where $a < b < c < d$) are positive integers not exceeding 2012, and they form an arithmetic sequence in this order. How many different sets of possible values of (a, b, c, d) are there?

Analysis. Although we need to choose 4 numbers from the first 2012 positive integers, there is nothing to do with $\binom{2012}{4}$. Indeed, the arithmetic sequence depends on only two parameters. In other words, if two of a, b, c, d are fixed, then the remaining numbers are uniquely determined (but they need not be integers and need not be smaller than or equal to 2012). To deal with the upper bound and the implicit lower bound $a > 0$, one way is to fix a and d first.

Solution. Suppose a, b, c, d form an arithmetic sequence in this order. Then we can write $b = a + s$, $c = a + 2s$ and $d = a + 3s$ for some positive integer s . Therefore, we must have $d \equiv a \pmod{3}$.

Conversely, for any fixed a and d satisfying $d \equiv a \pmod{3}$, we can define $b = \frac{2a+d}{3}$ and $c = \frac{a+2d}{3}$ so that both of them are integers and a, b, c, d form an arithmetic sequence.

Therefore, the problem is the same as finding distinct positive integers a and d not exceeding 2012 such that they leave the same remainder when divided by 3.

- If $a \equiv d \equiv 0 \pmod{3}$, they can be $3, 6, 9, \dots, 2010$. There are $\binom{670}{2}$ choices.
- If $a \equiv d \equiv 1 \pmod{3}$, they can be $1, 4, 7, \dots, 2011$. There are $\binom{671}{2}$ choices.
- If $a \equiv d \equiv 2 \pmod{3}$, they can be $2, 5, 8, \dots, 2012$. There are $\binom{671}{2}$ choices.

Note that combination instead of permutation is used in each case because the order of a and d is fixed (i.e. a is always less than d).

Thus, the total number of possible tuples is

$$\binom{670}{2} + \binom{671}{2} + \binom{671}{2} = \frac{670 \times (669 + 671 + 671)}{2} = 673685.$$

□

Sometimes when we try to count by dividing into a few cases, some scenarios may fall into more than one case. Then the addition principle cannot be applied. Indeed, we need to modify the formula by subtracting the overlapping cases. This is the spirit of the following theorem.

Theorem 1.4. (Inclusion-exclusion principle(容斥原理)) Let $S_1, S_2, \dots, S_n$ be any finite sets. Then

$$\left| \bigcup_{i=1}^n S_i \right| \leq \sum_{i=1}^n |S_i| - \sum_{i < j} |S_i \cap S_j| + \sum_{i < j < k} |S_i \cap S_j \cap S_k| - \dots + (-1)^{n+1} \left| \bigcap_{i=1}^n S_i \right|.$$

Proof. It suffices to show that every element in $S_1 \cup S_2 \cup \dots \cup S_n$ contributes exactly 1 to the right-hand side, since the elements not in $S_1 \cup S_2 \cup \dots \cup S_n$ contribute nothing to both sides. Indeed, suppose an element a belongs to exactly m of the sets $S_1, S_2, \dots, S_n$. Then its contribution to the first sum is m , its contribution to the second sum is $\binom{m}{2}$ since we have to choose 2 of those m sets and take the intersection. Similarly, its contribution to the i th sum is $\binom{m}{i}$ for $i = 0, 1, \dots, m$, and it contributes nothing to the remaining sums. Hence, the total contribution is

$$\binom{m}{1} - \binom{m}{2} + \binom{m}{3} - \dots + (-1)^{m+1} \binom{m}{m}. \quad (1)$$

Using the binomial theorem to expand $(1 - 1)^m = 0$, we find that

$$\binom{m}{0} - \binom{m}{1} + \binom{m}{2} - \dots + (-1)^m \binom{m}{m} = 0.$$

Therefore, (1) is equal to $\binom{m}{0} = 1$. □

Problem 1.4. (IMO HK TST 2014 Test 1 Problem 4) A number is *good* if the sum of its digits is 18 (for example 130518 is a good number). Find the number of good 6-digit numbers.

Analysis. This is equivalent to solving the equation $x_1 + x_2 + \dots + x_6 = 18$ where each variable is an integer with a certain bound. The solution using binomial coefficients is standard.

Solution. We first prove the following well-known fact.

Claim. Let n and k be positive integers. Then the number of nonnegative integer solutions to $x_1 + x_2 + \cdots + x_k = n$ is $\binom{n+k-1}{n} = \binom{n+k-1}{k-1}$.

Proof. Suppose there are n identical balls and $k-1$ identical sticks. We rearrange these $n+k-1$ objects. For each arrangement, the sticks partition the balls into k groups. For $i = 1, 2, \dots, k$, the number of balls in the i th group corresponds to the value of x_i . This corresponds to a solution to the Diophantine equation, and such a correspondence is a bijection (see subsection 1.3 for a more detailed discussion on bijection). From this combinatorial point of view, it is easy to count, since we only need to decide the places for which the n identical balls are. There are $\binom{n+k-1}{n}$ possibilities. $\square$

Back to the original problem. We are going to solve $x_1 + x_2 + \cdots + x_6 = 18$ in integers where $1 \leq x_1 \leq 9$ (since the leading digit cannot be 0) and $0 \leq x_i \leq 9$ for $i = 2, 3, 4, 5, 6$. We first use the substitution $y = x_1 - 1$ so that $0 \leq y \leq 8$. The equation becomes $y + x_2 + x_3 + \cdots + x_6 = 17$. The advantage is that every variable can be 0.

By the claim, there are $\binom{17+6-1}{6-1}$ solutions for which $y, x_2, \dots, x_6$ are nonnegative integers. However, those solutions for which some variables exceed their upper bounds should be rejected. For example, we eliminate those solutions satisfying $y \geq 9$ and $x_2, \dots, x_6 \geq 0$. Rewriting the equation in the form $(y-9) + x_2 + \cdots + x_6 = 8$, we know that there are $\binom{8+6-1}{6-1}$ such cases. Similarly, the number of solutions for a particular x_i ($2 \leq i \leq 6$) satisfying $x_i \geq 10$ is $\binom{7+6-1}{6-1}$.

By the inclusion-exclusion principle, we still need to see whether it is possible for two of the inequalities $y \geq 9$, $x_2 \geq 10$, $\dots$, $x_6 \geq 10$ to hold simultaneously. If this could happen, then those cases should be added up again. Indeed, such cases cannot happen due to the fact that $y + x_2 + x_3 + \cdots + x_6 = 17$. Thus, the final answer is

$$\binom{22}{5} - \binom{13}{5} - 5 \times \binom{12}{5} = 26334 - 1287 - 5 \times 792 = 21087.$$

$\square$

1.2 Recurrence Relations

In some cases, when we count step by step, we cannot simply count by using the multiplication principle as the steps are not independent. However, usually we can still obtain some relations between consecutive steps. For example, if a_n denotes the

number of ways to finish n steps, then we may set up a relation between a_n and a_{n+1} (or even more terms). This sets up a recurrence relation, and we can use some algebraic methods to find the general term a_n .

Before going through the application in counting, we first review the process of finding the general terms of some simple recurrence relations.

Proposition 1.3. Let $\{a_n\}$ be a sequence of real numbers.

(a) If $a_n = ca_{n-1} + f(n)$ for some constant c and some function f . Then

$$a_n = c^n a_0 + f(n) + cf(n-1) + c^2 f(n-2) + \cdots + c^{n-1} f(1).$$

(b) If $a_n = ta_{n-1} + ua_{n-2}$ for some constants t and u where $u \neq 0$, consider the *characteristic equation* (特徵方程)

$$\lambda^2 - t\lambda - u = 0.$$

- If it has two distinct (complex) roots α, β , then there exist constants A, B such that

$$a_n = A\alpha^n + B\beta^n.$$

- If it has a double root α , then there exist constants A, B such that

$$a_n = A\alpha^n + Bn\alpha^n.$$

Proof.

(a) Adding the equations

$$\begin{aligned} a_n &= ca_{n-1} + f(n), \\ ca_{n-1} &= c^2 a_{n-2} + cf(n-1), \\ c^2 a_{n-2} &= c^3 a_{n-3} + c^2 f(n-2), \\ &\vdots \\ c^{n-1} a_1 &= c^n a_0 + c^{n-1} f(1), \end{aligned}$$

we obtain the desired formula.

(b) See [proposition 4.6](#) of **The Very Basic Note in Algebra 1.5.1**.

□

Note that the constants A and B in the second result can be determined by the initial conditions, i.e. the values of a_0 and a_1 (or a_1 and a_2).

Problem 1.5. (IMO Prelim 2016 Problem 17) 10 boys are seated in a row, and each of them puts on a red, green or blue hat randomly. What is the probability that everyone finds at least one neighbour (another boy sitting on his left or right) who has put on a hat of the same colour as his own?

Analysis. We can generalize the same problem to n boys instead of 10. The condition does not allow us to count by assigning hats to the boys one by one. Indeed, we have to consider some neighbours as a whole and deduce some recurrence relations.

Solution. Let a_n be the number of ways to assign hats (in one of three given colours) to n boys in a row such that each boy finds at least one neighbour whose hat has the same colour as his. Clearly, we have $a_1 = 0$ and $a_2 = 3$.

Consider a row of n boys with $n \geq 3$. Note that the hat of the n th boy must have the same colour as that of the $(n-1)$ st boy. If this colour is not the same as that of the $(n-2)$ nd boy, then there are a_{n-2} ways to assign hats to the first $n-2$ boys by definition, and there are 2 ways to assign hats to the last 2 boys so that the colour is distinct from that of the $(n-2)$ nd boy.

If the colour of hats of the $(n-2)$ nd boy is the same as that of the $(n-1)$ st boy, then there are a_{n-1} ways to assign hats to the first $n-1$ boys. Then the colour of hat of the last boy is uniquely determined.

Therefore, we have established a recurrence relation $a_n = a_{n-1} + 2a_{n-2}$ for $n \geq 3$. Solving the characteristic equation $\lambda^2 - \lambda - 2 = 0$, we obtain two distinct roots 2 and -1 . This shows

$$a_n = A \cdot 2^n + B \cdot (-1)^n$$

for some constants A, B . Putting $n = 2$ and $n = 1$, we need to solve

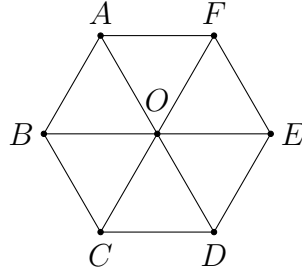
$$\begin{cases} 4A + B = 3, \\ 2A - B = 0. \end{cases}$$

This easily gives $A = \frac{1}{2}$ and $B = 1$. This means $a_n = 2^{n-1} + (-1)^n$ for all integers $n \geq 1$. As it is clear that there are 3^n ways to distribute the hats randomly without constraint, the probability is $\frac{2^{n-1} + (-1)^n}{3^n}$. In particular, when $n = 10$, we obtain

$$a_{10} = \frac{513}{3^{10}} = \frac{19}{2187}.$$

□

Problem 1.6. (CHKMO 2003 Problem 2) Let $ABCDEF$ be a regular hexagon of side length 1, and O be the centre of the hexagon. In addition to the sides of the hexagon, line segments are drawn from O to each vertex, making a total of twelve unit line segments. Find the number of paths of length 2003 along these line segments that start at O and terminate at O .



Analysis. This is another standard example which can be solved using recurrence relations. To construct a path that terminates at O , the second last step must be one of A, B, C, D, E, F . Therefore, in order to work out a recurrence relation of the number of paths of different lengths, we also need to know the number of such paths that terminate at a point different from O . Thus, it is more convenient to introduce two sequences corresponding to different terminating points.

Solution. For each integer $n \geq 0$, let a_n be the number of paths of length n starting at O and terminating at O . Also, let b_n be the number of paths of length n starting at O and terminating at A . By symmetry, b_n is also the number of those paths terminating at any of B, C, D, E, F .

To count a_n for $n \geq 1$, the $(n-1)$ st step must terminate at one of A, B, C, D, E, F . For each of these 6 points, there are b_{n-1} paths terminating at that point. This gives a total of $6b_{n-1}$ paths. In other words, we have

$$a_n = 6b_{n-1}. \quad (1)$$

To count b_n for $n \geq 1$, suppose the n th step terminates at A . Then the $(n-1)$ st step may terminate at B, F or O . If it is B or F , then there are b_{n-1} paths. If it is O , then there are a_{n-1} paths. Therefore, we have

$$b_n = 2b_{n-1} + a_{n-1}. \quad (2)$$

We now combine (1) and (2) to obtain a recurrence relation for the sequence $\{a_n\}$. This can be done by replacing every b_j by a_{j+1} in view of (1). We have

$$\frac{1}{6}a_{n+1} \stackrel{(1)}{=} b_n \stackrel{(2)}{=} 2b_{n-1} + a_{n-1} \stackrel{(1)}{=} \frac{1}{3}a_n + a_{n-1}.$$

This can be rewritten as $a_n = 2a_{n-1} + 6a_{n-2}$ for $n \geq 2$. The roots of the characteristic equation $\lambda^2 - 2\lambda - 6 = 0$ are $1 \pm \sqrt{7}$. Suppose $a_n = A(1 + \sqrt{7})^n + B(1 - \sqrt{7})^n$. Using the initial conditions $a_0 = 1$ and $a_1 = 0$, we solve

$$\begin{cases} A(1 + \sqrt{7}) + B(1 - \sqrt{7}) = 0, \\ A + B = 1. \end{cases}$$

This yields $A = \frac{7 - \sqrt{7}}{14}$ and $B = \frac{7 + \sqrt{7}}{14}$. Therefore, the answer is

$$a_{2003} = \frac{7 - \sqrt{7}}{14}(1 + \sqrt{7})^{2003} + \frac{7 + \sqrt{7}}{14}(1 - \sqrt{7})^{2003}.$$

□

Remarks. Note that the same recurrence relation holds for the sequence $\{b_n\}$, i.e. $b_n = 2b_{n-1} + 6b_{n-2}$. This is a common phenomenon of this kind of problem.

Sometimes we need not find the general term of a recurrence sequence. As long as we have obtained some recurrence relations, we may be able to deduce some useful relations between various quantities. We illustrate this idea in the following problem.

Problem 1.7. (APMO 2017 Problem 3) Let $A(n)$ denote the number of sequences $a_1 \geq a_2 \geq \dots \geq a_k$ of positive integers for which $a_1 + \dots + a_k = n$ and each $a_i + 1$ is a power of two ($i = 1, 2, \dots, k$). Let $B(n)$ denote the number of sequences $b_1 \geq b_2 \geq \dots \geq b_m$ of positive integers for which $b_1 + \dots + b_m = n$ and each inequality $b_j \geq 2b_{j+1}$ holds ($j = 1, 2, \dots, m-1$).

Prove that $A(n) = B(n)$ for every positive integer n .

Analysis. A natural strategy is to establish a certain recurrence relation for each of the sequences $\{A(n)\}$ and $\{B(n)\}$, and complete the proof by induction. Therefore, we first consider some small cases to look for some possible relations.

When $n = 1$, the only representation is $1 = 1$, and hence $A(1) = B(1) = 1$.

When $n = 2$, we have $A(2) = 1$, which corresponds to $1 + 1 = 2$, and we have $B(2) = 1$, which corresponds to $2 = 2$.

When $n = 3$, we have $A(3) = 2$, which corresponds to $1 + 1 + 1 = 3$ and $3 = 3$, and we have $B(3) = 2$, which corresponds to $2 + 1 = 3$ and $3 = 3$.

When $n = 4$, we have $A(4) = 2$, which corresponds to $1 + 1 + 1 + 1 = 4$ and $3 + 1 = 4$, and we have $B(4) = 2$, which corresponds to $3 + 1 = 4$ and $4 = 4$. Similarly, we find that $A(5) = B(5) = 2$.

When $n = 6$, we have $A(6) = 3$, which corresponds to $1 + 1 + 1 + 1 + 1 + 1 = 6$, $3 + 1 + 1 + 1 = 6$ and $3 + 3 = 6$, and we have $B(6) = 3$, which corresponds to $5 + 1 = 6$, $4 + 2 = 6$ and $6 = 6$.

When $n = 7$, we have $A(7) = 4$, which corresponds to $1 + 1 + 1 + 1 + 1 + 1 + 1 = 7$, $3 + 1 + 1 + 1 + 1 = 7$, $3 + 3 + 1 = 7$ and $7 = 7$, and we have $B(7) = 4$, which corresponds to $4 + 2 + 1 = 7$, $6 + 1 = 7$, $5 + 2 = 7$ and $7 = 7$.

We should now have some feeling on these sequences. Firstly, the solutions corresponding to $A(n)$ and $B(n)$ are not the same (although we need to show that the numbers of solutions are the same). Secondly, solutions corresponding to $A(n)$ with smaller number of variables is somehow related to solutions corresponding to $B(n)$ with larger number of variables. For example, when $n = 2^s - 1$ (e.g. $n = 7$), the special solution $n = n$ contributes 1 to $A(n)$, while the special solution (in the sense that each variable is twice of the next variable) $2^{s-1} + 2^{s-2} + \dots + 1 = n$ contributes 1 to $B(n)$. These contributions only arise when n changes from $2^s - 2$ to $2^s - 1$. Based on

this observation, we may try to connect the sequences $\{A(n)\}$ and $\{B(n)\}$ via the number of variables or the size of the variables.

Solution. For any positive integers n, s , let $A(n, s)$ be the number of solutions to $a_1 + \dots + a_k = n$ counted by $A(n)$ with the restriction that the largest variable a_1 is equal to $2^s - 1$. Let $B(n, s)$ be the number of solutions to $b_1 + \dots + b_m = n$ counted by $B(n)$ with the restriction that the number of variables m is equal to s . In view of the small cases, we prove the following.

Claim. For any positive integers n, s , we have $A(n, s) = B(n, s)$.

Proof. We use induction on n . When $n = 1$, we have $A(1, 1) = 1 = B(1, 1)$ and $A(1, s) = 0 = B(1, s)$ for $s \geq 2$. These can be verified directly.

Now, we proceed to the inductive step and work on the case n . Note that both $A(n, s)$ and $B(n, s)$ are zero if $n < 2^s - 1$. For the special case $n = 2^s - 1$, we have $A(n, s) = 1$ as the only solution is $2^s - 1 = 2^s - 1$, and $B(n, s) = 1$ as the only solution is $2^{s-1} + 2^{s-2} + \dots + 1 = 2^s - 1$. In the following, we consider the general case where $n > 2^s - 1$.

For $A(n, s)$, we first remove the term $a_1 = 2^s - 1$ on the left-hand side of $a_1 + \dots + a_k = n$. Then we have to solve $a_2 + \dots + a_k = n - (2^s - 1)$ under the given constraints and $a_2 \leq 2^s - 1$. This corresponds to the cases $A(n - 2^s + 1, t)$ with $t = 1, 2, \dots, s$. This gives the recurrence relation

$$A(n, s) = A(n - 2^s + 1, 1) + A(n - 2^s + 1, 2) + \dots + A(n - 2^s + 1, s). \quad (1)$$

For $B(n, s)$, in view of (1), we should reduce the equation $b_1 + \dots + b_s = n$ to a sum which is equal to $n - 2^s + 1$. Since the condition $b_j \geq 2b_{j+1}$ should be kept (and motivated by the special solution as discussed), we define $c_j = b_j - 2^{s-j}$ for $j = 1, 2, \dots, s$. Then the equation becomes $c_1 + \dots + c_s = n - 2^s + 1$. Note that $b_j \geq 2b_{j+1}$ if and only if (iff) $c_j \geq 2c_{j+1}$ for each $j = 1, 2, \dots, s - 1$. This shows we are reducing to cases counted in $B(n - 2^s + 1)$. Although we write $c_1 + \dots + c_s$ here, it does not mean that the equation consists of s variables. This is because each c_j here could be 0. If we remove the zero terms, the equation consists of at most s variables. This gives

$$B(n, s) = B(n - 2^s + 1, 1) + B(n - 2^s + 1, 2) + \dots + B(n - 2^s + 1, s). \quad (2)$$

Comparing (1) and (2), we find that $A(n, s) = B(n, s)$ by induction. $\square$

Finally, by the claim, we easily obtain

$$A(n) = A(n, 1) + A(n, 2) + \dots = B(n, 1) + B(n, 2) + \dots = B(n)$$

as desired. $\square$

1.3 Bijections

A bijection is a function for which every element in the codomain has a unique preimage in the domain. This means there is a correspondence between the elements in the domain and the elements in the codomain, or the function has an inverse. When the domain and the codomain are finite sets, a simple corollary is that they have the same size.

For the application in counting, we can try to rephrase the things that we need to count to some other things through a bijection. Sometimes the counting job becomes easier using different formulations. Indeed, we have seen some examples in previous subsections. For example, when we solve a Diophantine equation of the form $x_1 + x_2 + \cdots + x_k = n$ in [problem 1.4](#), we use balls and sticks to transform the problem. This reduces the problem to a counting problem which can be easily solved using binomial coefficients.

Problem 1.8. (Austria MO Final Round 2019 Problem 3) In the country of Oddland, there are stamps with values 1 cent, 3 cent, 5 cent, etc., one type for each odd number. The rules of Oddland Postal Services stipulate the following: for any two distinct values, the number of stamps of the higher value on an envelope must never exceed the number of stamps of the lower value.

In the country of Squareland, on the other hand, there are stamps with values 1 cent, 4 cent, 9 cent, etc., one type for each square number. Stamps can be combined in all possible ways in Squareland without additional rules.

Prove for every positive integer n : In Oddland and Squareland there are equally many ways to correctly place stamps of a total value of n cent on an envelope. Rearranging the stamps on an envelope makes no difference.

Analysis. Equivalently, we need to prove the following result: For every positive integer n , the number of nonnegative integer solutions to $a_1 + 3a_2 + 5a_3 + \cdots = n$ with $a_1 \geq a_2 \geq a_3 \geq \cdots$ is equal to the number of nonnegative integer solutions to $b_1 + 4b_2 + 9b_3 + \cdots = n$. Instead of counting the number of solutions to each equation, a much simpler idea is to find a bijection between the sets containing the solutions to the two equations. In other words, we have to find a certain way to pair up a solution $(a_1, a_2, \dots)$ with a solution $(b_1, b_2, \dots)$ such that all solutions are paired up. This idea is very similar to that in [problem 1.7](#), but this problem is much simpler. The key observation is the identity

$$1 + 3 + 5 + \cdots + (2k - 1) = k^2$$

that relates odd numbers with square numbers.

Solution. Suppose we use a_i stamps with values $2i - 1$ cent in Oddland for each i , and use b_j stamps with values j^2 cent in Squareland for each j .

Let

$$A = \left\{ (a_1, a_2, \dots) : a_i \in \mathbb{N}_0, \sum_{i \geq 1} (2i-1)a_i = n, a_1 \geq a_2 \geq \dots \right\},$$

$$B = \left\{ (b_1, b_2, \dots) : b_j \in \mathbb{N}_0, \sum_{j \geq 1} j^2 b_j = n \right\}.$$

We need to prove $|A| = |B|$. Indeed, for each $(b_1, b_2, \dots) \in B$, define $a_i = b_i + b_{i+1} + \dots$ for every i . Then $a_1 \geq a_2 \geq \dots$ and

$$\begin{aligned} a_1 + 3a_2 + 5a_3 + \dots &= (b_1 + b_2 + b_3 + \dots) + 3(b_2 + b_3 + \dots) + 5(b_3 + \dots) + \dots \\ &= b_1 + (1+3)b_2 + (1+3+5)b_3 + \dots \\ &= b_1 + 4b_2 + 9b_3 + \dots \\ &= n. \end{aligned}$$

(For a more rigorous justification, the coefficient of b_j is $1 + 3 + 5 + \dots + (2j-1) = j^2$.) This shows $(a_1, a_2, \dots) \in A$. Note that this mapping is bijective because it has an inverse

$$(a_1, a_2, \dots) \mapsto (b_1, b_2, \dots) = (a_1 - a_2, a_2 - a_3, \dots).$$

As it is obvious that A and B are finite sets, this implies $|A| = |B|$ as desired. $\square$

Problem 1.9. (International Zhautykov Olympiad 2015 Problem 5) Let A_n be the set of partitions of the sequence $1, 2, \dots, n$ into several subsequences such that every two neighbouring terms of each subsequence have different parities, and B_n the set of partitions of the sequence $1, 2, \dots, n$ into several subsequences such that all the terms of each subsequence have the same parity (for example, the partition $(1, 4, 5, 8), (2, 3), (6, 9), (7)$ is an element of A_9 , and the partition $(1, 3, 5), (2, 4), (6)$ is an element of B_6).

Prove that for every positive integer n the sets A_n and B_{n+1} contain the same number of elements.

Analysis. Again, it is unlikely that we can find $|A_n|$ and $|B_{n+1}|$ directly. Even if we can do so, it is usually not the best method. The best idea is still to find a bijection between A_n and B_{n+1} .

To find a suitable bijection, we may work on the small cases and look for any pattern. When $n = 1$, we have

$$A_1 = \{(1)\},$$

$$B_2 = \{(1)(2)\}.$$

This means (1) is mapped to $(1)(2)$. When $n = 2$, we have

$$A_2 = \{(1)(2), (1, 2)\},$$

$$B_3 = \{(1)(2)(3), (1, 3)(2)\}.$$

In view of the case $n = 1$, it may be more natural to map $(1)(2)$ to $(1)(2)(3)$ (since a single (a) is mapped to $(b)(c)$). Thus, $(1, 2)$ should be mapped to $(1, 3)(2)$. When $n = 3$, we have

$$A_3 = \{(1)(2)(3), (1, 2)(3), (2, 3)(1), (1, 2, 3)\},$$

$$B_4 = \{(1)(2)(3)(4), (1, 3)(2)(4), (2, 4)(1)(3), (1, 3)(2, 4)\}.$$

Again, $(1)(2)(3)$ is mapped to $(1)(2)(3)(4)$. In view of the case $n = 2$, $(1, 2)(3)$ should be mapped to $(1, 3)(2)(4)$, while $(2, 3)(1)$ should be mapped to $(2, 4)(1)(3)$. Then $(1, 2, 3)$ must be mapped to $(1, 3)(2, 4)$. By listing out more and more ‘natural’ pairs for larger n , we may eventually find a simple way to express this correspondence. This is the hint to the solution below.

Solution. To form a partition of $1, 2, \dots, n$ into subsequences, we can draw some arrows from a smaller number to a larger number that indicate pairs of consecutive terms in a subsequence. For example, when $n = 7$, the partition $(1, 2, 5, 6)(3, 4)(7)$ can be interpreted as

$$1 \longrightarrow 2 \longrightarrow 5 \longrightarrow 6, \quad 3 \longrightarrow 4, \quad 7.$$

Now, given any partition $x \in A_n$, we construct a partition y of $1, 2, \dots, n+1$ as follows: We draw an arrow from i to $j+1$ iff there is an arrow from i to j in the partition x . For example, if $x = (1, 2, 5, 6)(3, 4)(7)$ with arrows

$$1 \longrightarrow 2, 2 \longrightarrow 5, 5 \longrightarrow 6, 3 \longrightarrow 4,$$

the arrows in y are

$$1 \longrightarrow 3, 2 \longrightarrow 6, 5 \longrightarrow 7, 3 \longrightarrow 5.$$

This gives $y = (1, 3, 5, 7)(2, 6)(4)(8)$. We check that this always gives a partition in B_{n+1} . Firstly, each arrow in x is of the form $i \longrightarrow j$ where $1 \leq i < j \leq n$, and i and j have different parities. Therefore, $1 \leq i < j+1 \leq n+1$, and i and $j+1$ have the same parity. Secondly, since each i in the partition x can only be followed by at most one term, each i in y can only be followed by at most one term. Similarly, each i in y can only follow at most one other term. From these, we see that y is a partition of $1, 2, \dots, n+1$ in B_{n+1} .

It is clear that the above mapping is bijective, as the inverse is to map each arrow $i \longrightarrow j+1$ in y to the arrow $i \longrightarrow j$ in x (note that we must have $i < j$ since i and $j+1$ have the same parity and $i < j+1$). Therefore, we have $|A_n| = |B_{n+1}|$. $\square$

In the next example, we are not using bijection to show that two sets have equal size. Rather than that, we use bijection to rephrase the question to a simpler question to work on.

Problem 1.10. (CMO 2001 Problem 5) The circumference of a circle with perimeter 24 is divided into 24 equal parts. We choose 8 points from these 24 points of division so that the arc length formed between any two points is not equal to 3 or 8. How many different ways are there to choose the 8 points satisfying the constraint?

Analysis. A natural way is to use numbers to restate the problem. This means we need to choose 8 numbers from $0, 1, \dots, 23$ so that no 2 chosen numbers differ by 3 or 8 (mod 24). Since a difference of 3 or 8 is important, we may further rearrange the numbers so that neighbouring numbers differ by 3 or 8. This reduces the problem to a simple counting problem.

Solution. Label the points from 0 to 23 in this order. We rearrange these 24 numbers as follows:

0	3	6	9	12	15	18	21
8	11	14	17	20	23	2	5
16	19	22	1	4	7	10	13

The key feature of this table is that any two numbers from 0 to 23 differ by 3 or 8 (mod 24) iff they are neighbours in this table. Here, 0, 21 are neighbours, 0, 16 are neighbours, etc. Hence, we only need to count the number of ways to choose 8 integers from this table so that no two chosen numbers are neighbours.

First of all, we can only choose one number from each column because any two numbers in the same column are neighbours. As there are 8 columns and we have to choose 8 numbers, we need to choose exactly one number from each column. The problem can now be solved easily using recurrence relation.

For any positive integer n , consider a $3 \times n$ table. Let a_n be the number of ways to choose one cell from each column so that no two chosen cells are neighbours of each other (recall our convention of neighbours). We have to find a_8 .

Suppose $n \geq 3$. If the chosen cell in column $n - 1$ lies in a different row from the chosen cell in column 1, then there are a_{n-1} ways to choose the cells in the first $n - 1$ columns. In order that the condition is satisfied, the chosen cell in column n is uniquely determined.

If the chosen cell in column $n - 1$ lies in the same row as the chosen cell in column 1, then the chosen cell in column $n - 2$ lies in a different row from the chosen cell in column 1. There are a_{n-2} ways to choose the cells in the first $n - 2$ columns (and hence $n - 1$ columns). Also, there are 2 ways to choose the cell in column n .

We obtain the recurrence relation $a_n = a_{n-1} + 2a_{n-2}$ for $n \geq 3$. The initial conditions are $a_1 = 0$ (since the first column is a neighbour of the last column which is itself) and $a_2 = 6$. Since we are only interested in a_8 , there is no need to find the general term. We have

$$\{a_n\}_{n \geq 1} = 0, 6, 6, 18, 30, 66, 126, 258, \dots$$

The answer is $a_8 = 258$. □

Problem 1.11. (IMO Shortlist 2002 C3) Let n be a positive integer. A sequence of n positive integers (not necessarily distinct) is called *full* if it satisfies the following condition: for each positive integer $k \geq 2$, if the number k appears in the sequence

then so does the number $k-1$, and moreover the first occurrence of $k-1$ comes before the last occurrence of k . For each n , how many full sequences are there?

Analysis. It is not hard to guess there are $n!$ full sequences by considering small cases. Since $n!$ counts the number of permutations of $1, 2, \dots, n$, we may try to think of a bijection between the two kinds of sequences. In other words, we have to label the terms in a full sequence from 1 to n so that two full sequences are labelled differently and every label corresponds to a full sequence. From the condition, such a labelling should be related to the size and order of the integers in the sequence.

For example, when $n = 3$, the full sequences are

$$(1, 1, 1), (1, 1, 2), (1, 2, 1), (1, 2, 2), (2, 1, 2), (1, 2, 3).$$

It is natural to label the sequence $(1, 2, 3)$ by $(1, 2, 3)$. Thus, we may try to label the terms in increasing order. For the sequence $(1, 1, 2)$, the largest term 2 should be labelled by 3. As the label $(1, 2, 3)$ has been used, we can only label this sequence by $(2, 1, 3)$. Similarly, the only natural way to label the sequence $(1, 2, 2)$ is $(1, 3, 2)$. Afterwards, $(1, 2, 1)$ can only be labelled by $(2, 3, 1)$ (as $(1, 3, 2)$ has been used), while $(2, 1, 2)$ can only be labelled by $(3, 1, 2)$ (as $(2, 1, 3)$ has been used). Finally, $(1, 1, 1)$ is labelled by $(3, 2, 1)$.

Based on these trials, we may notice that the same number should be labelled from right to left. This leads to the following bijection.

Solution. Let S be the set of all full sequences and let T be the set of all permutations of $1, 2, \dots, n$. For any full sequence $(a_1, a_2, \dots, a_n)$, we consider a function f defined by $f(a_1, a_2, \dots, a_n) = (b_1, b_2, \dots, b_n)$, where $b_1, b_2, \dots, b_n$ is obtained by labelling the integers in $a_1, a_2, \dots, a_n$ using 1 to n in ascending order of size, and from right to left if the integers are equal. For example, the full sequence $(2, 1, 1, 2, 3)$ becomes $(4, 2, 1, 3, 5)$, while the full sequence $(2, 1, 2, 3, 1)$ becomes $(4, 2, 3, 5, 1)$. This clearly gives a mapping $f : S \rightarrow T$. We shall show that f is a bijection, so that $|S| = |T| = n!$.

It suffices to find an inverse for f . For any $(b_1, b_2, \dots, b_n) \in T$, we define another function g as follows. We first partition $1, 2, \dots, n$ into several (possibly only one) groups

$$(1, 2, \dots, x_1), (x_1 + 1, x_1 + 2, \dots, x_2), \dots, (x_{s-1} + 1, x_{s-1} + 2, \dots, x_s)$$

so that for any two numbers in the same group, the larger number appears on the left of the smaller number in the sequence $(b_1, b_2, \dots, b_n)$, and the number $x_i + 1$ appears on the right of x_i for $i = 1, 2, \dots, s-1$. For example, the sequence $(4, 2, 3, 5, 1)$ is put in the groups $(1, 2), (3, 4), (5)$. Next, for each $j = 1, 2, \dots, n$, if b_j belongs to the m th group, we define $a_j = m$. Then we define $g(b_1, b_2, \dots, b_n) = (a_1, a_2, \dots, a_n)$. For example, $g(4, 2, 3, 5, 1) = (2, 1, 2, 3, 1)$.

From the construction, it is obvious that the first occurrence of $k-1$ comes before the last occurrence of k in the sequence $(a_1, a_2, \dots, a_n)$. Therefore, $(a_1, a_2, \dots, a_n) \in S$. We now show that g is the inverse of f .

Suppose $f(a_1, a_2, \dots, a_n) = (b_1, b_2, \dots, b_n)$ and $g(b_1, b_2, \dots, b_n) = (c_1, c_2, \dots, c_n)$. Suppose $a_{i_1} = a_{i_2} = \dots = a_{i_v} = m$ with $i_1 < i_2 < \dots < i_v$ and all other terms are not m . Then $b_{i_1}, b_{i_2}, \dots, b_{i_v}$ are consecutive integers in descending order. Therefore, they belong to the same group in the definition of $g(b_1, b_2, \dots, b_n)$. Also, $b_{i_1} + 1$ appears on the right of b_{i_1} , while $b_{i_v} - 1$ (if exists) appears on the left of b_{i_v} . This shows $b_{i_1}, b_{i_2}, \dots, b_{i_v}$ are the only numbers in this group. Therefore, the terms for $m = 1$ correspond to group 1, the terms for $m = 2$ correspond to group 2, etc. This yields $c_{i_1} = c_{i_2} = \dots = c_{i_v} = m$. As this is true for any m , this gives $(a_1, a_2, \dots, a_n) = (c_1, c_2, \dots, c_n)$ and hence $g \circ f$ is the identity on S .

Suppose $g(b_1, b_2, \dots, b_n) = (a_1, a_2, \dots, a_n)$ and $f(a_1, a_2, \dots, a_n) = (d_1, d_2, \dots, d_n)$. We consider any term b_j which is the ℓ th term in the group $(x_{i-1} + 1, x_{i-1} + 2, \dots, x_i)$ (suppose $x_0 = 0$). Then $a_j = i$ by definition. Also, there are x_{i-1} terms in $(a_1, a_2, \dots, a_n)$ smaller than i , and ℓ terms in $(a_1, a_2, \dots, a_n)$ which are equal to i with indices larger than or equal to j . By definition of f , we have $d_j = x_{i-1} + \ell = b_j$. Hence, $f \circ g$ is the identity on T .

These yield $g = f^{-1}$ so that $|S| = |T| = n!$. □

Remarks. The rigorous proof that the function is bijective is a bit annoying, mainly due to the notations. However, it is rather obvious if one really understands the labelling.

Here is a summary of the counting methods covered in this section.

- addition principle (problems 1.1, 1.3)
- multiplication principle (problems 1.1, 1.2)
- permutation and combination (problems 1.3, 1.4)
- inclusion-exclusion principle (problem 1.4)
- recurrence relation (problems 1.5, 1.6, 1.7, 1.10)
- bijection (problems 1.3, 1.4, 1.7, 1.8, 1.9, 1.10, 1.11)

2 Extremal Combinatorics and Existence Problems

Extremal combinatorics refers to a kind of problems which ask for the optimal value of a quantity such that a certain thing can occur. Such a problem usually consists of two parts. Firstly, we need to show that the value cannot exceed the bound as claimed. Secondly, we have to show that such a bound is attainable. This second part is related to existence problems, which usually only requires a simple construction. Such a construction can often be inspired by small cases.

The following theorem may be useful in obtaining some bounds, though we will not provide any example using this result later on.

Theorem 2.1. (Bonferroni inequalities) Let $S_1, S_2, \dots, S_n$ be any finite sets. Then

$$\left| \bigcup_{i=1}^n S_i \right| \leq \sum_{i_1=1}^n |S_{i_1}| - \sum_{i_1 < i_2} |S_{i_1} \cap S_{i_2}| + \sum_{i_1 < i_2 < i_3} |S_{i_1} \cap S_{i_2} \cap S_{i_3}| - \dots + \sum_{i_1 < \dots < i_k} |S_{i_1} \cap \dots \cap S_{i_k}|$$

for odd $k \leq n$, and

$$\left| \bigcup_{i=1}^n S_i \right| \geq \sum_{i_1=1}^n |S_{i_1}| - \sum_{i_1 < i_2} |S_{i_1} \cap S_{i_2}| + \sum_{i_1 < i_2 < i_3} |S_{i_1} \cap S_{i_2} \cap S_{i_3}| - \dots - \sum_{i_1 < \dots < i_k} |S_{i_1} \cap \dots \cap S_{i_k}|$$

for even $k \leq n$.

This is the inequality version of the inclusion-exclusion principle, where we only consider the intersection of at most k sets on the right. The same method can be used to prove this result.

Proof. Consider any element a in $\bigcup_{i=1}^n S_i$. Suppose a belongs to exactly m of the sets $S_1, S_2, \dots, S_n$. We move everything to the left-hand side of the inequality. Then the contribution of a to the quantity is

$$f(m) = 1 - \binom{m}{1} + \binom{m}{2} - \binom{m}{3} + \dots + (-1)^k \binom{m}{k}.$$

By expanding the binomial coefficients, we find that f is a polynomial in m of degree k and leading coefficient $\frac{(-1)^k}{k!}$. Using the binomial theorem, we find that $f(m) = 0$ for $m = 1, 2, \dots, k$. Therefore, these are the only roots of f , and hence

$$f(m) = \frac{(-1)^k}{k!} (m-1)(m-2)\dots(m-k).$$

For odd k , we have $f(m) \leq 0$ for $m = 1, 2, \dots, n$, while for even k , we have $f(m) \geq 0$ for $m = 1, 2, \dots, n$. Also, it is clear that equality holds iff m is one of $1, 2, \dots, k$. This means every element in the union belongs to at most k sets. $\square$

2.1 Induction

Induction is the most basic technique which is commonly used in combinatorics, especially in some easier problems. Here we shall go through a few problems to demonstrate its usage.

Problem 2.1. (European Mathematical Cup 2021 Senior Problem 1) Alice drew a regular 2021-gon in the plane. Bob then labelled each vertex of the 2021-gon with a real number, in such a way that the labels of consecutive vertices differ by at most 1. Then, for every pair of non-consecutive vertices whose labels differ by at most 1, Alice drew a diagonal connecting them. Let d be the number of diagonals Alice drew. Find the least possible value that d can obtain.

Analysis. Usually, the large number 2021 has no important meaning. The problem is essentially the same if we replace 2021 by an arbitrary integer $n \geq 3$ (or sometimes odd integer $n \geq 3$). For problems whose answers depend on a parameter n , induction is always our first approach.

For example, to reduce the case from n to $n - 1$, what we need to do is to remove a vertex such that the labels of its two adjacent vertices differ by at most 1 (otherwise the condition is no longer satisfied after removing this vertex). Therefore, the key step is to show the existence of such a vertex. It is not hard to see how we should choose this vertex by writing down some examples (which we should do to construct an optimal example).

Solution. Instead of a regular 2021-gon, we consider an arbitrary convex polygon with $n \geq 3$ vertices. Consider any convex polygon $P_1P_2\cdots P_n$ where $n \geq 4$. Let a_j be the label of P_j for each j . Without loss of generality (WLOG), we may assume a_2 is the smallest. By the given condition, both a_1 and a_3 belong to the interval $[a_2, a_2 + 1]$. Thus, we must have $|a_1 - a_3| \leq 1$. This shows there is a diagonal joining P_1 and P_3 . If we remove the vertex P_2 , we are left with a convex polygon with $n - 1$ vertices such that the labels of consecutive vertices differ by at most 1 (including the pair a_1 and a_3). In other words, there is at least one more diagonal (P_1P_3) for the case n than the case $n - 1$.

For the base case $n = 3$, the number of diagonals must be 0. Therefore, it is easy to prove by induction that there are at least $n - 3$ diagonals for the case n using the above observation. This establishes the lower bound.

To prove that $n - 3$ is the upper bound, it suffices to provide an example of labelling such that exactly $n - 3$ diagonals are drawn. It is not particularly hard to construct an example. We can start from some small cases to look for some simple pattern. Alternatively, the above inductive step also provides some hint to the construction, i.e. the vertex with the smallest label should not be joined to any other vertex. For example, if the smallest label is 0, then the labels other than the adjacent two labels must be larger than 1.

Here gives one example for the case $n = 2021$ (which easily generalizes to other cases). The labels of the vertices are

$$0, 1, 2, \dots, 1009, 1010, 1009.1, \dots, 2.1, 1.1, 0.1$$

in this order. Then only the diagonals joining the pair $(k, k + 0.1)$ and the pair $(k, k - 0.9)$ (where $k = 1, 2, \dots, 1009$) are drawn. There are exactly $2018 = 2021 - 3$ diagonals.

Therefore, the answer is 2018. □

Problem 2.2. Each cell of an $m \times n$ board is in one of k given colours so that all colours are used. A *quadruple* is a group of four cells which are the intersections of any two rows and two columns. Find the maximum possible value of k such that there exists a colouring of the board for which no quadruple contains four distinct colours.

Analysis. If $m = 1$, then we can take $k = n$. If $m = 2$, then a simple way to make all quadruples contain a repeated colour is to use the same colour for the second row. Similarly, if we have more rows, we can use a single colour to colour an additional row. This provides a lower bound on k , which is very likely to be the optimal case as there does not seem to be any natural construction for the general case other than this. Using induction is convenient since the conditions for smaller cases form a subset of the conditions for larger cases.

This time we demonstrate the use of a two-dimensional induction, which means there are two parameters (m, n) . A typical inductive step is to assume all cases (m, n) with $m \leq m_0$, $n \leq n_0$ and $(m, n) \neq (m_0, n_0)$ hold, and work on the case (m_0, n_0) . The corresponding base cases are $(1, n)$ and $(m, 1)$ for arbitrary m, n .

Solution. We claim that the maximum possible value of k is $m + n - 1$.

We first provide a possible construction for $k = m + n - 1$. Suppose the colours are $C_1, C_2, \dots, C_{m+n-1}$. For the first row, we use the colours $C_1, C_2, \dots, C_n$ to colour the n cells respectively. Then for the i th row where $i = 2, 3, \dots, m$, we use the colour C_{n+i-1} to colour the whole row. All the colours are used. For any quadruple, at least two of its cells must lie in the same row which is not the first row. Then these two cells have the same colour, and hence the quadruple cannot contain four distinct colours.

Next, we prove that the number of colours cannot exceed $m + n - 1$ by induction. When $m = 1$ or $n = 1$, there are only $m + n - 1$ cells and hence this is trivially satisfied. Assume this bound holds for all cases (m, n) with $m \leq m_0$, $n \leq n_0$ and $(m, n) \neq (m_0, n_0)$. We consider the case (m_0, n_0) for some $m_0, n_0 \geq 2$.

Suppose on the contrary that at least $m_0 + n_0$ colours are used. WLOG assume $n_0 \geq m_0$ (we shall find this assumption useful later on). If we delete any column, then

we obtain an $m_0 \times (n_0 - 1)$ sub-board which satisfies the same condition. Therefore, we can apply the inductive hypothesis to show that at most $m_0 + n_0 - 2$ colours are used in this sub-board. This implies the deleted column contains at least 2 colours which are not used in any other columns. We say that a cell is *special* if the colour used in that cell is not used in any other columns. This means there are at least 2 special cells in each column. These special cells are useful because the colours of these special cells cannot be used in many other cells. It is likely that we can derive some contradiction by considering some quadruples with many special cells.

As there are at least $2n_0 \geq 2m_0$ special cells, there must be 2 special cells lying in the same row. WLOG assume cells $(1, 1)$, $(1, 2)$ are special (here, cell (x, y) is the cell in row x and column y). We can also assume another special cell in column 1 is $(2, 1)$. By the given constraint, the other special cell in column 2 cannot be $(2, 2)$. Thus, we may assume it is $(3, 2)$.

(1, 1)	(1, 2)
(2, 1)	(2, 2)
(3, 1)	(3, 2)

Consider the quadruple with cells $(1, 1)$, $(1, 2)$, $(2, 1)$, $(2, 2)$. The colour of $(2, 2)$ has to be the same as the colour of one of $(1, 1)$, $(1, 2)$, $(2, 1)$. But since $(1, 1)$ and $(2, 1)$ are special cells in column 1, the colour of $(2, 2)$ must be the same as that of $(1, 2)$. Similarly, by considering the quadruple with cells $(1, 1)$, $(1, 2)$, $(3, 1)$, $(3, 2)$, we conclude that the colour of $(3, 1)$ is the same as that of $(1, 1)$. But then $(2, 1)$, $(2, 2)$, $(3, 1)$, $(3, 2)$ will be a quadruple having four distinct colours. This is a contradiction.

Therefore, at most $m_0 + n_0 - 1$ colours can be used for the case (m_0, n_0) . By induction, we must have $k \leq m + n - 1$. Thus, the maximum possible value of k is $m + n - 1$. $\square$

Problem 2.3. (Czech-Polish-Slovak Match 2015 Problem 2 modified) A family of sets F is called perfect if the following condition holds: For every triple of sets $X_1, X_2, X_3 \in F$, at least one of the sets $(X_1 - X_2) \cap X_3$, $(X_2 - X_1) \cap X_3$ is empty. If F is a perfect family consisting of some subsets of $\{1, 2, \dots, n\}$, find the largest possible number of elements in F .

Analysis. The original problem provides the answer and only asks one to prove the upper bound. The problem is modified so that it is now a problem in extremal combinatorics. Apart from proving the bound, one also needs to guess the answer first.

Usually, the optimal case is an extreme case that is easy to construct. For example, in order that the the intersections are empty, the simplest case is that all pairs of sets in F are disjoint. To maximize $|F|$, we may put a single element in each set in

F . Together with the empty set, this gives a total of $n + 1$ pairwise disjoint sets. In other words, the answer is at least $n + 1$.

Since the above construction is one of the most extreme case, we may first assume it gives the optimal value, and attempt to prove $|F| \leq n + 1$. (However, we should be flexible and should not stick to one single guess when working on a problem.) Then the next question is how to establish this upper bound.

How should we proceed if we attempt to use induction? If we remove one or more elements from $\{1, 2, \dots, n\}$, we also need to remove all sets in F that contain some of these removed elements. Since we are going to establish an upper bound for $|F|$, we should remove some elements which only belong to a small number of sets in F . This can be done by using the given condition, since it allows us to conclude that some sets are disjoint.

Solution. The answer is $n + 1$. For example, F can be $\{\emptyset, \{1\}, \{2\}, \dots, \{n\}\}$. The condition is trivially satisfied, and we have $|F| = n + 1$. So it remains to prove $|F| \leq n + 1$ in general.

We shall prove by induction that $|F| \leq n + 1$ if F is a perfect family of some subsets of an n -element set S_n . When $n = 1$, there are only two subsets of S_1 , and so we must have $|F| \leq 2$. Now, assume this holds for all smaller cases.

If every pair of sets in F is disjoint, it is easy to see that there can be at most $n + 1$ sets, since there is at most one empty set and every nonempty set consists of a distinct element. Thus, we may assume not every pair of sets in F is disjoint. In other words, there exists $A \in S_n$ (not necessarily belonging to F) which is a subset of at least two elements in F . We may choose such a set A for which $|A|$ is maximized. For example, if $F = \{\{1, 2\}, \{1, 2, 3\}, \{1, 2, 4\}, \{1\}, \{3, 4, 5\}\}$, then we can choose $A = \{1, 2\}$ since it is a subset of three sets in F , and we cannot find 3 elements which belong to two sets in F .

Suppose the only elements in F containing A are $A \cup B_j$ for $j = 1, 2, \dots, k$ where $k \geq 2$. Then every pair of B_i and B_j is disjoint due to the maximality of $|A|$. For any other set $C \neq A \cup B_j$ in F , consider $(X_1, X_2, X_3) = (A \cup B_2, C, A \cup B_1)$. If $(A \cup B_2) - C$ and $A \cup B_1$ are disjoint, then C must contain everything in A as A belongs to both $A \cup B_2$ and $A \cup B_1$. However, this is impossible since otherwise C should be one of the sets $A \cup B_j$. Therefore, $C - (A \cup B_2)$ and $A \cup B_1$ must be disjoint. It follows that C cannot contain any element in B_1 as B_1 is disjoint with $A \cup B_2$. By symmetry, C is disjoint with every B_j for $j = 1, 2, \dots, k$. (Note that we can only conclude this given that $k \geq 2$.)

Based on our idea, we should now remove all elements in $B_1 \cup B_2 \cup \dots \cup B_k$. Suppose there are m elements in total which are removed.

- If none of the B_j 's is empty, then we must have $m \geq k$ as the B_j 's are pairwise disjoint. From above, only the sets $A \cup B_j$ in F contain some of the removed elements. Therefore, by removing these k sets from F , we are left with a family

of $|F| - k$ sets which are subsets of S_{n-m} . This family is perfect because the conditions are still satisfied. By induction (as $n-m < n$), the new family contains at most $n-m+1$ sets. Thus, we have $|F| \leq (n-m+1) + k \leq (n-m+1) + m = n+1$.

- If one of the B_j 's is empty, say B_1 , then we must have $m \geq k-1$. This time we only need to remove the sets $A \cup B_j$ for $j = 2, 3, \dots, k$ from F , since $A \cup B_1 = A$ does not contain any of the removed elements. We now have a perfect family whose elements are subsets of S_{n-m} . By induction, it contains at most $n-m+1$ sets, and hence $|F| \leq (n-m+1) + (k-1) \leq (n-m+1) + m = n+1$.

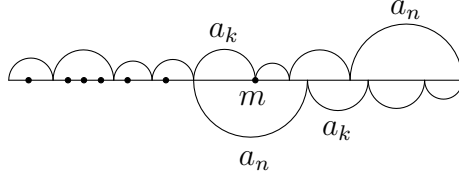
In both cases, we have $|F| \leq n+1$. So this upper bound holds for all n by induction. The proof is complete. $\square$

Problem 2.4. (IMO Shortlist 2009 C7/Problem 6) Let $a_1, a_2, \dots, a_n$ be distinct positive integers and let M be a set of $n-1$ positive integers not containing $s = a_1 + a_2 + \dots + a_n$. A grasshopper is to jump along the real axis, starting at the point 0 and making n jumps to the right with lengths $a_1, a_2, \dots, a_n$ in some order. Prove that the order can be chosen in such a way that the grasshopper never lands on any point in M .

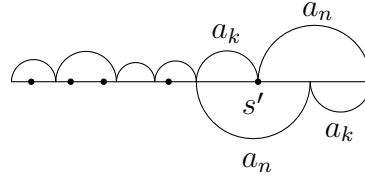
Analysis. It is natural to use induction to solve this problem. If we can use one step to jump over a point in M , then the claim follows from the inductive hypothesis easily. Therefore, it is natural to consider the largest jump first. We have to modify the jumps to handle the exceptional cases.

Solution. Suppose $a_1 < a_2 < \dots < a_n$. For $n = 1$, the statement is trivially satisfied since $|M| = 0$. Assume the statement holds for all smaller n . Firstly, we tend to use a_n as the last step.

- If $s' = a_1 + a_2 + \dots + a_{n-1} \notin M$ and there exists an $m \in M$ such that $s' < m < s$, then we can remove the step a_n and the points in M which lie between s' and s . Then we are reduced to the case $n-1$ since the size of the new set M' is at most $n-2$. If $|M'| < n-2$, we can add some points to M' arbitrarily to make it have size $n-2$. Therefore, we can apply the inductive hypothesis to choose an order of $a_1, \dots, a_{n-1}$ in such a way that the grasshopper can avoid the points in M' and hence the points in M .
- If $s' \notin M$ but there is no point in M lying between s' and s , we remove the step a_n and the largest point $m \in M$ and apply the inductive hypothesis. This gives a certain order of $a_1, \dots, a_{n-1}$ for which all the points in M other than m are avoided. If the path does not pass through m , then we are done. Otherwise, suppose the jump with step a_k lands on m for some $k = 1, 2, \dots, n-1$. Then we use a_n to replace a_k in that jump. As $a_n > a_k$, such a jump must jump over point m . Due to the maximality of m , the grasshopper will not land on any point in M afterwards. Thus, the remaining jumps can be done arbitrarily and we are done.



- If $s' \in M$ and s' is the largest point in M , we can argue in the same way as the second case. Indeed, we remove the step a_n and the point s' and apply the inductive hypothesis. This gives a path which is close to what we need, except that the second last jump with step a_k lands on s' . We can use a_n to replace a_k in that jump so that the point s' is jumped over. Next, we can use a_k as the last jump. This avoids every point in M .



- If $s' \in M$ and there exists another point $m \in M$ lying strictly between s' and s , we try to modify the jumps. We would like to use a_n as the second last step and some other a_j as the last step. Then the last two steps will jump over two points s' and m and hence we can apply the inductive hypothesis. This means it suffices to find an index j such that both $s - a_j$ and $s - a_j - a_n$ do not lie in M . Suppose on the contrary that at least one of them lies in M for each $j = 1, 2, \dots, n-1$. Note that

$$\begin{aligned} s - a_1 &> s - a_2 > \dots > s - a_{n-1} > s - a_n \\ &> s - a_1 - a_n > s - a_2 - a_n > \dots > s - a_{n-1} - a_n. \end{aligned}$$

Therefore, we have found n distinct points in M , one corresponding to each $j = 1, 2, \dots, n-1$ together with the point $s' = s - a_n$. This contradicts $|M| = n-1$. Therefore, such an index j must exist. Removing the steps a_n and a_j , and removing the points s' and m , we are in a position of the case $n-2$ and hence the inductive hypothesis can be applied. This case is thus solved.

By induction, we can always choose a path which avoids every point in M . This completes the proof. $\square$

Remarks. Except for the first case, we seem to get stuck in every other case in the inductive step. However, we can actually modify the jumps in some ways to meet the goal. This shows we should not reject an idea easily even though it cannot be applied in a straightforward way.

2.2 Pigeonhole Principle

Theorem 2.2. (**Pigeonhole principle**(抽屜原理/鴿巢原理)) Suppose there are n elements each of which belongs to exactly one of the sets $A_1, A_2, \dots, A_m$. Then there exist $\left\lceil \frac{n}{m} \right\rceil$ of these elements which belong to the same set.

The pigeonhole principle is rather a common sense. Another common proving technique, i.e. proof by contradiction, can essentially provide the same result. Because of this, some standard usages of the pigeonhole principle are less interesting. Usually, the hardest step in using the pigeonhole principle is to construct the sets and the elements in an appropriate way.

Problem 2.5. For any positive integer n , find the smallest m such that among any m integers, there must be some of them which add up to a multiple of n .

Analysis. It suffices to consider modulo n . In order to avoid the sums being multiples of n , the most natural way is to set all integers to be congruent modulo n . For example, if all chosen integers are of the form $kn + 1$, then we need at least n of them to give a sum of a multiple of n . This gives a lower bound for m , which seems to be the optimal value with some trial and error.

To prove the upper bound $m \leq n$, we need to construct some pigeonholes. Since we are considering modulo n , and we wish to show that something is congruent to 0 modulo n , it may be natural to use the n residue classes as the pigeonholes. If we can show by the pigeonhole principle that two objects (numbers), x_i and x_j , are congruent modulo n , then we have $n \mid x_i - x_j$. This suggests we should choose some objects such that each difference $x_i - x_j$ represents a sum of the given numbers. The following gives a classical construction for this purpose.

Solution. Consider the $n - 1$ integers $1, n + 1, \dots, n(n - 2) + 1$. Each of them is congruent to 1 modulo n . Therefore, the sum of any of these integers must congruent to k modulo n for some $k = 1, 2, \dots, n - 1$, which cannot be 0. Therefore, m is at least n .

For any n integers $a_1, a_2, \dots, a_n$, consider $s_1, s_2, \dots, s_n$ defined by

$$s_j = a_1 + a_2 + \dots + a_j.$$

If one of the s_j 's is a multiple of n , then we are done by choosing $a_1, a_2, \dots, a_j$. Otherwise, these n integers must fall in the other $n - 1$ residue classes modulo n . By the pigeonhole principle, there exist indices $i < j$ such that $s_i \equiv s_j \pmod{n}$. Then

$$a_{i+1} + a_{i+2} + \dots + a_j = s_j - s_i \equiv 0 \pmod{n},$$

and hence we can choose $a_{i+1}, a_{i+2}, \dots, a_j$. This shows n is the optimal value. $\square$

Problem 2.6. (CHKMO 2006 Problem 1) Let M be a subset of $\{1, 2, \dots, 2006\}$ with the following property: For any three elements x, y and z ($x < y < z$) of M , $x + y$ does not divide z . Determine the largest possible size of M .

Analysis. One of the optimal cases is to choose all the large elements, since we just need to ensure $x + y > z$ for all distinct $x, y, z \in M$. On the other hand, to show the existence of $x, y, z \in M$ satisfying $x + y \mid z$ for a certain size of M , it suffices to find $x, y, z \in M$ satisfying $x + y = z$. It may be even more convenient if we restrict z to be the largest element in M (so that there are more possible ways for two elements to sum up to z). Although these impose extra restrictions, it turns out that the bound is the same, but it is easier to work out a proof.

Solution. When $M = \{1003, 1004, \dots, 2006\}$, the sum of any two elements in M is at least 2007, which is larger than the largest element in M . Therefore, $x + y \nmid z$ for any $x, y, z \in M$. This gives a possible case for $|M| = 1004$.

Consider any subset M satisfying the constraint. Suppose the largest element of M is a . Then M contains at most one element in each of the pairs

$$(1, a-1), (2, a-2), \dots, \left(\left\lfloor \frac{a}{2} \right\rfloor, \left\lceil \frac{a}{2} \right\rceil\right)$$

(note that the last pair may contain the same number, which does not affect the validity of our claim). Therefore, M contains at most $\frac{a}{2}$ elements smaller than a . Since a is the largest element, M cannot contain elements larger than a . Therefore,

$$|M| \leq \frac{a}{2} + 1 \leq 1003 + 1 = 1004.$$

Hence, the largest possible size of M is 1004. $\square$

Problem 2.7. (CHKMO 2012 Problem 2) Some of the lattice points (x, y) , with $1 \leq x \leq 101$ and $1 \leq y \leq 101$ are marked so that no 4 marked points form the vertices of an isosceles trapezoid with bases parallel to the x -axis or the y -axis (a rectangle is counted as an isosceles trapezoid). Determine the maximum number of marked points. (A lattice point is a point with integral coordinates.)

Analysis. In establishing the upper bound, we only focus on those isosceles trapezoids with bases parallel to the x -axis. Each such isosceles trapezoid has vertices (x_1, y_1) , (x_2, y_1) , (x_3, y_2) and (x_4, y_2) with $x_1 + x_2 = x_3 + x_4$. Therefore, the objects in the application of the pigeonhole principle may be the sum of x -coordinates of a pair of points on the same horizontal line.

The construction for the optimal case is slightly harder than usual. We can get some hints by looking into the small cases.

Solution. Consider pairs of marked points with the same y -coordinates. If there exist two pairs of marked points with the same sum of x -coordinates and different y -coordinates, then the 4 points in these pairs are the vertices of an isosceles trapezoid.

Suppose there are $a_1, a_2, \dots, a_{101}$ marked points in the 101 horizontal lines respectively (this means there are a_1 marked points with y -coordinate equal to 1, etc.). For the horizontal line $y = 1$, suppose the marked points are $(x_1, 1), (x_2, 1), \dots, (x_{a_1}, 1)$ where $x_1 < x_2 < \dots < x_{a_1}$. Since

$$x_1 + x_2 < x_1 + x_3 < \dots < x_1 + x_{a_1} < x_2 + x_{a_1} < \dots < x_{a_1-1} + x_{a_1},$$

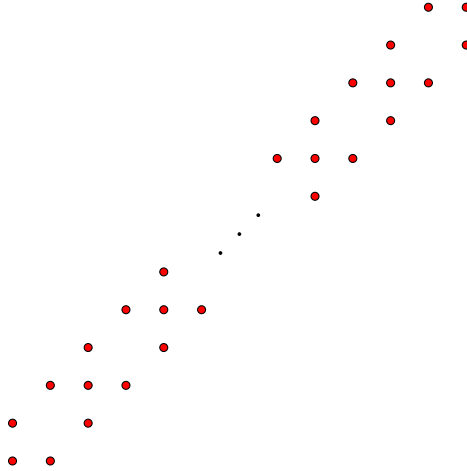
there are at least $2a_1 - 3$ distinct sums formed from the x -coordinates. This only holds for $a_1 \geq 2$. For $a_1 = 0$ or $a_1 = 1$, clearly, no sum can be formed. Therefore, there are at least

$$f(a_1) + f(a_2) + \dots + f(a_{101}) \tag{1}$$

sums where $f(n) = \max \{2n - 3, 0\}$. Each of these sums lies between $1 + 2 = 3$ and $100 + 101 = 201$. There are 199 possibilities. If (1) exceeds 199, then two of the sums are equal by the pigeonhole principle. Since the equal sums must correspond to distinct horizontal lines from the construction, we can find an isosceles trapezoid as suggested. Therefore, we may assume (1) is at most 199. Then

$$199 \geq f(a_1) + f(a_2) + \dots + f(a_{101}) \geq (2a_1 - 3) + (2a_2 - 3) + \dots + (2a_{101} - 3).$$

This implies $a_1 + a_2 + \dots + a_{101} \leq 251$. In other words, the number of marked points is at most 251. This can be attained, for example, from the following construction.



It is not hard to see that there is no isosceles trapezoid (a rigorous proof is to show that the sums of x -coordinates of pairs of points in the same horizontal line are all different, and the same holds for y -coordinates by symmetry). There are

$$3 + 5 + 49 + 3 = 251$$

points in total. □

2.3 Colouring

Colouring refers to the method of partitioning a certain set into several subsets and use the distinct properties of these subsets to proceed, usually using the pigeonhole principle to deduce useful results. It is a useful tool to solve many tiling problems.

Problem 2.8. Let a be a positive integer. Find all $m \times n$ rectangles which can be tiled by $1 \times a$ (and $a \times 1$) rectangles completely.

Analysis. For a tiling, we mean every cell is covered by exactly one rectangle. Therefore, an immediate necessary condition for this to hold is that the area of the $m \times n$ rectangle is a multiple of the area of the small rectangle. However, this is not sufficient. For example, we cannot use 1×4 rectangles to tile a 2×6 rectangle.

Indeed, it is only clear that a tiling is possible if one of m, n is divisible by a . Therefore, it may make a better sense that the answer is $a \mid m$ or $a \mid n$.

To use colouring, a standard way is to colour the cells using different colours such that each small rectangle must cover some cells of certain colours. For example, we may think of a colouring such that all the a cells covered by each small rectangle are of distinct colours.

Solution. If $a \mid n$, we can put $\frac{n}{a}$ copies of $1 \times a$ rectangles in each row to tile the whole rectangle. Similarly, a tiling exists when $a \mid m$. Now, assume $a \nmid m$ and $a \nmid n$.

We use (i, j) to denote the cell in row i and column j where $0 \leq i \leq m-1$ and $0 \leq j \leq n-1$. We use a colours $C_0, C_1, \dots, C_{a-1}$ to colour the cells of an $m \times n$ rectangle such that the cell (i, j) is coloured by C_r where r is the unique index satisfying $i + j \equiv r \pmod{a}$. No matter horizontal or vertical rectangles are used, each $1 \times a$ rectangle must cover a cells with pairwise distinct colours. Therefore, if a tiling exists, then the number of cells of each colour should be the same. We claim that this is not the case.

C_0	C_1	C_2	C_3	$\dots$
C_1	C_2	C_3	C_4	$\dots$
C_2	C_3	C_4	C_5	$\dots$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\ddots$

Let $m = q_1a + r_1$ and $n = q_2a + r_2$ where $1 \leq r_1, r_2 \leq a-1$. Clearly, there is an equal number of cells of each colour in the first q_1a rows. After removing these rows, there is an equal number of cells of each colour in the first q_2a columns. We further remove these columns as they will not affect the result. In other words, we may assume $q_1 = q_2 = 0$.

WLOG assume $r_1 \leq r_2$. The bottom right corner cell is of colour $C_{r_1+r_2-2}$ (where all indices are taken modulo a). Since $r_1 + r_2 - 2 < a + r_1 - 2$, the only cells of colour

C_{r_1-2} are $(0, r_1 - 2), (1, r_1 - 3), \dots, (r_1 - 2, 0)$, and the only cells of colour C_{r_1-1} are $(0, r_1 - 1), (1, r_1 - 2), \dots, (r_1 - 1, 0)$. Thus, there are $r_1 - 1$ cells of colour C_{r_1-2} and r_1 cells of colour C_{r_1-1} . The numbers are distinct, which means there cannot be any tiling.

C_0	C_1	C_2	$\dots$	C_{r_1-1}	$\dots$	C_{r_2-1}
C_1	C_2	C_3	$\dots$	C_{r_1}	$\dots$	C_{r_2}
C_2	C_3	C_4	$\dots$	C_{r_1+1}	$\dots$	C_{r_2+1}
$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$	$\ddots$	$\vdots$
C_{r_1-1}	C_{r_1}	C_{r_1+1}	$\dots$	C_{2r_1-2}	$\dots$	$C_{r_1+r_2-2}$

Therefore, a tiling exists iff $a \mid m$ or $a \mid n$. □

Problem 2.9. (IMO HK TST 2015 Test 2 Problem 4) A 11×11 grid is to be covered completely without overlapping by some 2×2 squares and L -shapes each composed of three unit cells. Determine the smallest number of L -shapes used. (Each shape must cover some grids entirely and cannot be placed outside the 11×11 grid. The L -shapes may be reflected or rotated when placed on the grid.)



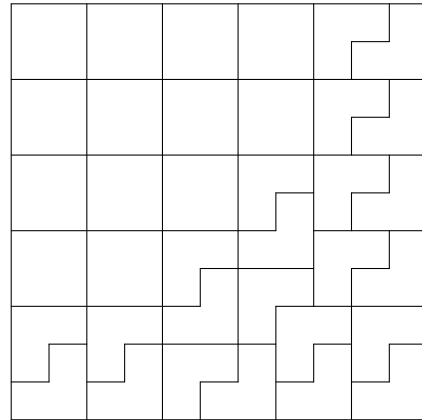
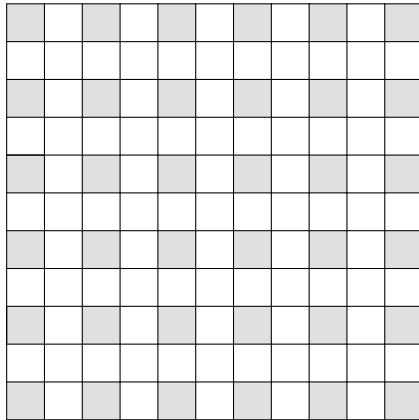
Analysis. Suppose x squares and y L -shapes are used. A simple counting gives $4x + 3y = 121$. This gives $y = 3, 7, 11, \dots$ and correspondingly, $x = 28, 25, 22, \dots$. To minimize y , we have to maximize x . However, it seems that the grid cannot contain too many squares, or otherwise it cannot be tiled completely. This suggests using colouring to find another constraint which has to be satisfied by x and y .

The square shape is less flexible. It is easy to colour the grid using four colours such that the four cells covered by each square are of different colours like the previous problem. Although each L -shape does not cover cells of all colours, we can still tell it covers cells of different colours. By counting the number of cells of each colour, we already get what we need. The following solution looks simpler as we only need to focus on one of the four colours (or one group of cells). But the idea is based on this.

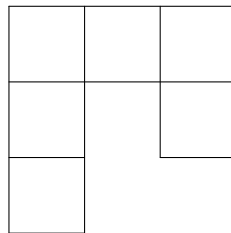
Solution. Suppose x squares and y copies of L -shapes are used. Then the total number of cells is $4x + 3y$, which should be equal to $11^2 = 121$.

We colour the cells as shown. Then every square and every L -shape covers at most one blackened cell. As there are 36 blackened cells, we have $x + y \geq 36$. Therefore,

$y = 4(x + y) - (4x + 3y) \geq 4(36) - 121 = 23$. The figure on the right gives one example using 23 copies of L -shapes. $\square$



Problem 2.10. (IMO Shortlist 2004 C7/Problem 3) Define a ‘hook’ to be a figure made up of six unit squares as shown below in the picture, or any of the figures obtained by applying rotations and reflections to this figure.

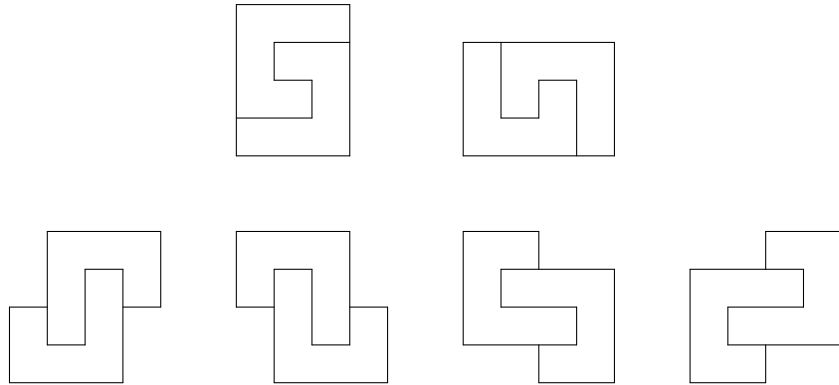


Determine all $m \times n$ rectangles that can be covered without gaps and without overlaps with hooks such that

- the rectangle is covered without gaps and without overlaps
- no part of a hook covers area outside the rectangle.

Analysis. Firstly, it is obvious that the hooks must come in pairs to form one of two possible shapes. Therefore, by counting the number of cells, we immediately know that mn is divisible by 12. However, this is not a sufficient condition for such a tiling to exist. Indeed, we shall use various colouring to obtain more information. The only idea is to colour the cells in various ways and see whether we can conclude anything on the number of coloured cells covered by each shape.

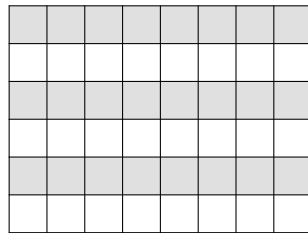
Solution. Note that the centre of a hook must be occupied by another hook. Thus, the hooks come in pairs in one of the following ways.



Thus, we may simply consider tiling using these shapes. We use R_1 and R_2 respectively to denote the shapes in the first row, and use S_1, S_2, S_3, S_4 respectively to denote the shapes in the second row.

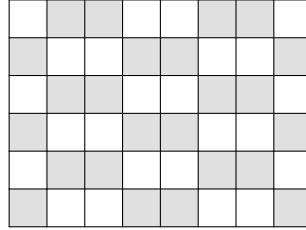
Since each shape consists of 12 cells, we must have $12 \mid mn$ if the tiling is possible. We first consider the hardest case $2 \mid m$ and $6 \mid n$. We would like to show that one of m, n is divisible by 4 so that this case is indeed contained in the other cases that we consider later.

We colour all the cells in the odd rows as shown below. Under this colouring, each R_1, S_1, S_2, S_3, S_4 occupies exactly 6 coloured cells, while each R_2 occupies 4 or 8 coloured cells. As m is even, the number of coloured cells is equal to the number of uncoloured cells. Hence, the number of R_2 's occupying 4 coloured cells is the same as the number of R_2 's occupying 8 coloured cells. This shows the number of R_2 's is even. If we colour the odd columns instead of rows, then the same argument shows that the number of R_1 's is even.



We further consider a colouring as shown below. This means for the odd rows, only cells in the k th column with $k \equiv 2, 3 \pmod{4}$ are coloured, while for the even rows, only cells in the k th column with $k \equiv 0, 1 \pmod{4}$ are coloured. Then each of R_1, R_2, S_1, S_2 occupies 6 coloured cells. For S_3 and S_4 , those which are at an even distance from the leftmost edge (and hence the rightmost edge) each occupies 6 coloured cells, while each of the remaining occupies 4 or 8 coloured cells. Therefore, there is an even number of shape of the second kind. Similarly, if we translate this

colouring to the left by 1 column (meaning that a cell (ℓ, k) is coloured if ℓ is odd and $k \equiv 1, 2 \pmod{4}$, or ℓ is even and $k \equiv 0, 3 \pmod{4}$), then the number of S_3 's and S_4 's which are at even distances from the leftmost edge is even. By combining these results, we find that the total number of S_3 's and S_4 's is even. By symmetry, the total number of S_1 's and S_2 's is also even.



Now, we have found that in the case $2 \mid m$ and $6 \mid n$, the number of shapes is even. Therefore, the number of cells is a multiple of 24. This shows at least one of m, n is divisible by 4. This will be covered in the following cases.

If $4 \mid m$ and $3 \mid n$, then we can use only R_1 to tile the rectangle in the natural way. Up to symmetry, it remains to consider the case where $12 \mid m$ and n is not divisible by 3 or 4. Since any $m \times 3$ or $m \times 4$ rectangle can be tiled, a tiling exists if n can be written in the form $3a + 4b$ for some nonnegative integers a, b . It is known that every n which is at least $(3 - 1)(4 - 1) = 6$ can be written in this form (indeed, for this particular case, it suffices to check that all residue classes $1, 2, 5, 7, 10, 11 \pmod{12}$ greater than 5 have such representations). Therefore, the only cases left are $n = 1, 2, 5$.

Since every shape occupies at least 3 columns, the cases $n = 1, 2$ cannot be tiled. For $n = 5$, no matter which shape we use to cover the $(1, 1)$ cell, that shape must occupy the $(3, 3)$ cell. Due to symmetry, the shape covering the $(1, 5)$ cell must also occupy the $(3, 3)$ cell. Thus, these two shapes overlap but they cannot be the same shape. This shows no tiling is possible.

We conclude that an $m \times n$ rectangle can be tiled if it belongs to at least one of the following cases:

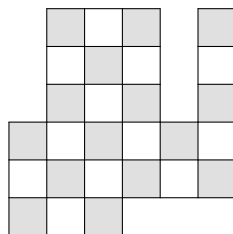
- $4 \mid m$ and $3 \mid n$; or
- $3 \mid m$ and $4 \mid n$; or
- $12 \mid m$ and $n \geq 7$; or
- $m \geq 7$ and $12 \mid n$.

□

Problem 2.11. (EMMO 2016) We use 1×2 dominoes to create a polygon which has angles 90° or 270° and does not have holes. Prove that it has at least one side of even length.

Analysis. Colouring is useful not only in tiling problems. We can also colour some objects in some ways to obtain additional information. For problems involving dominoes, it is useful to colour the cells black and white alternately such that adjacent cells have different colours. An immediate fact is that the number of black cells and the number of white cells must be the same. However, by drawing some polygons with interior angles 90° or 270° and odd side lengths only, we notice that such polygons should have more cells of one colour.

Solution. Suppose on the contrary that all sides have odd length. The polygon can be partitioned into unit cells so that each domino covers two adjacent cells. We colour all the cells black and white alternately. Suppose a corner cell is black. Since all sides have odd length, all the corner cells are black.



As each domino covers exactly one black cell and one white cell, the number of black cells and the number of white cells should be the same. However, we are going to show that the number of black cells in such polygons is indeed larger.

One way is to count the number of sides of black cells minus the number of sides of white cells. The key observation is that the interior black sides cancel out with the interior white sides because only cells of different colours are adjacent to each other. Thus, it suffices to count the sides that lie on the boundary of the polygon. For each side of the polygon, there is one more black side than white side because the colour changes alternately and the sides at the two ends are both black. Therefore, it is clear that there are more black sides than white sides in total, which means there are more black cells than white cells. This is a contradiction as explained. In other words, there must be a side of the polygon of even length. $\square$

2.4 Double Counting

Double counting is not a counting method. Indeed, it is a useful tool in solving combinatorial problems with the help of counting. The method is to count a certain quantity in two or more different ways to obtain a nontrivial equality. More generally, we can obtain bounds for that quantity to derive an inequality that is useful for our purpose. Many identities involving the binomial coefficients can be proved in this way. We have seen such an example when we prove Pascal's identity. We first go

through one more example which has nothing to do with existence problems, just to illustrate the idea of double counting.

Problem 2.12. Let n and m be positive integers satisfying $n \geq m$. Prove that

$$\sum_{i=0}^n \binom{n}{i} \binom{n-i}{m-i} = 2^m \binom{n}{m}.$$

Analysis. We first show how to use a combinatorial argument to prove an algebraic identity. Note that $\binom{n}{m}$ counts the number of ways to choose m objects from n objects. Also, $\binom{n}{i}$ and $\binom{n-i}{m-i}$ have the combinatorial meaning of choosing i and $m-i$ objects respectively, which add up to m objects. This gives a linkage between the two sides. We only need to see how to deal with the extra factor 2^m . For example, we may assign some colours to the m chosen objects.

Solution. Suppose we want to choose m objects from n objects and use one of the two colours, black and white, to colour each chosen object. There are obviously $\binom{n}{m} \times 2^m$ ways to do so.

Another way to count is as follows. We first choose i objects from the n objects and colour them black. Then we choose $m-i$ objects from the remaining $n-i$ objects and colour them white. There are $\binom{n}{i} \binom{n-i}{m-i}$ ways to do so for each fixed $i = 0, 1, \dots, n$. Summing over these i 's, we obtain the left-hand side of the identity. This establishes the identity. $\square$

Problem 2.13. (**Sperner's theorem**(斯佩納定理)) Let $\mathcal{F}$ be a family of subsets of $\{1, 2, \dots, n\}$ such that there do not exist two distinct elements $S, T \in \mathcal{F}$ satisfying $S \subset T$. Find the greatest possible value of $|\mathcal{F}|$.

Analysis. Recall that a family of sets means a set whose elements are sets. For example, $\{\{3\}, \{1, 2\}, \{2, 3\}\}$ is a family of subsets of $\{1, 2, 3\}$. But it does not satisfy the condition as $\{3\} \subset \{2, 3\}$.

To guess the answer, one may find a sufficient condition for two sets S, T not being subsets of each other. Indeed, if $|S| = |T|$ and they are distinct subsets, then we cannot have $S \subset T$ or $T \subset S$. Therefore, we can choose all the k -element subsets of $\{1, 2, \dots, n\}$. The number $\binom{n}{k}$ is maximized when $k = \left\lfloor \frac{n}{2} \right\rfloor$. This is indeed the answer.

We present an interesting proof to this famous theorem below, which demonstrates how to use double counting to establish an inequality.

Solution. We construct some permutations of $1, 2, \dots, n$ as follows. For each set $S \in \mathcal{F}$, we construct all those permutations such that the first $|S|$ elements are exactly the elements in S . For example, if $n = 5$ and $S = \{1, 2, 3\}$, we construct the following permutations of $1, 2, 3, 4, 5$:

1, 2, 3, 4, 5;	1, 2, 3, 5, 4;
1, 3, 2, 4, 5;	1, 3, 2, 5, 4;
2, 1, 3, 4, 5;	2, 1, 3, 5, 4;
2, 3, 1, 4, 5;	2, 3, 1, 5, 4;
3, 1, 2, 4, 5;	3, 1, 2, 5, 4;
3, 2, 1, 4, 5;	3, 2, 1, 5, 4.

Note that the permutations can be formed by permuting the $|S|$ elements in S , and the $n - |S|$ elements not in S . Therefore, each set S corresponds to $|S|!(n - |S|)!$ different permutations.

Note that all permutations of $1, 2, \dots, n$ constructed in this way are pairwise distinct. Indeed, suppose there are two permutations corresponding to the sets $S, T \in \mathcal{F}$ which are identical. WLOG assume $|S| \leq |T|$. Then the first $|S|$ elements in this permutation are exactly the elements in S , but they also belong to T by construction. This means $S \subset T$, which is a contradiction.

It follows that the total number of such permutations cannot exceed the total number of all possible permutations of $1, 2, \dots, n$. Clearly, there are $n!$ such permutations. Therefore, we have

$$\sum_{S \in \mathcal{F}} |S|!(n - |S|)! \leq n!.$$

This can be rewritten as

$$\sum_{S \in \mathcal{F}} \binom{n}{|S|}^{-1} \leq 1.$$

As $\binom{n}{|S|} \leq \binom{n}{\lfloor \frac{n}{2} \rfloor}$, this yields

$$\binom{n}{\lfloor \frac{n}{2} \rfloor} \geq \sum_{S \in \mathcal{F}} 1 = |\mathcal{F}|.$$

This gives an upper bound for $|\mathcal{F}|$. The upper bound can be attained when, for example, $\mathcal{F}$ contains all $\lfloor \frac{n}{2} \rfloor$ -element subsets of $\{1, 2, \dots, n\}$. Clearly, there do not exist two distinct S, T among these subsets such that $S \subset T$. $\square$

Problem 2.14. We fill in the numbers $1, 2, \dots, 101$ in a 101×101 table such that each number appears exactly 101 times. Prove that there exists a row or a column consisting of at least 11 distinct numbers.

Analysis. A standard way to use double counting is to count the number of pairs (a, b) with some special properties, where a belongs to a certain set A and b belongs to a certain set B . For each fixed $a \in A$, we count the number of $b \in B$ such that (a, b) has the special properties. Next, for each fixed $b \in B$, we count the number of $a \in A$ such that (a, b) has the special properties. This allows us to set up an identity, or sometimes an inequality if only bounds can be obtained.

In this problem, we need to work on numbers in some rows or columns. Therefore, one possible idea is to choose A to be a set of numbers and choose B to be a set of rows or columns.

Solution. We label the 101 rows and 101 columns $\ell_1, \ell_2, \dots, \ell_{202}$ (in any order). Consider some pairs (a, ℓ_j) where a and j are integers from 1 to 101. We say that a pair (a, ℓ_j) is *special* if the number a appears in the line (row or column) ℓ_j . Let N be the number of special pairs.

Firstly, for each fixed a , we wish to count the number of special pairs (a, ℓ_j) . Clearly, there is no way for us to work out the exact value as the table is arbitrary. However, we can find some bounds for this value. Suppose there are exactly x rows and y columns which contain at least one copy of a . Clearly, there can be at most xy copies of a in the whole table. It follows that

$$101 = \text{number of } a\text{'s in the table} \leq xy \leq \left(\frac{x+y}{2}\right)^2.$$

This implies $x+y \geq 2\sqrt{101} > 20$, or $x+y \geq 21$ since x and y are integers. (Note that double counting is also used to derive the above inequality as we have used two ways to count the number of a 's in the table.)

From above, we know that there are at least 21 special pairs (a, ℓ_j) for each fixed $a = 1, 2, \dots, 101$. This implies

$$N \geq 21 \times 101 = 2121. \quad (1)$$

Secondly, if we suppose on the contrary that each row or column consists of at most 10 distinct numbers, then there are at most 10 special pairs (a, ℓ_j) for each fixed ℓ_j . This implies

$$N \leq 10 \times 200 = 2000. \quad (2)$$

Now, inequalities (1) and (2) contradict with each other. This means there must be a row or a column consisting of at least 11 distinct numbers. $\square$

Problem 2.15. (CHKMO 2007 Problem 3) In a school there are 2007 male and 2007 female students. Each student joins not more than 100 clubs in the school. It is known that any two students of opposite genders have joined at least one common club. Show that there is a club with at least 11 male and 11 female members.

Analysis. Analogous to the previous problem, one may tend to count the number of pairs (S, C) where S is a student joining a club C . But then the condition that any two students of opposite genders have joined at least one common club cannot be used easily. Rather than that, we may count some pairs of students of opposite genders who have joined the same club. To make it simple, we may count some triples instead.

Solution. Suppose on the contrary that no club contains at least 11 male and 11 female members. We shall count the number of triples (M, F, C) where M is a male student, F is a female student and C is a club joined by these two students.

Firstly, for each of the 2007 choices of M and each of the 2007 choices of F , the condition shows that there is at least one club C for which (M, F, C) should be counted. This gives at least 2007^2 triples.

Secondly, we can count according to the clubs. We label the clubs as $C_1, C_2, \dots, C_k$. For each $i = 1, 2, \dots, k$, let m_i and f_i be the number of male students and female students in the club C_i respectively. Then the number of triples (M, F, C) is

$$\sum_{i=1}^k m_i f_i.$$

It is likely that we can work out an upper bound for this sum, since we know that at least one of m_i and f_i cannot be too large for each i . Indeed, by the assumption, we can divide these clubs into two groups. The first group consists of those clubs with at most 10 male members (say $C_1, C_2, \dots, C_\ell$). The second group consists of those clubs with at least 11 male members, and hence at most 10 female members (say $C_{\ell+1}, C_{\ell+2}, \dots, C_k$). Then

$$\begin{aligned} \sum_{i=1}^k m_i f_i &= \sum_{i=1}^{\ell} m_i f_i + \sum_{i=\ell+1}^k m_i f_i \\ &\leq 10 \sum_{i=1}^{\ell} f_i + 10 \sum_{i=\ell+1}^k m_i \\ &\leq 10 \sum_{i=1}^k f_i + 10 \sum_{i=1}^k m_i. \end{aligned}$$

Note that the sum over f_i counts the total number of female members in all clubs. By another simple double counting, this is equal to the total number of clubs joined by all the female students. Therefore, we have

$$\sum_{i=1}^k f_i \leq 100 \times 2007.$$

The same upper bound holds for the sum over m_i . Combining all these, we obtain

$$2007^2 \leq 10 \times 100 \times 2007 + 10 \times 100 \times 2007 = 2000 \times 2007.$$

This is a contradiction. Hence, there must be a club with at least 11 male and 11 female members. $\square$

Problem 2.16. (IMO Shortlist 2001 C8/Problem 3) Twenty-one girls and twenty-one boys took part in a mathematical contest.

- Each contestant solved at most six problems.
- For each girl and each boy, at least one problem was solved by both of them.

Prove that there was a problem that was solved by at least three girls and at least three boys.

Analysis. The setting is exactly the same as the previous problem. However, the same solution does not apply directly since we would obtain the inequality

$$21^2 \leq 2 \times 6 \times 21 + 2 \times 6 \times 21 = 24 \times 21$$

if we follow the same steps. Therefore, we need to modify the arguments to strengthen the bounds.

Solution. Suppose on the contrary that no problem was solved by at least 3 girls and 3 boys. We count the number of triples (P, G, B) where P is a problem solved by a girl G and a boy B .

Firstly, for each of the 21 choices of G and each of the 21 choices of B , there exists a P such that (P, G, B) should be counted. This gives at least 21^2 triples.

Secondly, we divide the problems into two groups. Group I consists of those problems which were solved by at most 2 girls. Group II consists of those problems which were solved by at least 3 girls, and hence at most 2 boys. We first consider those triples (P, G, B) for which each problem belongs to group I. We call these *special* triples.

Intuitively, since these problems were solved by a small number of girls, each such girl was involved in many triples. This in turn means that each such problem has to be solved by many boys, which may contradict the given conditions. This is the idea to proceed.

For each boy B , we claim that B solved at most 5 group I problems. Indeed, if B solved 6 group I problems, then he cannot solve any more problem due to the constraint. For each of these 6 problems, there are at most $6 \times 2 = 12$ triples to be counted, since there are at most 2 choices of G for each P . This contradicts the fact that there are at least 21 triples containing B . This proves the claim. Therefore, we have at most $5 \times 2 = 10$ special triples containing B . Since this holds for each boy, there can be at most $10 \times 21 = 210$ special triples.

Similarly, there can be at most 210 triples consisting of problems which belong to group II. This yields the contradiction

$$441 = 21^2 \leq 210 + 210 = 420.$$

Hence, there was a problem that was solved by at least 3 girls and at least 3 boys. $\square$

2.5 Extremal Principle

Sometimes we may consider an extremal situation. This means we may choose a situation for which a certain quantity is maximized or minimized. This usually allows us to have more assumptions to start with compared to working on an arbitrary situation.

Problem 2.17. Let n be a positive integer. Suppose $3n$ cells of a $2n \times 2n$ table are blackened. Prove that one can choose n rows and n columns such that all blackened cells lie in a chosen row or a chosen column.

Analysis. Since we want the chosen rows and columns to contain many blackened cells, we may simply consider a maximal case and claim that it satisfies the condition. On average, every 1.5 of the blackened cells should lie in a chosen row or a chosen column. Thus, it suffices to have half of the chosen rows and columns each of which contains 2 blackened cells.

Solution. Firstly, we choose n rows which contain the greatest number of blackened cells. Such a choice may not be unique but it does not matter. Secondly, we choose n columns which contain the greatest number of blackened cells which do not lie in any of the chosen rows. We claim that this choice satisfies the condition.

Suppose on the contrary that there is a blackened cell not lying in the chosen rows and columns. Then there are at least $n + 1$ blackened cells not lying in any of the chosen rows (and those cells should lie in different columns). Since only n rows are not chosen, 2 of these blackened cells must belong to the same row by the pigeonhole principle. By the maximality of the first step, every chosen row must contain at least 2 blackened cells. This shows at most $3n - 2n = n$ blackened cells do not lie in the chosen rows, which is a contradiction. Therefore, our choice does meet the condition. $\square$

Remarks. When using the extremal principle, one has to ensure that the maximum or minimum value does exist. In particular, this is always the case if there are only finitely many possibilities (like finitely many ways to choose n rows from $2n$ rows).

Problem 2.18. Let m and n be positive integers. There is a real number in each cell of an $m \times n$ table. In each step, we can change the signs of all the numbers in a row or in a column. Prove that it is possible to make the sum of every row and every column nonnegative after finitely many steps.

Analysis. Since the sum of every row and every column has to be nonnegative, the sum of all the numbers should be large. We may pick the case corresponding to the largest sum. To do so, we first need to ensure that there are only finitely many possibilities for the sum.

Solution. If we change the signs of a certain row or column twice, then the effects are cancelled out. Therefore, we may assume the sign of each row and each column is changed at most once. Hence, there are only finitely many possible configurations by applying the steps. Among all configurations which can be attained, we choose one for which the sum of all numbers is the largest. We claim that the condition is satisfied under this configuration.

Suppose on the contrary (and WLOG) that the sum $-S$ of the first row is negative. If we change the signs of numbers in the first row, then this sum is changed to S , which is greater than $-S$. Thus, the total sum is increased. This contradicts the maximal choice. Therefore, the sum of any row and any column is nonnegative and we are done. $\square$

Problem 2.19. (APMO 2008 Problem 2) Students in a class form groups each of which contains exactly three members such that any two distinct groups have at most one member in common. Prove that, when the class size is 46, there is a set of 10 students in which no group is properly contained.

Analysis. Since we want to find a subset of the class which does not contain any group, we may choose such a subset with the largest size. If the size exceeds 10, we may simply remove the extra students. Therefore, the key point is to derive a contradiction by assuming the size is smaller than 10.

Solution. Consider any set of students which does not contain a group. We choose such a set S with the largest size. It suffices to show that the size is at least 10. Suppose on the contrary that $|S| \leq 9$. By the maximality, we cannot add any student to this set. Therefore, for any student not in S , he/she must form a group with two other students in S . We count the number N of groups with students a, b, c such that $a, b \in S$ and $c \notin S$.

On the one hand, any $c \notin S$ corresponds to at least one such group as mentioned. Therefore, $N \geq 46 - |S|$. On the other hand, every pair of $a, b \in S$ corresponds to at most one such group since two distinct groups have at most one common member. This gives $N \leq \binom{|S|}{2}$. Combining the inequalities, we obtain $|S|(|S| + 1) \geq 92$. This contradicts the assumption

$$|S|(|S| + 1) \leq 9 \times 10 = 90.$$

Therefore, we must have $|S| \geq 10$. Thus, we can choose any subset of S with size 10 to meet the condition. $\square$

Remarks. A proof using the extremal principle can usually be replaced by a proof using induction. For example, in this problem, one can extend the set S by adding more and more students until the size becomes 10 using basically the same argument.

Problem 2.20. (CHKMO 2015 Problem 4) Let $n \geq 2$ be an integer. There are n distinct circles on the plane such that any two circles have two distinct intersections and no three circles have a common intersection. Initially there is a coin on each of the intersection points of the circles. Starting from X , players X and Y alternately take away a coin, with the restriction that one cannot take away a coin lying on the same circle as the last coin just taken away by the opponent in the previous step. The one who cannot do so will lose. In particular, one loses when there is no coin left. For what values of n does Y has a winning strategy?

Analysis. Since the number of coins is even, Y is the one who takes the last coin if the restriction is ignored. Therefore, we believe that Y should be the winner. It suffices to divide the intersection points into pairs such that the two intersection points in the same pair do not lie on the same circle (and thus we need $n \geq 4$ to apply this argument). One can give a construction directly. However, we shall use the extremal principle as an alternative solution. This shows such a pairing exists without providing the explicit construction.

Solution. We use (i, j) to denote a coin on the intersection of circles i and j . Thus, the two coins on the intersection of circles i and j are (i, j) and (j, i) for $i \neq j$.

For $n = 2$ or $n = 3$, X wins by taking away $(1, 2)$ as Y cannot take the other coins. For $n = 4$, WLOG assume X takes away $(1, 2)$ in the first step. Then Y should take away $(3, 4)$. Afterwards, X can only take away $(2, 1)$, and then Y can only take away $(4, 3)$. At this stage, X cannot take away any coin and hence Y wins.

Now, we claim that Y has a winning strategy for $n \geq 5$. It suffices to pair up the $n(n-1)$ intersection points such that the two points in the same pair do not lie on the same circle. If this can be done, whenever X takes away any coin on a point, Y then takes away the coin on the point which is in the same pair as that. This shows Y can always take away a coin. The game will end when X cannot take a coin, which must happen since there is a finite number of coins. Then Y will win the game.

Consider all pairings of the $n(n-1)$ intersection points which do not contain a pair $\{(i, j), (j, i)\}$. Such pairings exist for $n \geq 3$. For each pairing, we count the number of *bad pairs*, meaning pairs of intersection points lying on the same circle, and denote the number by N . Among all possible pairings, we choose one for which N is minimized. So we have to prove $N = 0$.

Suppose on the contrary that $N \geq 1$. WLOG assume there is a pair $\{(a, b), (a, c)\}$ where $b \neq c$. We say that an intersection point (i, j) is *special* if $i = a$ or $j = a$ or $\{i, j\} = \{b, c\}$. We claim that there is another pair containing two intersection points which are not special. Indeed, there are $\frac{n(n-1)}{2}$ pairs in total, and there are only $(n-1) + (n-1) + 2 = 2n$ special points. Since $\frac{n(n-1)}{2} \geq 2n$ for $n \geq 5$ and there exists a pair containing two special points as assumed, this claim is true. There are two possibilities for this non-special pair.

- Suppose there is a pair $\{(w, x), (y, z)\}$ of non-special points where w, x, y, z are pairwise distinct. WLOG we may assume w, x are distinct from b , and y, z are distinct from c (since (b, c) and (c, b) are special). We swap the points (a, c) and (w, x) . We have the new pairs $\{(a, b), (w, x)\}$ and $\{(a, c), (y, z)\}$. From our assumption, both new pairs are not bad. Therefore, N is decreased by 1, which contradicts the minimality of our choice.
- Suppose there is a bad pair $\{(w, x), (w, y)\}$ of non-special points where w, x, y are pairwise distinct. WLOG we may assume w, x are distinct from b . (If $w = b$, then we may assume w, x are distinct from c .) We swap the points (a, c) and (w, x) . We have the new pairs $\{(a, b), (w, x)\}$ and $\{(a, c), (w, y)\}$. The first pair is not bad from the assumption. Therefore, N is decreased by at least 1. Again, this is a contradiction.

Therefore, we must have $N = 0$. Thus, Y can apply the strategy to win the game for $n \geq 5$. To conclude, Y wins iff $n \geq 4$. $\square$

2.6 Majorization Inequality

In this subsection, we introduce an inequality which is useful in solving some combinatorics problems.

In the following, we say that a sequence $x_1 \geq x_2 \geq \dots \geq x_n$ *majorizes* (優於) another sequence $y_1 \geq y_2 \geq \dots \geq y_n$ if they satisfy the following conditions:

$$\begin{aligned}
x_1 &\geq y_1, \\
x_1 + x_2 &\geq y_1 + y_2, \\
&\dots \\
x_1 + x_2 + \dots + x_{n-1} &\geq y_1 + y_2 + \dots + y_{n-1}, \\
x_1 + x_2 + \dots + x_n &= y_1 + y_2 + \dots + y_n.
\end{aligned}$$

We write $(x_1, x_2, \dots, x_n) > (y_1, y_2, \dots, y_n)$ or $(y_1, y_2, \dots, y_n) < (x_1, x_2, \dots, x_n)$ in that case. These are not the inequality signs as the symbols bend a little bit. Note that the sums of the two sequences must be equal in view of the definition.

The main result we need is the following.

Theorem 2.3. (Majorization inequality/Karamata's inequality (蓋不等式)) Let f be a convex function defined on an interval I . For any real numbers $x_1, x_2, \dots, x_n, y_1, y_2, \dots, y_n \in I$ with $(x_1, x_2, \dots, x_n) > (y_1, y_2, \dots, y_n)$, we have

$$f(x_1) + f(x_2) + \dots + f(x_n) \geq f(y_1) + f(y_2) + \dots + f(y_n).$$

We are not going to prove this inequality. Also, we do not bother to define a convex function here as we are only interested in its application in combinatorics. Indeed, we will only apply this inequality to the binomial function $\binom{k}{2}$, which is well-known to be convex. In other words, if $(x_1, x_2, \dots, x_n) \succ (y_1, y_2, \dots, y_n)$, then we have

$$\binom{x_1}{2} + \binom{x_2}{2} + \dots + \binom{x_n}{2} \geq \binom{y_1}{2} + \binom{y_2}{2} + \dots + \binom{y_n}{2}.$$

Problem 2.21. (China TST 1992 Problem 1) 16 students took part in a competition. All problems were multiple choice style. Each problem had four choices. It was said that any two students had at most one answer in common, find the maximum number of problems.

Analysis. The nature of this problem is similar to that of [problem 2.16](#). So it is likely that we can use double counting here. For example, we can count the number of pairs of common answers in two ways.

Solution. We count the number of pairs of common answers given by the students. Denote this number by N .

Firstly, since each pair of the 16 students gave at most one pair of common answer, we easily obtain

$$N \leq \binom{16}{2} = 120. \quad (1)$$

Secondly, suppose there are n problems, and suppose there are a, b, c, d students answering the four choices of a particular problem respectively. Then there are exactly

$$\binom{a}{2} + \binom{b}{2} + \binom{c}{2} + \binom{d}{2}$$

pairs of common answers arising from this problem. This time we need to find a lower bound. Since $(a, b, c, d) \succ \left(\frac{a+b+c+d}{4}, \frac{a+b+c+d}{4}, \frac{a+b+c+d}{4}, \frac{a+b+c+d}{4} \right)$, by the majorization inequality, we have

$$\binom{a}{2} + \binom{b}{2} + \binom{c}{2} + \binom{d}{2} \geq 4 \binom{\frac{1}{4}(a+b+c+d)}{2} = 4 \binom{4}{2} = 24.$$

Since this holds for every problem, we obtain

$$N \geq 24n. \quad (2)$$

Comparing (1) and (2), we need $24n \leq 120$, i.e. $n \leq 5$.

It remains to provide a construction with exactly 5 problems. While it is not too easy to work out a construction, there are some hints that we can observe from the above proof. Indeed, all equalities in the above proof should hold. In particular, each

pair of students gave exactly one common answer, and there are exactly 4 students choosing each choice in every problem. With this in mind and with some trial and error, we are able to work out the following construction, for example (where A, B, C, D denote the 4 choices of each of the problems P1, P2, P3, P4, P5).

Student	P1	P2	P3	P4	P5
1	A	A	A	A	A
2	A	B	B	B	B
3	A	C	C	C	C
4	A	D	D	D	D
5	B	A	B	C	D
6	B	C	D	A	B
7	B	D	C	B	A
8	B	B	A	D	C
9	C	A	C	D	B
10	C	D	B	A	C
11	C	B	D	C	A
12	C	C	A	B	D
13	D	A	D	B	C
14	D	B	C	A	D
15	D	C	B	D	A
16	D	D	A	C	B

□

Remarks. If f is a convex function, we always have

$$f(x_1) + f(x_2) + \cdots + f(x_n) \geq nf\left(\frac{x_1 + x_2 + \cdots + x_n}{n}\right).$$

This special case of the majorization inequality is called Jensen's inequality.

Problem 2.22. (CHKMO 2011 Problem 4) Let $A_1A_2\cdots A_n$ be a cyclic polygon ($n \geq 3$). Find the maximum number of distinct acute-angled triangles whose vertices are chosen from $A_1, A_2, \dots, A_n$.

Analysis. We shall count the number of non-acute triangles, which means a triangle with an angle $\geq 90^\circ$. An angle $\angle ABC$ is at least 90° iff points A, B, C lie on the same semicircle in this order. This suggests a simple way to count the total number in terms of the binomial coefficient. It remains to use algebra to find an upper bound.

Solution. We first count the number N_1 of non-acute triangles with A_1 as one vertex which is $< 90^\circ$. Let ℓ be the diameter of the circle which passes through A_1 . Then for points B, C on the circle, $\triangle A_1BC$ is not acute with $\angle BA_1C < 90^\circ$ iff B and C lie on the same side of ℓ (possibly on ℓ). Suppose there are a points on one side of ℓ and b points on the other side of ℓ other than A_1 . Note that the two sides share at most

one common point which is the other intersection of ℓ and the circle. Therefore, we must have $a + b = n - 1$ or $a + b = n$. We find that

$$N_1 = \binom{a}{2} + \binom{b}{2}.$$

We have to minimize N_1 . Note that $b \geq n - 1 - a$ so that $\binom{b}{2} \geq \binom{n-1-a}{2}$. WLOG assume $a \geq n - 1 - a$. If n is odd, then $(a, n - 1 - a) > \left(\frac{n-1}{2}, \frac{n-1}{2}\right)$. Using the majorization inequality, we obtain

$$N_1 \geq \binom{a}{2} + \binom{n-1-a}{2} \geq \binom{\frac{n-1}{2}}{2} + \binom{\frac{n-1}{2}}{2} = \frac{(n-1)(n-3)}{4}.$$

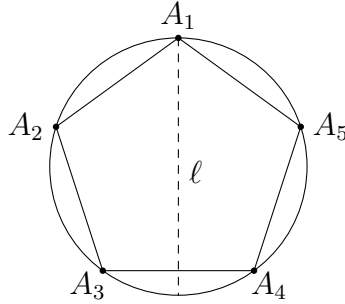
The same bound holds if we replace A_1 by any other vertex A_j . We sum up all the lower bounds. Note that each non-acute triangle is counted twice in this sum (one corresponding to each acute angle). Therefore, the number of non-acute triangles is at least

$$n \cdot \frac{(n-1)(n-3)}{4} \cdot \frac{1}{2} = \frac{n(n-1)(n-3)}{8}.$$

Thus, the number of acute triangles is at most

$$\binom{n}{3} - \frac{n(n-1)(n-3)}{8} = \frac{n(n-1)(n+1)}{24}.$$

This can be attained when $A_1 A_2 \dots A_n$ is a regular polygon (since in that case all inequalities used become equalities).



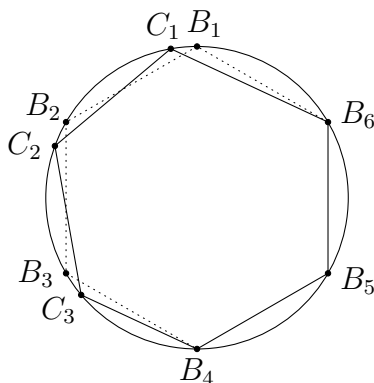
Similarly, if n is even, then $(a, n - 1 - a) > \left(\frac{n}{2}, \frac{n-2}{2}\right)$ (this only works for the even case as $\frac{n}{2}$ and $\frac{n-2}{2}$ are integers). Using the majorization inequality, we obtain

$$N_1 \geq \binom{a}{2} + \binom{n-1-a}{2} \geq \binom{\frac{n}{2}}{2} + \binom{\frac{n-2}{2}}{2} = \frac{(n-2)^2}{4}.$$

Thus, the number of acute triangles is at most

$$\binom{n}{3} - n \cdot \frac{(n-2)^2}{4} \cdot \frac{1}{2} = \frac{n(n-2)(n+2)}{24}.$$

The regular polygon case does not yield equality. Indeed, we first consider a regular polygon $B_1B_2\cdots B_n$. Then we rotate half of the points, say $B_1, B_2, \dots, B_{\frac{n}{2}}$, anticlockwise about the centre of the circle with a sufficiently small angle θ to points $C_1, C_2, \dots, C_{\frac{n}{2}}$. Then $C_1C_2\cdots C_{\frac{n}{2}}B_{\frac{n}{2}+1}B_{\frac{n}{2}+2}\cdots B_n$ is one possible case which gives the upper bound $\frac{n(n-2)(n+2)}{24}$.



□

The following lists out the techniques that can be used to solve extremal combinatorics and existence problems.

- induction (problems 2.1, 2.2, 2.3, 2.4)
- pigeonhole principle (problems 2.5, 2.6, 2.7)
- colouring (problems 2.8, 2.9, 2.10, 2.11)
- double counting (problems 2.12, 2.13, 2.14, 2.15, 2.16, 2.19, 2.21)
- extremal principle (problems 2.6, 2.17, 2.18, 2.19, 2.20)
- majorization inequality (problems 2.21, 2.22)

3 Games and Operations

Operation problems refer to those combinatorial problems which define certain rules of operations and ask whether something can be done by applying the operations repeatedly. Induction is still a helpful tool in solving some basic problems of this kind. In general, the main idea is to think of some algorithms to apply the operations. To do so, one can play around with the small cases to familiarize with the operations, and figure out the crucial component in the setting. Another useful idea is to look for some invariances, which mean quantities or properties which are unchanged throughout the process.

For game problems, they share some similarities with operation problems. The difference is that such problems usually involve two players and ask if one player has a winning strategy. To show that one player has a winning strategy, we need to provide an algorithm for that player, and consider all possible ways that the opponent plays. This is because we cannot control how the opponent reacts.

3.1 Algorithm

When we construct an algorithm so as to meet our goal, we may first consider some small cases and look for patterns. If some patterns can be found, then usually the same ideas can be generalized to all cases, or we can use induction to work out a proof. However, sometimes cases corresponding to different parities may give different results, and the few smallest cases may also give different results than the general case. So we should be flexible and should not stick to a certain observation.

If there seems to be no hints observed from the small cases, then we cannot avoid working on the general case. But we can still simplify the problem by transforming the general case to certain special cases for consideration. The exact trick in solving a particular problem greatly depends on the rules of the operations, so we can only learn these by going through some problems.

Problem 3.1. Let $n \geq 3$ be an integer. Initially, a number 1 and a number 2 are written on the blackboard. Players X and Y alternately write down a number on the blackboard according to the following rules:

- (1) if k is a number on the blackboard, one may write down $k + 1$; or
- (2) if k and ℓ are two numbers on the blackboard, one may write down $k\ell$.

Note that the same integer may appear on the blackboard more than once. In operation (2), k and ℓ must be different copies of numbers on the blackboard but they may be equal. For example, player X may write down an extra 2 in the first

step, and then player Y may write down 4 in the next step as $4 = 2 \times 2$. However, X cannot write down 4 in the first step because there is only one copy of 2.

The game will end when one of the following is satisfied:

- (i) there are n numbers having the same remainder when divided by n ; or
- (ii) there are n numbers having pairwise distinct remainders when divided by n .

The one who writes down the last number will win. Find all n for which player X has a winning strategy.

Analysis. If one tries to work on the small cases, one would find that player X has a winning strategy for $n = 3, 4, 5, 6, 7$. This may lead to a wrong feeling that player X always has a winning strategy, which is indeed not the case. This shows we should not easily believe in any pattern observed from small cases.

To find out who is the potential winner, we may simplify the rules of the game. For example, if the players can write down any integer on the blackboard without any constraint, then the players only need to prevent the number of distinct residue classes to exceed $n - 2$ and prevent the number of integers in each residue class to exceed $n - 2$. In other words, the players only have $(n - 2)^2$ chances in total. This shows the potential winner should depend on the parity of n . The rules of the game only affect the results of small cases. The potential winner only needs to guarantee that the residue class 1 can be written down, since all other numbers can always be written down as long as they appear for the first time.

Solution. In view of the rules, we shall consider everything modulo n .

For $n = 3$, player X wins immediately by writing down 3.

For $n = 4$, the one who writes down 3 or 4 will lose because the opponent will complete all residue classes. Thus, both players will only write down 2 (because 1 cannot be written down before 3 and 4 appear). By considering the parity, player X is the winner.

For odd $n \geq 5$, the number $(n - 2)^2 - 2$ is odd so that player X is the potential winner. The strategy is to write down $3, 4, \dots, \frac{n+1}{2}$ one by one regardless of what player Y has written down. Meanwhile, player Y has written down $\frac{n-5}{2}$ numbers. So there can be at most $\frac{n-3}{2} \leq n-2$ numbers of the same residue class, and at most $2 + \frac{n-3}{2} + \frac{n-5}{2} = n-2$ numbers of different residue classes. Thus, player Y cannot win at this stage. One can write down the residue class 1 as $2 \times \frac{n+1}{2} = n+1$. Therefore, any residue class which has appeared can be written down.

To prevent losing the game, both players will avoid to have $n - 1$ distinct residue classes and avoid to have $n - 1$ numbers in the same residue classes. Hence, there

are only $(n-2)^2$ chances for them to write the numbers (note that one can always add a new residue class due to operation (1)). Considering the parity, player X is the winner.

For $n = 6$, player X can still win the game. He can first write down another 2. If player Y does not add a new residue class, then the two players can only write down 2 alternately, and player X will win. If player Y writes down 3 or 4, then player X writes down 4 or 3 correspondingly. Afterwards, both players will avoid writing down 5 or 6, and hence they can only add more 2, 3, 4's. There are only 13 chances in this case (including the initial number 1). Thus, player X will win.

For even $n \geq 8$, we claim that player Y is the potential winner. No matter player X writes down 2 or 3 initially, player Y can always write down 4. Afterwards, player Y only needs to write down $5, 6, \dots, \frac{n}{2}$ and n one by one regardless of what player X has written down. Meanwhile, player X has written down $\frac{n-4}{2}$ numbers. So there can be at most $\frac{n-2}{2} \leq n-2$ numbers of the same residue class, and at most $2 + \frac{n-4}{2} + \frac{n-4}{2} = n-2$ numbers of different residue classes. Thus, player X cannot win at this stage. One can write down the residue class 1 by operation (1). Therefore, any residue class which has appeared can be written down. As in the odd case, player Y will win.

Therefore, player X wins iff n is odd or $n = 4, 6$. □

Problem 3.2. (CMO 2010 Problem 5) We carry out some operations on the cards placed on the points $A_1, A_2, \dots, A_n$ and the point O ($n \geq 3$). An operation refers to one of the following:

- (1) If the number of cards on a certain point A_i is at least 3, then we can take away 3 of them and put one on each of the points A_{i-1}, A_{i+1} and O (here $A_0 = A_n, A_{n+1} = A_1$); or
- (2) If the number of cards on the point O is at least n , then we can take away n of them and put one on each of the points $A_1, A_2, \dots, A_n$.

Prove that if the total number of cards placed on these $n+1$ points is at least n^2+3n+1 , then we can always apply some operations so that the number of cards on each point is not less than $n+1$.

Analysis. We would like to handle as few cases as possible, so that it is easier to study the problem. Due to the special point O , we may first apply only operations (1) to move most of the cards to point O . If the number of cards on O is large enough, then we can apply operations (2) to get what we want. This leaves us only a case when the number of cards on O lies in a certain range.

Afterwards, we can try to reduce the total number of cards on the A_i 's since this is the same as moving more cards to O . Because of the initial setting, we may assume each A_i contains 0, 1 or 2 cards only. There are not too many possibilities. By working on some examples, we may be able to work out an algorithm to meet this goal.

Solution. We apply operation (1) repeatedly in an arbitrary way until this cannot be done. This must come to an end since the number of cards on O is increased by 1 in each operation. Then each point A_i contains at most 2 cards, so that O contains at least

$$n^2 + 3n + 1 - 2n = n^2 + n + 1$$

cards. We focus on this state and let $a_1, a_2, \dots, a_n, b$ be the number of cards on $A_1, A_2, \dots, A_n, O$ respectively. The above says $a_i \leq 2$ and $b \geq n^2 + n + 1$.

If $b \geq n^2 + 2n + 1$, then we can apply $n + 1$ operations (2) so that each point has at least $n + 1$ cards. Therefore, we may now assume $n^2 + n + 1 \leq b \leq n^2 + 2n$.

At this stage, we can only apply operation (2). We apply operation (2) once so that each a_i satisfies $1 \leq a_i \leq 3$ and b satisfies $n^2 + 1 \leq b \leq n^2 + n$. Note that we would like to make $S = a_1 + a_2 + \dots + a_n$ small, so that b exceeds its upper bound and we are done. Therefore, we should reduce the number of points A_i with 3 cards. Thus, we apply operation (1) on those points. Since the positions of these points are interdependent, we should do so step by step. There are a few cases to consider. We claim that we can always decrease the value of S while keeping the condition $1 \leq a_i \leq 3$ (which ensures that we can argue repeatedly).

- If each a_i is at least 2, then since $b \geq n^2 + 3n + 1 - 3n = n^2 + 1$, we can apply $n - 1$ operations (2) so that each a_i becomes at least $n + 1$ and b is at least $n + 1$. Thus we are done.
- If $a_2 = a_n = 3$, $2 \leq a_3, a_4, \dots, a_{n-1} \leq 3$ and $a_1 = 1$, we apply operation (1) to each $A_2, A_3, \dots, A_n$ in that order. For $k = 2, 3, \dots, n - 1$, after the operation to A_k , the value of a_{k+1} is increased by 1 and hence it is at least 3. Therefore, we can apply the operations as stated. After these operations, we have $a_1 = 3$, $a_2 = a_n = 1$ and $1 \leq a_3, a_4, \dots, a_{n-1} \leq 2$. The condition $1 \leq a_i \leq 3$ is preserved and S is decreased (since S is decreased by 1 during each operation (1)).
- If $a_2 = a_k = 3$, $2 \leq a_3, a_4, \dots, a_{k-1} \leq 3$ and $1 \leq a_1, a_{k+1} \leq 2$ for some $3 \leq k \leq n - 1$, we apply operation (1) to each $A_2, A_3, \dots, A_k$ in that order. Then we have $a_2 = a_k = 1$, $1 \leq a_3, a_4, \dots, a_{k-1} \leq 2$ and $2 \leq a_1, a_{k+1} \leq 3$. The condition $1 \leq a_i \leq 3$ is preserved and S is decreased.

Following any of these cases repeatedly, we must come to an end since S cannot decrease indefinitely. This means the configuration does not contain any of the specified cases up to symmetry. Indeed, the last two cases mean that there are two distinct points A_2, A_k each containing 3 cards so that every point between them (in

one of the directions) contains either 2 or 3 cards. (The restrictions $1 \leq a_1, a_{k+1} \leq 2$ can be ignored since we can replace k by $k+1$ if $a_{k+1} = 3$, etc.)

Suppose among $a_1, a_2, \dots, a_n$, there are x, y, z of them equal to 1, 2, 3 respectively. If $z > x$, then either $z = 1$ or there are two distinct $a_2, a_k = 3$ with $a_3, \dots, a_{k-1} \geq 2$ by the pigeonhole principle. In the former case, we have $x = 0$ so that we are done as discussed. In the latter case, we can decrease the value of S and repeat the arguments. Therefore, it suffices to consider $z \leq x$. Indeed, at this stage the value of S is small enough for our purpose. We find that

$$b \geq n^2 + 3n + 1 - x - 2y - 3z \geq n^2 + 3n + 1 - 2(x + y + z) = n^2 + n + 1.$$

As explained, we simply need to apply n operations (2) so that each point contains at least $n+1$ cards. This completes the proof. $\square$

Problem 3.3. (IMO Shortlist 2009 C8) For any integer $n \geq 2$, we compute the integer $h(n)$ by applying the following procedure to its decimal representation. Let r be the rightmost digit of n .

- (1) If $r = 0$, then the decimal representation of $h(n)$ results from the decimal representation of n by removing this rightmost digit 0.
- (2) If $1 \leq r \leq 9$ we split the decimal representation of n into a maximal right part R that solely consists of digits not less than r and into a left part L that either is empty or ends with a digit strictly smaller than r . Then the decimal representation of $h(n)$ consists of the decimal representation of L , followed by two copies of the decimal representation of $R - 1$. For instance, for the number $n = 17,151,345,543$, we will have $L = 17,151$, $R = 345,543$ and $h(n) = 17,151,345,542,345,542$.

Prove that, starting with an arbitrary integer $n \geq 2$, iterated application of h produces the integer 1 after finitely many steps.

Analysis. Induction seems to be a useful tool to solve this problem. However, if one tries to work on the small cases, one will realize that even the case $n = 3$ is nontrivial and the case $n = 4$ is almost nonsense. Indeed, one should note that the process is tedious if there exist some large digits. Conversely, if n only consists of digits 0 and 1, then it is relatively simpler. Therefore, we may try to use a different inductive approach in contrast to the usual ordering.

Another useful observation is that if there is a digit 0, then every digit on the left of this 0 remains unchanged before this 0 is deleted. This allows us to break an integer into several parts each of which does not contain the digit 0. The same applies to larger digits using induction.

Solution. We say that a positive integer n is *good* if the problem statement holds. Firstly, suppose n has the form $\overline{X0Y}$ where X and Y are strings of digits. We note

that the string X remains unchanged before the 0 is deleted using operation (1). If we can prove that both X and Y are good, then we can first apply the operations repeatedly until Y becomes 1. From the above observation, the integer becomes $\overline{X00\cdots 01}$. After one more operation, we obtain $\overline{X00\cdots 0}$. By deleting the zeros, we are left with the case X , which is good by assumption. Therefore, the integer n is also good. From this, it suffices to show that every n without the digit 0 is good.

Next, we consider $n = \overline{X1}$ where X does not contain the digit 0. After one more operation, the integer becomes $\overline{X0X0}$. Breaking the integer into two parts, it suffices to show X is good. In other words, we can actually delete the unit digit 1 from $\overline{X1}$ to obtain X . This can be used to replace the original operation (1) (as the unit digit will never be 0 under our consideration).

In the same way, suppose n has the form $\overline{X1Y}$ where X and Y do not contain 0. Suppose both X and Y are good. We apply the operations repeatedly using our new operation (1). Then the digit 0 never appears and hence X remains unchanged throughout the operations. The same holds until Y is transformed to an integer consists of only the digit 1, so that the integer becomes $\overline{X11\cdots 1}$. By deleting the 1's and using the fact that X is good, we conclude that n is good. In other words, we may now break the integer into several parts using the digits 1. Then we only need to show that each such part is good.

We can use the same argument repeatedly, or argue neatly using induction. At the end, it suffices to show that all integers containing only the digit 9 are good, which is simple. Indeed, consider $n = \underbrace{99\cdots 9}_m$ where $m \geq 1$. Applying the operation once,

we obtain $\underbrace{99\cdots 9}_{(m-1)} 8 \underbrace{99\cdots 9}_{(m-1)} 8$. We can delete the unit digit 8 and break this integer

into two parts using the middle 8. It suffices to show that $\underbrace{99\cdots 9}_{(m-1)}$ is good. By the

method of descent or by induction, it is rather obvious that the integer can eventually be transformed to 9. After one more operation, this becomes 88. The unit digit is deleted and leaves a single 8. Inductively, the integer will be transformed to 7, 6, $\dots$, 1 step by step. This shows any n is good as desired. $\square$

Problem 3.4. (IMO Shortlist 2010 C4/Problem 5) In each of six boxes $B_1, B_2, B_3, B_4, B_5, B_6$ there is initially one coin. There are two types of operation allowed:

Type 1: Choose a nonempty box B_j with $1 \leq j \leq 5$. Remove one coin from B_j and add two coins to B_{j+1} .

Type 2: Choose a nonempty box B_k with $1 \leq k \leq 4$. Remove one coin from B_k and exchange the contents of (possibly empty) boxes B_{k+1} and B_{k+2} .

Determine whether there is a finite sequence of such operations that results in boxes B_1, B_2, B_3, B_4, B_5 being empty and box B_6 containing exactly $2010^{2010^{2010}}$ coins.

Analysis. At first glance, it seems impossible to create so many coins since only type 1 operation may add one extra coin in total, and there are very few boxes to carry out the operations. One may then try to prove that the total number of coins should be bounded by some number M , which is less than $2010^{2010^{2010}}$. However, one will get into trouble in this direction because such a sequence of operations actually exists! Many contestants were trapped during the IMO contest because of this.

In order not to get into this trap, one should always think in different directions and not to stick to a particular answer. For this problem, one should indeed ask a question before guessing the answer. Why should the number be $2010^{2010^{2010}}$? It seems like this only stands for an extremely large number. But why doesn't the proposer set the number to 2010^{2010} , which is also a very large number? If you still believe the answer is 'no', then there are only two possible reasons for this:

- it is easier to obtain an upper bound M which is less than $2010^{2010^{2010}}$, but not 2010^{2010} ; or
- there exists a sequence of operations that produces at least 2010^{2010} coins in total but not $2010^{2010^{2010}}$ coins.

In view of the second possibility, it makes sense to look for a sequence that can create a large number of coins. One may work on a similar problem with fewer boxes first. If there are $n = 2, 3, 4$ boxes respectively, the maximum number of coins that we can obtain in box B_n is 3, 7, 28 respectively. In particular, when $n = 4$, one way to obtain 28 coins is as follows (where red colours denote type 1 operations and blue colours denote type 2 operations):

$$\begin{aligned} 1, \textcolor{red}{1}, 1, 1 &\longrightarrow 1, 0, \textcolor{red}{3}, 1 \longrightarrow 1, 0, \textcolor{red}{2}, 3 \longrightarrow 1, 0, \textcolor{red}{1}, 5 \longrightarrow \textcolor{red}{1}, 0, 0, 7 \\ &\longrightarrow 0, \textcolor{blue}{2}, 0, 7 \longrightarrow 0, 1, \textcolor{red}{7}, 0 \longrightarrow 0, \textcolor{blue}{1}, 0, 14 \longrightarrow 0, 0, \textcolor{red}{14}, 0 \longrightarrow 0, 0, 0, 28. \end{aligned}$$

This example tells us a crucial observation in solving this problem. To create more coins, we should make three consecutive boxes contain $a, 0, b$ coins respectively, and apply a type 2 operation to get $a - 1, b, 0$ coins. The following algorithm is mainly based on this observation.

Solution. Suppose there are $a_1, a_2, \dots, a_j$ coins respectively in j consecutive boxes. If it is possible to change the number of coins to $a'_1, a'_2, \dots, a'_j$, we shall write

$$a_1, a_2, \dots, a_j \longrightarrow a'_1, a'_2, \dots, a'_j.$$

We first provide some algorithms that allow us to create many coins within 3 or 4 consecutive boxes.

Claim 1. $a, 0, 0 \longrightarrow 0, 2^a, 0$ for any $a \geq 1$.

Proof. We use red and blue colours to denote type 1 and type 2 operations respectively. Firstly, we have

$$b, \textcolor{red}{c}, 0 \longrightarrow \textcolor{blue}{b}, 0, 2c \longrightarrow b - 1, 2c, 0.$$

This shows we can use one coin in the first box to double the number of coins in the second box. As $a, 0, 0 \rightarrow a-1, 2, 0$, it is then obvious that we can use $a-1$ coins in the first box to obtain $2 \cdot 2^{a-1} = 2^a$ coins in the second box. $\square$

Claim 2. Define $x_1 = 2$ and $x_{n+1} = 2^{x_n}$ for $n \geq 1$. Then $a, 0, 0, 0 \rightarrow 0, x_a, 0, 0$ for any $a \geq 1$.

Proof. We use a green colour to denote the sequence of operations in **claim 1**. Then we have

$$\begin{aligned} a, 0, 0, 0 &\rightarrow a-1, 2, 0, 0 \rightarrow a-1, 0, x_2, 0 \\ &\rightarrow a-2, x_2, 0, 0 \rightarrow a-2, 0, x_3, 0 \\ &\rightarrow a-3, x_3, 0, 0. \end{aligned}$$

By noticing the pattern or using induction, we eventually obtain $0, x_a, 0, 0$ as desired. $\square$

Note that the sequence x_n increases rapidly. So we can already use **claim 2** to create many coins. For example, using a purple colour to denote the sequence of operations in **claim 2**, we have

$$\begin{aligned} 1, 1, 1, 1, 1, 1 &\rightarrow 1, 1, 1, 1, 0, 3 \rightarrow 1, 1, 1, 0, 3, 0 \rightarrow 1, 1, 0, 3, 0, 0 \\ &\rightarrow 1, 0, 3, 0, 0, 0 \rightarrow 0, 3, 0, 0, 0, 0 \rightarrow 0, 0, 16, 0, 0, 0 \rightarrow 0, 0, 0, x_{16}, 0, 0. \end{aligned}$$

The number x_{16} is much larger than $2010^{2010^{2010}}$. For example, we have

$$2010^{2010^{2010}} < 2^{11 \cdot 2010^{2010}} < 2^{2010^{2011}} < 2^{2^{11 \cdot 2011}} < 2^{2^{2^{15}}} < x_{16}.$$

Therefore, we can apply type 2 operation to box B_4 finitely many times until the number of coins becomes $0, 0, 0, \frac{2010^{2010^{2010}}}{4}, 0, 0$. Then we easily obtain

$$0, 0, 0, \frac{2010^{2010^{2010}}}{4}, 0, 0 \rightarrow 0, 0, 0, 0, \frac{2010^{2010^{2010}}}{2}, 0 \rightarrow 0, 0, 0, 0, 0, 2010^{2010^{2010}}.$$

This shows the desired sequence of operations exists. $\square$

Remarks. If one attempts to use the method of exhaustion to prove that such a sequence does not exist, one may come up with a similar algorithm. This shows it is not a bad idea to use exhaustion to work on an IMO problem!

Problem 3.5. (IMO Shortlist 2014 N3/Problem 5) For each positive integer n , the Bank of Cape Town issues coins of denomination $\frac{1}{n}$. Given a finite collection of such coins (of not necessarily different denominations) with total value at most $99 + \frac{1}{2}$, prove that it is possible to split this collection into 100 or fewer groups, such that each group has total value at most 1.

Analysis. The first simple idea that may come to our mind is to use the greedy algorithm. This means we first put the coins of larger denomination to the groups until this cannot be done, and try to squeeze the remaining coins of smaller denomination to the space. To do so, we need to know how small the denomination can be for the remaining coins.

For example, suppose we have already put some coins into 100 groups. Since the total value is at most $99 + \frac{1}{2}$ and there are 100 groups of capacity 1, there must be a group which still has a capacity of at least $\frac{100 - 99\frac{1}{2}}{100} = \frac{1}{200}$. In other words, we can always squeeze some coins of denomination at most $\frac{1}{200}$ into the groups. Now the only question left is how to allocate the coins of denomination at least $\frac{1}{200}$. It turns out that the greedy algorithm is sufficient for this purpose if we argue in a suitable way.

Solution. First of all, we merge every two coins of denomination $\frac{1}{2n}$ to form a coin of denomination $\frac{1}{n}$. This does not change the conditions, as it simply means we assume the two coins are always put together. Since there are finitely many coins, we must terminate after finitely many mergings. Afterwards, we may assume there is at most one coin of denomination $\frac{1}{2n}$ for each $n \geq 1$.

Secondly, if there are at least n coins of denomination $\frac{1}{n}$, we can put n of these coins in a group so that its total value is 1. By ignoring all these groups, we are left with some coins with total value at most $m + \frac{1}{2}$ for some $m \geq 0$, such that the number of coins of denomination $\frac{1}{n}$ is at most $n - 1$ for every $n \geq 1$. We want to put these coins into at most $m + 1$ groups, each with total value at most 1.

For each $k = 1, 2, \dots, m + 1$, we put all coins of denomination $\frac{1}{2k-1}$ or $\frac{1}{2k}$ into the same group. This is possible since the total denomination of the k th group is at most

$$\frac{1}{2k-1} \cdot (2k-2) + \frac{1}{2k} < \frac{2k-2}{2k-1} + \frac{1}{2k-1} = 1.$$

In this way, we can put all coins of denomination at least $\frac{1}{2(m+1)}$ into $m + 1$ groups.

Finally, we can put all remaining coins one by one. Indeed, since the total value of the coins is at most $m + \frac{1}{2}$, and there are $m + 1$ groups each with total value at most

1, there must be a group which still has a capacity of at least

$$\frac{(m+1) - (m + \frac{1}{2})}{m+1} = \frac{1}{2(m+1)}.$$

As every remaining coin has denomination less than $\frac{1}{2(m+1)}$, we can always put this coin into a group. Since there are finitely many coins, the process must eventually end, and we have formed an admissible grouping. $\square$

3.2 Invariance

Invariance refers to a quantity or a property which remains unchanged when operations are carried out. Some simple invariances can be easily observed, like the total number of coins, the parity of a value, etc. To look for nontrivial invariances, one may look for the common properties of all possible outcomes under the operations, especially those of the desired outcome.

Problem 3.6. The first $2n$ positive integers lie in a row in an arbitrary order. In each step, we can swap two adjacent numbers, or choose three adjacent numbers a, b, c in this order and replace them by c, a, b (but not b, c, a). Find the smallest number of steps required so that we can always rearrange the positive integers in ascending order regardless of the initial order.

Analysis. The ‘worst’ situation is probably the case when the integers are in descending order. With some trial, one would guess the answer is n^2 . It is not hard to use induction to prove that n^2 steps are always sufficient. However, induction could not be used in a straightforward way to prove the lower bound. This is because when there are more numbers, there are more flexibilities for us to choose the operations. Thus, the inductive hypothesis cannot be applied (which probably requires us to fix and ignore 2 numbers).

To obtain the lower bound, it suffices to show that at least n^2 steps are needed for a particular permutation, such as $2n, 2n-1, \dots, 1$. This permutation has the feature that the numbers are arranged in exactly the opposite order compared to the final one. Therefore, every pair of numbers has to switch their order at least once. If we swap two numbers in a step, we can only switch the order of one pair. If we permute three numbers cyclically in a step, we can switch the order of at most two pairs. Using these observations, we immediately know that at least

$$\left\lceil \frac{1}{2} \binom{2n}{2} \right\rceil = \left\lceil \frac{n(2n-1)}{2} \right\rceil = n^2 + \left\lceil -\frac{n}{2} \right\rceil$$

steps are needed. However, this bound is too weak for our purpose. The reason is that we can only use the second kind of operations in each step. For example, we

need to move the number 1 to the left by $2n - 1$ positions. This cannot be done by changing $a, b, 1$ to $1, a, b$ every time by considering the parity. Therefore, there must be a step which cannot switch the order of two pairs. This suggests it may be better to work on each number individually.

Solution. The answer is n^2 . We first prove by induction that we can always rearrange the numbers in ascending order in n^2 steps. This is trivial when $n = 1$. Suppose this holds for some $n = k$. When $n = k + 1$, there are $2k + 2$ numbers. Let x and y be the positions of 1 and 2 respectively.

If $x \geq 3$, we can choose 3 adjacent numbers $a, b, 1$ and replace them by $1, a, b$ in one step. Thus, in each step the position of 1 can be moved by 2 steps to the left, until $x = 1$ or $x = 2$. In the latter case, we can swap 1 and the number on its left. So 1 can be moved to the expected position in at most $\left\lceil \frac{2k + 2 - 1}{2} \right\rceil = k + 1$ steps. Afterwards, 2 can similarly be moved to the expected position in at most $\left\lceil \frac{2k + 2 - 2}{2} \right\rceil = k$ steps. After fixing 1 and 2, we can apply the inductive hypothesis to rearrange the remaining $2k$ numbers in k^2 steps. So at most

$$(k + 1) + k + k^2 = (k + 1)^2$$

steps are required to rearrange the numbers in ascending order. By induction, we get the upper bound n^2 for the number of steps required.

Next, consider the initial ordering $2n, 2n - 1, \dots, 1$. We claim that at least n^2 steps are needed in this case. Consider any number k . We need to move it to the left by $2n + 1 - 2k$ positions (note that this also holds when $k > n$, with the convention that moving -1 step to the left means moving 1 step to the right). It is clear that there must be some numbers which are moved to the right in each step. We first examine how many times the number k should move to the right throughout the process.

For every $m < k$, the pair (m, k) is in a wrong order initially in the sense that m lies on the right of k . Since m has to lie on the left of k in the end, there must be an occasion for which their order is changed. This only happens in an operation for which m moves to the left while k moves to the right (possibly together with another number). Since only one number is moved to the left in each step, there is at most one such m in one operation. Therefore, k has to move to the right in at least $k - 1$ operations (and hence at least $k - 1$ positions), and it needs to move to the left for at least

$$(2n + 1 - 2k) + (k - 1) = 2n - k$$

positions (which is nonnegative). As k can move at most two positions to the left each time, there are at least $\left\lceil \frac{2n - k}{2} \right\rceil$ operations for which k is the number moving to the left. Again, since there is exactly one number moving to the left in each operation, there should be at least

$$\sum_{k=1}^{2n} \left\lceil \frac{2n - k}{2} \right\rceil = n + (n - 1) + (n - 1) + (n - 2) + (n - 2) + \dots + 1 + 1 + 0 = n^2$$

operations in total.

Therefore, the minimum number of steps is n^2 . $\square$

Remarks. The invariances that we have considered in this solution are the following:

- the number of positions that each number should move to the left; and
- the number of pairs (m, k) which have to switch their order (for a fixed k).

Problem 3.7. Initially, there are three positive integers with sum S written on the blackboard. An operation consists of the following two steps in order. Suppose the positive integers on the blackboard are a, b, c .

- Firstly, we replace two of the integers a and b with difference at least 2 by $\left\lfloor \frac{a+b}{2} \right\rfloor$ and $\left\lceil \frac{a+b}{2} \right\rceil$ respectively.
- Secondly, we choose one of the integers $\left\lfloor \frac{a+b}{2} \right\rfloor$, $\left\lceil \frac{a+b}{2} \right\rceil$ or c (we denote the chosen integer by d) and replace this integer d by $S - d$.

For example, if the initial numbers are 1, 3, 7 with $S = 11$, then we can first replace these numbers by 4, 3, 4 since $\frac{1+7}{2} = 4$, and then replace these numbers by 4, 3, 7 since $7 = 11 - 4$. Note that only a single operation has been carried out. Also, the value of S is fixed throughout the process (although the sum of numbers may change). Prove that the process must come to an end no matter how the operations are carried out.

Analysis. The process will end when any two of the numbers differ by at most 1. It is not simple to find an invariance since the sum of the numbers may vary greatly, although it is not hard to prove that the numbers must be smaller than S throughout the process. We would like to find a function which keeps on decreasing. In view of our goal, such a function should be related to the differences of the numbers. Any such function must be decreasing in the first step. However, it does not make sense when a number d is replaced by $S - d$, unless S is involved in the differences in a certain way. This suggests the following choice.

Solution. Suppose the three numbers are a, b, c . We define

$$f = \left| a - \frac{S}{2} \right| + \left| b - \frac{S}{2} \right| + \left| c - \frac{S}{2} \right|.$$

We claim that f is monotonic decreasing throughout the process. Suppose we carry out the first step on a and b where $a \geq b + 2$.

- If $\frac{S}{2} \geq a > b$, then $\frac{S}{2} \geq \left\lfloor \frac{a+b}{2} \right\rfloor, \left\lceil \frac{a+b}{2} \right\rceil$. Note that the sum of these two numbers remains unchanged in the first step. Therefore,

$$f = S - a - b + \left| c - \frac{S}{2} \right|$$

remains unchanged.

- Similarly, if $a > b \geq \frac{S}{2}$, then f remains unchanged.
- Suppose $a > \frac{S}{2} > b$. Since $\left\lfloor \frac{a+b}{2} \right\rfloor, \left\lceil \frac{a+b}{2} \right\rceil$ differ by at most 1, they are both at least $\frac{S}{2}$ or both at most $\frac{S}{2}$. Therefore, the value of f is changed to

$$|S - (a+b)| + \left| c - \frac{S}{2} \right|.$$

In this case, one can show that

$$|S - (a+b)| < a - b = \left| a - \frac{S}{2} \right| + \left| b - \frac{S}{2} \right|.$$

Thus, f is strictly decreasing.

Next, in the second step, suppose we replace a number a by $S - a$. Then f remains unchanged since

$$\left| a - \frac{S}{2} \right| = \left| (S - a) - \frac{S}{2} \right|.$$

This shows f is monotonic decreasing. Since f can only take nonnegative rational values with denominator 2, it cannot decrease indefinitely. Thus, either the process terminates, or f eventually becomes a constant. From our discussion, this can only happen if we only apply the first step to two numbers which are both $\geq \frac{S}{2}$ or both $\leq \frac{S}{2}$. We shall show that this is impossible, meaning that the two numbers on the same side of $\frac{S}{2}$ must become closer and closer to each other.

To do so, we further define

$$g = |d_a - d_b| + |d_b - d_c| + |d_c - d_a|$$

where $d_x = \left| x - \frac{S}{2} \right|$ for $x = a, b, c$. This describes how close the differences of the numbers to $\frac{S}{2}$ are. WLOG assume we apply the first step to two numbers a, b with $a > b \geq \frac{S}{2}$. Then $|d_a - d_b| = |a - b|$ must decrease. Also,

$$|d_b - d_c| + |d_c - d_a| = \left| b - \frac{S}{2} - d_c \right| + \left| a - \frac{S}{2} - d_c \right|$$

is monotonic decreasing as before (here we may assume $\frac{S}{2} + d_c$ is a constant and argue in the same way). In the second step, the value of g remains unchanged. Therefore,

if we always choose two numbers on the same side of $\frac{S}{2}$, then g is strictly decreasing, which is impossible if there are infinitely many steps. Therefore, the process must come to an end as desired. $\square$

Problem 3.8. (APMO 2009 Problem 1) Consider the following operation on positive real numbers written on a blackboard: Choose a number r written on the blackboard, erase that number, and then write a pair of positive real numbers a and b satisfying the condition $2r^2 = ab$ on the board.

Assume that you start out with just one positive real number r on the blackboard, and apply this operation $k^2 - 1$ times to end up with k^2 positive real numbers, not necessarily distinct. Show that there exists a number on the board which does not exceed kr .

Analysis. The strategy is to find a function f such that $f(r)$ and $f(a) + f(b)$ have some relations under the constraint $2r^2 = ab$. As it seems that the optimal operation is to take $a = b = \sqrt{2}r$, we would like to have $f(r) = 2f(\sqrt{2}r)$. The variable r can be ignored if f is homogeneous. This suggests the choice of f in the solution.

Solution. Let $f(x) = \frac{1}{x^2}$. Whenever $2r^2 = ab$, we have

$$f(r) = \frac{1}{r^2} = \frac{2}{ab} \leq \frac{1}{a^2} + \frac{1}{b^2} = f(a) + f(b).$$

Therefore, the sum of the function values is monotonic increasing. Suppose the k^2 numbers obtained in the end are $r_1 \leq r_2 \leq \dots \leq r_{k^2}$. Then

$$\frac{1}{r^2} \leq \sum_{i=1}^{k^2} \frac{1}{r_i^2} \leq \frac{k^2}{r_1^2}.$$

This gives $r_1 \leq kr$ as desired. $\square$

Problem 3.9. (IMO Shortlist 2005 C5) There are n markers, each with one side white and the other side black. In the beginning, these n markers are aligned in a row so that their white sides are all up. In each step, if possible, we choose a marker whose white side is up (but not one of the outermost markers), remove it, and reverse the closest marker to the left of it and also reverse the closest marker to the right of it. Prove that, by a finite sequence of such steps, one can achieve a state with only two markers remaining if and only if $n - 1$ is not divisible by 3.

Analysis. To show that the final state can be attained, one simply needs to find an algorithm, probably with the use of induction. The harder part is to show that for the case $3 \mid n - 1$, the final state cannot be attained no matter how we operate. We have to find a function which is fixed throughout the process. If this can be done, then we know more about the configurations when the number of markers decreases.

The first simple idea may be assigning 1 for a white marker and 0 for a black marker and consider the sum of all values (in other words, we count the number of white markers). However, this is not an invariance, even if we only consider modulo 3 (which should be the case from the answer). Indeed, the sum modulo 3 is fixed only when we remove a white marker with both neighbours being white. In other cases, the number of white markers is decreased or increased by 1. It seems that only the parity is related, but not modulo 3.

We can try to modify the assignment of values by considering the operations. Suppose we remove the middle white marker in WWB (with obvious meaning of the labels). They become BW and all other markers are unaffected. The number of white markers is decreased by 1, but we would like the values assigned to the two white markers in WWB to have the same sum as that assigned to the white marker in BW . This suggests the relative position of the white and black markers should be taken into account. That is how we think of the solution given below.

Solution. For $n = 2$ and $n = 3$, it is obvious that the statement is true. Suppose we can achieve the final state (the state with only two markers remaining) for some n not congruent to 1 (mod 3). Consider the case $n + 3$. We use W to denote a marker with the white side facing upwards and B for black similarly. We apply the operations as follows. A marker with an underscore is the marker to be removed in the next step.

$$W\underline{W}WW\dots W \longrightarrow BB\underline{W}W\dots W \longrightarrow B\underline{W}BW\dots W \longrightarrow WW\dots W.$$

Since 3 steps are carried out, the number of markers is decreased by 3. Thus, we easily see that we can achieve the final state whenever $n \not\equiv 1 \pmod{3}$ by induction.

Suppose $n \equiv 1 \pmod{3}$. The case $n = 1$ is trivial. For the general case, we give a score to each marker for any given configuration. For each black marker, the score is 0. For each white marker, the score is 1 if there is an even number of black markers to its left, and the score is -1 if there is an odd number of black markers to its left. For example, the scores of the markers in $WWBWBWW$ are $1, 1, 0, -1, 0, 1, 1$ respectively. We claim that the sum S of all the scores assigned modulo 3 is an invariance.

- If $W\underline{W}W$ is changed to BB , then the scores of all markers not shown are unaffected. The scores of the white markers WW are the same. Therefore, their contribution to $S \pmod{3}$ is 0. As black markers do not contribute to the sum, $S \pmod{3}$ remains unchanged.
- If $W\underline{W}B$ is changed to BW , then the scores of all markers not shown are unaffected. The scores of the markers involved change either from $1, 1, 0$ to $0, -1$ or from $-1, -1, 0$ to $0, 1$. In both cases, $S \pmod{3}$ remains unchanged.
- If $B\underline{W}W$ is changed to WB , then the scores of all markers not shown are unaffected. The scores of the markers involved change either from $0, 1, 1$ to $-1, 0$ or from $0, -1, -1$ to $1, 0$. In both cases, $S \pmod{3}$ remains unchanged.

- If BWB is changed to WW , then the scores of all markers not shown are unaffected. The scores of the markers involved change either from $0, 1, 0$ to $-1, -1$ or from $0, -1, 0$ to $1, 1$. In both cases, $S \pmod{3}$ remains unchanged.

This proves the claim. Initially, we have $S = n \equiv 1 \pmod{3}$. Assume it is possible to have two markers remaining. Then the sum of their scores is $\equiv 1 \pmod{3}$. This can happen only when the two markers are WB . However, we also note that the parity of the number of black markers is fixed in each operation. Therefore, it cannot happen that there is exactly one black marker. This completes the proof. $\square$

Problem 3.10. (IMO Shortlist 2000 A5/Problem 3) k is a positive real. N is an integer greater than 1. N points are placed on a line, not all coincident. A *move* is carried out as follows. Pick any two points A and B which are not coincident. Suppose that A lies to the right of B . Replace B by another point B' to the right of A such that $AB' = kBA$. For what values of k can we move the points arbitrarily far to the right by repeated moves?

Analysis. When $N = 2$, we only need to see when $1 + k + k^2 + \dots$ is divergent. The goal is met when $k \geq 1$. When $N = 3$, it is natural to consider the unique operation applied on the two points with the greatest distance. Suppose the initial points are A, B, C from left to right. Let $AB = a$ and $BC = b$. After one operation, the distances change from a, b to $b, k(a + b)$. We want to find a certain function in terms of the distances a, b . If the value of the function does not decrease after an operation, then we can move the points arbitrarily far to the right. Although we cannot conclude anything if the function may decrease, it gives us a good hint to the answer.

Indeed, we consider $f = \alpha a + \beta b$ for some constants α, β . After one operation as specified, we have $f = \alpha b + \beta k(a + b) = \beta k a + (\alpha + \beta k)b$. If k is the critical value, then f should remain unchanged. This suggests we should have $\alpha = \beta k$ and $\beta = \alpha + \beta k$ by comparing coefficients. This gives $\beta = 2\alpha$. So we can simply choose $f = a + 2b$. From this, we also believe that the answer for the case $N = 3$ is $k \geq \frac{1}{2}$. Indeed, the function $f = a + 2b = (a + b) + b$ has a special meaning in our situation. It simply means the sum of distances of the points from the rightmost point. This can easily be generalized to the case N .

Solution. Suppose the points are $a_1 \geq a_2 \geq \dots \geq a_N$ where $a_1 > a_N$. Define

$$f = (a_1 - a_2) + (a_1 - a_3) + \dots + (a_1 - a_N).$$

Consider the case $k \geq \frac{1}{N-1}$. In each operation, we choose A to be the rightmost point and B to be the leftmost point. Then a_N is replaced by $a_1 + k(a_1 - a_N)$. However, the order of the points also changes. Therefore, the value of f becomes

$$\tilde{f} = \sum_{j=1}^{N-1} (a_1 + k(a_1 - a_N) - a_j) = k(N-1)(a_1 - a_N) + \sum_{j=2}^{N-1} (a_1 - a_j).$$

Since $k \geq \frac{1}{N-1}$ and $a_1 > a_N$, we have $\tilde{f} \geq f$. Since f is monotonic increasing and $a_1 - a_N \geq \frac{1}{N-1}f$, the distance between the two furthest points is bounded below by a positive number, say C_0 . In each step, the rightmost point is increased by at least C_0k . Therefore, after finitely many steps, the rightmost point, and hence all the points, can be made sufficiently far to the right.

We now turn to the case $k < \frac{1}{N-1}$. Although the above function is decreasing in this case, it does not mean the points cannot be made sufficiently large (for example, $1 + \frac{1}{2} + \frac{1}{3} + \dots$ is divergent). Thus, the absolute size of the points should also be taken into account. This suggests we should consider

$$g = a_1 + C_1((a_1 - a_2) + (a_1 - a_3) + \dots + (a_1 - a_N))$$

for some positive constant C_1 . We claim that g is monotonic decreasing if we choose C_1 appropriately. Suppose the operation is applied on the points $A = a_u$ and $B = a_t$. This means a_t is replaced by $a_u + k(a_u - a_t)$. If $a_u + k(a_u - a_t) < a_1$, then it is obvious that g decreases. If $a_u + k(a_u - a_t) \geq a_1$, then the order of the numbers is changed. Therefore, the value of g becomes

$$\tilde{g} = a_u + k(a_u - a_t) + C_1 \sum_{\substack{j=1 \\ j \neq t}}^N (a_u + k(a_u - a_t) - a_j).$$

Note that

$$\begin{aligned} & \tilde{g} \leq g \\ \Leftrightarrow & (a_u - a_1 + k(a_u - a_t)) + C_1((N-1)(a_u - a_1 + k(a_u - a_t)) - a_1 + a_t) \leq 0 \\ \Leftrightarrow & (1 + C_1(N-1))(a_u - a_1 + k(a_u - a_t)) - C_1(a_1 - a_t) \leq 0. \end{aligned}$$

Note that $a_u - a_1 + k(a_u - a_t) \leq k(a_1 - a_t)$ since this is equivalent to $(a_1 - a_u)(k+1) \geq 0$. Thus, if we choose C_1 such that $(1 + C_1(N-1))k = C_1$, i.e. $C_1 = \frac{k}{1 - (N-1)k}$, then

we obtain $\tilde{g} \leq g$. Note that C_1 is positive since $k < \frac{1}{N-1}$.

As g is monotonic decreasing, the value of g is bounded above. Since $a_1 < g$, the rightmost point a_1 is also bounded. This shows we cannot move the points to arbitrarily right. Thus, the answer is $k \geq \frac{1}{N-1}$. $\square$

Here is a summary of the two major ideas introduced in this section.

- providing algorithms ([problems 3.1, 3.2, 3.3, 3.4, 3.5, 3.7, 3.9, 3.10](#))
- finding invariances ([problems 3.6, 3.7, 3.8, 3.9, 3.10](#))

4 Graph Theory

Many mathematical problems can be rephrased in terms of graph theory. It is sometimes easier for us to think in this direction as there are some standard results and terminologies that we can use in this area. However, due to the many well-known facts discovered in this topic, the popularity of graph theory problems has declined in Olympiad. But it is still necessary for us to learn those terminologies and facts. In addition, we shall learn how to use graph theory in problems which do not seem to be related to graphs directly.

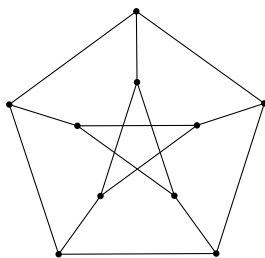
4.1 Concepts in Graph Theory

Firstly, we provide the usual terminologies in graph theory.

Definition 4.1. A *simple graph*(簡單圖) $G = (V, E)$ consists of a nonempty set of *vertices*(頂點) V and a set of *edges*(邊) E containing some unordered pairs of distinct vertices, such that there is at most one edge between every pair of vertices.

Let $v, w \in V$. Note that $\{v, w\}$ and $\{w, v\}$ represent the same edge if they belong to E . We may also use vw or wv to represent this edge. If $vw \in E$, we say that v and w are *adjacent*(相鄰) or *neighbouring* vertices. Also, we say that v is *incident with* vw (or vw is incident with v). In most cases, we only work on simple graphs of finitely many vertices.

We can visualize a simple graph geometrically. The vertices are represented by points, and edges are represented by segments joining the vertices. Since only the endpoints matter, we may use curves instead of segments. For example, the figure below shows the *Petersen graph*(佩特森圖), which has 10 vertices and 15 edges. It often serves as an example in graph theory since it has several special properties.



In the definition of a simple graph, we require that every edge joins two distinct vertices, and there is at most one edge between every pair of vertices. We can relax

these conditions and consider more general graphs. For example, we can have some *loops*(自環) in a graph, where a loop is an edge that joins a vertex with itself. We can allow more than one edge between two vertices to form a *multigraph*(多重圖). It is also possible to give directions to the edges, which gives a *directed graph*(有向圖). In other words, the edges vw and wv are not the same. In a simple directed graph, it is possible to have both edges vw and wv .

Definition 4.2. Let $G = (V, E)$ be a simple graph. The *degree*(度) of a vertex $v \in V$ is the number of edges incident with v , and is denoted by $\deg(v)$.

For a simple graph with n vertices, the degree of each vertex can only be $0, 1, \dots, n-1$. For a directed graph, we may define *in-degree*(入度) $\deg^-(v)$ (number of edges with v as the terminal vertex) and *out-degree*(出度) $\deg^+(v)$ (number of edges with v as the initial vertex) analogously.

Definition 4.3. A *complete graph*(完全圖) of n vertices is a simple graph for which every pair of vertices is joined by an edge. It is denoted by K_n .



Clearly, the degree of each vertex of K_n is $n-1$, and there are $\binom{n}{2}$ edges in total.

Definition 4.4. Let $G = (V, E)$ be a simple graph. A *subgraph*(子圖) is a graph $H = (V', E')$ such that $V' \subset V$ and $E' \subset E$.

This means a subgraph is a graph that contains some (possibly all) of the vertices and edges of the original graph. In particular, an *induced subgraph*(導出子圖) of $G = (V, E)$ is a subgraph $H = (V', E')$ such that E' is the set of all edges in E that join two vertices in V' . Equivalently, H can be obtained from G by removing some vertices together with all edges incident with some removed vertices. In addition, a *spanning subgraph*(生成子圖) is a subgraph that contains all vertices (and some edges) of the original graph.

We first go through some basic results about simple graphs which can be derived using double counting.

Theorem 4.1. ([Handshaking theorem](#)(握手定理)) Let $G = (V, E)$ be a simple graph. Then we have

$$\sum_{v \in V} \deg(v) = 2|E|.$$

Proof. For any edge in E , it contributes 2 to the sum of degrees as it has 2 endpoints. The result follows immediately. $\square$

This gives a simple relationship between the degrees and the number of edges.

Theorem 4.2. ([Mantel's theorem](#)(曼特爾定理)) Let $G = (V, E)$ be a simple graph with n vertices which does not contain any triangle. Then we have

$$|E| \leq \left\lfloor \frac{n^2}{4} \right\rfloor$$

Proof. A triangle means three edges vw, wx, xv . For each edge vw , we count the number of edges of the form vx or wx where $x \neq v, w$. Since there is no triangle, we cannot have both edges vx and wx . As there are $n - 2$ other vertex x , there are at most $n - 2$ such edges. On the other hand, the number of such edges must be

$$(\deg(v) - 1) + (\deg(w) - 1) = \deg(v) + \deg(w) - 2.$$

Summing over all edges vw , we obtain

$$\sum_{vw \in E} (\deg(v) + \deg(w) - 2) \leq \sum_{vw \in E} (n - 2),$$

or

$$\sum_{vw \in E} (\deg(v) + \deg(w)) \leq \sum_{vw \in E} n = n|E|.$$

For the left-hand side, note that $\deg(v)$ is counted $\deg(v)$ times since there are $\deg(v)$ edges incident with v by definition. It follows that

$$\sum_{vw \in E} (\deg(v) + \deg(w)) = \sum_{v \in V} \deg(v)^2 \geq \frac{1}{n} \left(\sum_{v \in V} \deg(v) \right)^2 = \frac{1}{n} (2|E|)^2 = \frac{4|E|^2}{n}$$

by the QM-AM inequality and the handshaking theorem. Combining these, we deduce

$$\frac{4|E|^2}{n} \leq n|E|.$$

This implies $|E| \leq \frac{n^2}{4}$, and hence $|E| \leq \left\lfloor \frac{n^2}{4} \right\rfloor$ as $|E|$ is an integer. $\square$

There are some Olympiad problems which can be rephrased as problems in graph theory in the obvious manner. We mainly use the terminologies in graph theory to solve such problems for convenience.

Problem 4.1. (APMO 2003 Problem 5) Given two positive integers m and n , find the smallest positive integer k such that among any k people, either there are $2m$ of them who form m pairs of mutually acquainted people or there are $2n$ of them forming n pairs of mutually unacquainted people.

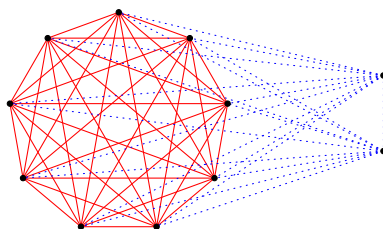
Analysis. WLOG assume $m \geq n$. We first start with some small cases. When $n = 1$, the condition is satisfied whenever there are 2 unacquainted people. So we may assume every pair of people is acquainted. In that case, we clearly have $k = 2m$.

When $n = 2$, in order to avoid having 2 pairs of mutually unacquainted people, it is not hard to see that the optimal case happens when there exists a person who is unacquainted to any other people, and these are the only pairs of unacquainted people. In that case, we should have $k = 2m + 1$. We can indeed confirm this by some exhaustion arguments.

From these small cases, we may observe that k is increased by 1 when n is increased by 1. This allows us to guess the answer and we can try to prove this using induction. Since there are two parameters, choosing the appropriate quantity to carry out induction is important so as to simplify the arguments.

Solution. WLOG assume $m \geq n$. We claim that the smallest k is $2m + n - 1$. In the following, we identify the people as vertices of a complete graph. Two vertices are joined by a red edge if the two people they represent are acquainted, and they are joined by a blue edge otherwise.

Firstly, we claim that $k \geq 2m + n - 1$. It suffices to give a counterexample when there are exactly $2m + n - 2$ vertices. Inspired by the construction in the special case $n = 2$, we give the following construction. Let $v_1, v_2, \dots, v_{2m+n-2}$ be the vertices. We colour the edges between any two of $v_1, v_2, \dots, v_{2m-1}$ red, and all other edges blue (the figure below corresponds to the case $m = 5$ and $n = 3$). Since only the $2m - 1$ vertices $v_1, v_2, \dots, v_{2m-1}$ are joined by some red edges, it is clear that we cannot find m disjoint pairs of red edges.



Suppose there are n disjoint pairs of blue edges. As there are

$$(2m + n - 2) - (2m - 1) = n - 1$$

vertices not belonging to the red K_{2m-1} , the n pairs of blue edges must be formed by some vertices including at least $2n - (n - 1) = n + 1$ of the vertices $v_1, v_2, \dots, v_{2m-1}$. By

the pigeonhole principle, 2 of them must form a blue edge, which is a contradiction. So this is indeed a counterexample.

Secondly, we prove that the condition is satisfied when $k = 2m + n - 1$. When $n = 1$, it is obvious that any $k = 2m$ vertices must contain a red K_{2m} or a blue K_2 , since a blue K_2 merely means a blue edge. We can easily extract m pairs of red edges from the former case if there is no blue edge. This serves as the base case of the induction on n for a fixed difference $m - n$.

Assume the claim holds for some pair (m, n) . For the inductive step, we work on the case $(m + 1, n + 1)$. In other words, we need to show that for any colouring of K_{2m+n+2} , there must be $m + 1$ disjoint pairs of red edges or $n + 1$ disjoint pairs of blue edges. If there exist 3 vertices v, w, x such that vw is red and vx is blue, we remove these 3 points and apply the inductive hypothesis to the remaining $2m + n - 1$ vertices. This either gives m disjoint pairs of red edges or n disjoint pairs of blue edges not involving v, w, x . By including the edge vw or the edge vx , we easily get what we want.

So it suffices to consider the case when every 3 vertices form a monochromatic triangle. If vw, wx, xv are red, then by considering v, w, y for any other vertex y , the edges vy and wy must be red. By considering v, y, z for any other vertex z , the edge yz must be red. Therefore, all edges in the graph are red. Similarly, if vw, wx, xv are blue, then all edges in the graph are blue. Both cases are trivial as

$$2m + n + 2 \geq 2(m + 1) \geq 2(n + 1).$$

The inductive step is complete.

By induction, the condition is satisfied when $k = 2m + n - 1$. So the minimum k is $\max\{m, n\} + m + n - 1$ in general. $\square$

Next, we introduce some more terminologies which are commonly used.

Definition 4.5. A *walk*(路徑) of length s is a sequence of edges $e_1, e_2, \dots, e_s$ in a simple graph such that the terminal vertex of e_j coincides with the initial vertex of e_{j+1} for each $j = 1, 2, \dots, s - 1$.

In particular, a *trail*(軌跡) is a walk for which every edge is used at most once, while a *path*(簡單路徑) is a walk for which every vertex is visited at most once (possibly except the initial vertex and the terminal vertex).

We sometimes use a sequence of vertices $v_1, v_2, \dots, v_{s+1}$ to denote a walk of length s , where v_j and v_{j+1} are adjacent for each $j = 1, 2, \dots, s$. The concept of paths is the most useful one among the above three concepts. Using these concepts, we can define the *distance*(距離) between two vertices to be the minimum length of a path that joins these two vertices.

Definition 4.6. A *circuit*(迴路) of length s is a walk $e_1, e_2, \dots, e_s$ in a simple graph such that the terminal vertex of e_s also coincides with the initial vertex of e_1 .

In particular, a *cycle*(圈) is a circuit for which every vertex is visited at most once except the initial vertex and the terminal vertex.

In fact, we do not normally emphasize the initial vertex of a circuit. For example, the circuit e_1, e_2, e_3, e_4 is the same as the circuit e_2, e_3, e_4, e_1 .

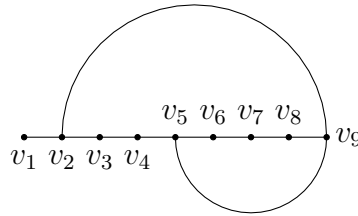
For a directed graph, the above definitions apply but the directions of the edges in a walk or a circuit should be consistent.

Problem 4.2. There are $n \geq 4$ people in a meeting. Suppose everyone knows at least 3 other people in the meeting. Prove that we can find k people for some integer $k \not\equiv 0 \pmod{5}$ with $k \geq 3$, and arrange them in a circle such that each person knows his two neighbours.

Analysis. The first question is how can we form a cycle (of an arbitrary length)? A simple idea is to add more and more vertices to form a longer path, until we have to go back to some vertices which have been used. This necessarily forms a cycle (which may not contain all vertices we have used in the argument). It remains to see why we can always find a cycle of length k which is not divisible by 5.

Solution. Consider the graph where vertices represent the people. Two vertices are joined by an edge if they know each other. We are given that $\deg(v) \geq 3$ for each v , and we need to prove that there exists a cycle of length $k \not\equiv 0 \pmod{5}$.

Consider a longest path $v_1, v_2, \dots, v_m$ such that $v_1 \neq v_m$. Such a path exists because there are finitely many vertices in the graph. (Alternatively, we can build up a maximal path using the idea mentioned above.) Note that we must have $m \geq 4$ since $\deg(v_m) \geq 3$ (thus we can find a new neighbour of v_m if $m \leq 3$). As the length is maximized, all neighbours of v_m belong to this path. As $\deg(v_m) \geq 3$, we can pick two neighbours v_i and v_j of v_m where $i < j < m - 1$.



This already gives us three cycles:

$$\begin{aligned} &v_i, v_{i+1}, \dots, v_{m-1}, v_m; \\ &v_j, v_{j+1}, \dots, v_{m-1}, v_m; \\ &v_i, v_{i+1}, \dots, v_{j-1}, v_j, v_m. \end{aligned}$$

Their lengths are $m - i + 1$, $m - j + 1$ and $j - i + 2$ respectively. Since

$$(m - i + 1) - (m - j + 1) - (j - i + 2) = -2,$$

it is impossible that all of $m - i + 1$, $m - j + 1$ and $j - i + 2$ are multiples of 5. Therefore, one of these cycles has a length which is not a multiple of 5. $\square$

Problem 4.3. (CMO 1987 Problem 3) In a sport tournament, every two players play against once. There is a winner for each match. After the tournament, we select some outstanding players. A player A is said to be outstanding if for any other player B , either A wins over B or there exists a player C such that C wins over B and A wins over C .

Suppose there is only one outstanding player according to the above rule. Prove that this player wins over all other players.

Analysis. Since the outstanding player should win over all other players, it is natural to consider the player who wins the greatest number of matches. We shall show that this player is the outstanding player, and he cannot lose to other players.

Solution. Consider a directed graph where vertices represent the players, and an edge is drawn from v to w if player v wins over player w . An outstanding vertex v now means a vertex such that the distance from v to w is at most 2 for any other vertex w .

Let v be a vertex with the largest out-degree. Suppose $vw_1, vw_2, \dots, vw_k$ are the only edges from v . For any other vertex x (if exists), since $\deg^+(x) \leq k$ and xv is an edge (otherwise x is one of the w_j 's), we cannot have all edges $xw_1, xw_2, \dots, xw_k$. WLOG assume w_kx is an edge. Then we have the path vw_k, w_kx of length 2. As this holds for any vertex $x \neq w_k$, this shows v is outstanding by definition. Thus, v is the only outstanding vertex.

Suppose on the contrary that v does not win every match. We expect that there should be an outstanding player among those who win over v . Thus, we consider the induced subgraph obtained by removing $v, w_1, \dots, w_k$. Using exactly the same argument as above, we can find an outstanding vertex y in this subgraph. Since yv is an edge, we can form the path yv, vw_j for each j . This shows y is outstanding even in the original graph. This contradicts the given fact that there is only one outstanding vertex. Therefore, v should win over all other players. $\square$

Remarks. In graph theory, it is always useful to consider a vertex with the smallest or the greatest degree.

Definition 4.7. A *connected graph*(*連通圖*) is a simple graph for which every two vertices are connected by a path. A *component*(*分量*) is a maximal connected subgraph. A directed graph is *strongly connected*(*強連通*) if there exist paths of both directions between every two vertices.

In other words, two vertices in the same component can be connected by a path, while vertices in different components cannot be connected by a path. Clearly, every simple graph of finitely many vertices can be decomposed into a finite number of components. The following gives a more specific result.

Proposition 4.1. Let $G = (V, E)$ be a simple graph of finitely many vertices such that $\deg(v) = 2$ for all $v \in V$. Then G can be partitioned into a finite number of disjoint cycles.

Proof. Let $v_1, v_2, \dots, v_k$ be a longest path with $v_1 \neq v_k$. Since $\deg(v_k) = 2$, v_k must be adjacent to a vertex different from v_{k-1} . As the path is the longest one, v_k must be adjacent to some v_m where $m < k - 1$. However, the two neighbours of v_i are v_{i-1} and v_{i+1} for $i = 2, 3, \dots, k - 2$. Therefore, we must have $m = 1$. This gives a cycle $v_1, v_2, \dots, v_k$, which must be a component as the neighbours of all these vertices are confirmed. By removing this cycle, we can prove by induction that the graph can be partitioned into finitely many disjoint cycles. $\square$

Proposition 4.2. Let $G = (V, E)$ be a connected graph with $|V| = n$. Then $|E| \geq n - 1$.

Proof. We prove this by induction on n . The base case $n = 2$ is trivial as we need an edge to connect two vertices. Assume the result holds for the case $n - 1$. For the case n , suppose on the contrary that there are at most $n - 2$ edges. Then the sum of degrees is at most $2(n - 2)$ by the handshaking theorem. By the pigeonhole principle, there is a vertex v such that

$$\deg(v) \leq \frac{2(n - 2)}{n} < 2.$$

As the graph is connected and $|V| \geq 2$, we must have $\deg(v) = 1$. By removing v and the edge incident with v , we obtain a connected graph of $n - 1$ vertices having at most $n - 3$ edges. This contradicts the inductive hypothesis. So the result holds by induction. $\square$

Problem 4.4. There are $n \geq 4$ railway stations in a city. For any two different stations, either there is a rail joining them directly or not. Suppose for any pair of stations, they are either joined directly by a rail or there exists another station to which both of them are joined. Also, there is no central station in the city, i.e. a station which is joined to all other stations directly. Find the minimum number of rails that are running in the city.

Analysis. Since the distance between any two vertices is either 1 or 2, there should be some vertices with large degrees. The condition prohibits the existence of a vertex with degree $n - 1$. So it is natural to consider some cases when there exist some vertices with degrees $n - 2$ or $n - 3$. By working out a few possibilities and

comparing these, it is not hard to find an optimal case to run the smallest number of rails.

To establish the lower bound, we may first consider a vertex v with the smallest degree. This is because there is only a few possibilities for the degree of v , or otherwise the number of edges satisfies the desired bound. Thus, we only need to work on a few cases. Also, the condition guarantees that all vertices are either adjacent to v or adjacent to a neighbour of v . This allows us to put all vertices into suitable groups for consideration.

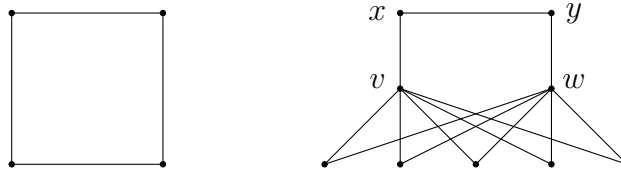
Solution. The answer is $\begin{cases} 4 & \text{if } n = 4, \\ 2n - 5 & \text{if } n \geq 5. \end{cases}$

In the following, we consider the graph $G = (V, E)$ where vertices represent the railway stations and edges represent the rails.

For $n = 4$, a cycle of length 4 satisfies the conditions. If $|E| \leq 3$, then there exists a vertex v such that

$$\deg(v) \leq \left\lfloor \frac{2 \times 3}{4} \right\rfloor = 1$$

by the pigeonhole principle. As the graph must be connected, we have $\deg(v) = 1$. Suppose w is the vertex adjacent to v . As the distance between v and any other vertex is 2, w must be adjacent to all vertices. This contradicts the given condition. Therefore, the minimum value of $|E|$ is 4 when $n = 4$.



For $n \geq 5$, we claim that the minimum value is $2n - 5$. We first provide a construction. Suppose we have the edges vx, xy, yw , and each of the other $n - 4$ vertices is adjacent to both v and w . There are

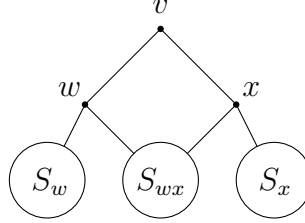
$$3 + 2(n - 4) = 2n - 5$$

edges in total. One can check that all conditions are satisfied. So $2n - 5$ edges suffice.

To prove the lower bound, let v be a vertex with the smallest degree. Clearly, its degree is positive.

- If $\deg(v) = 1$, say w is adjacent to v , then w should be adjacent to all other vertices in order that the distance between v and any other vertex is at most 2. This is a contradiction.
- If $\deg(v) = 2$, say w and x are adjacent to v , we define the following.

- S_w : the set of remaining vertices adjacent to w but not x
- S_x : the set of remaining vertices adjacent to x but not w
- S_{wx} : the set of remaining vertices adjacent to both w and x



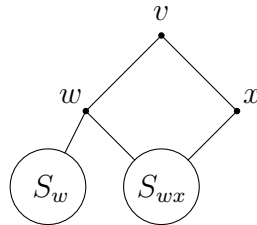
We first consider the general case where S_w and S_x are nonempty. For any $a \in S_w$ and $b \in S_x$, either ab is an edge or there exists $c \neq v, w, x$ (could be in S_w , S_x or S_{wx}) which is adjacent to both a and b . Consider the induced subgraph H formed by all such vertices c (a vertex that is adjacent to some $a \in S_w$ and $b \in S_x$ where a, b are non-adjacent) and all vertices in S_w and S_x . Note that H is connected since there exists a path between each $a \in S_w$ and each $b \in S_x$, and every other vertex c is adjacent to some $a \in S_w$ and $b \in S_x$ by definition. By [proposition 4.2](#), H contains at least

$$|S_w| + |S_x| - 1$$

edges. Together with those $|S_w| + |S_{wx}| + 1$ edges incident with w and those $|S_x| + |S_{wx}| + 1$ edges incident with x , we find that

$$\begin{aligned} |E| &\geq (|S_w| + |S_x| - 1) + (|S_w| + |S_{wx}| + 1) + (|S_x| + |S_{wx}| + 1) \\ &= 2(|S_w| + |S_x| + |S_{wx}|) + 1 \\ &= 2(n - 3) + 1 \\ &= 2n - 5. \end{aligned}$$

This establishes the lower bound if S_w and S_x are nonempty.



Next, WLOG assume S_x is empty. Then w and x are not adjacent or otherwise w is adjacent to all vertices. For any $a \in S_w$, there exists $b \in S_{wx}$ which is

adjacent to both a and x . This means there is an edge joining a and a vertex in S_{wx} for each $a \in S_w$. Thus, there are at least $|S_w|$ such edges in total. Therefore,

$$\begin{aligned} |E| &\geq |S_w| + (|S_w| + |S_{wx}| + 1) + (|S_{wx}| + 1) \\ &= 2(|S_w| + |S_{wx}|) + 2 \\ &= 2(n - 3) + 2 \\ &= 2n - 4. \end{aligned}$$

- If $\deg(v) \geq 3$, then the degree of all vertices are at least 3. We can use double counting to reject this case. Indeed, we count the number of pairs of vertices in two ways. On the one hand, there are exactly $\binom{n}{2}$ pairs. On the other hand, each pair of vertices w, x either corresponds to an edge wx or a pair of edges wy, yx for some vertex y . For each vertex y with degree d , there are $\binom{d}{2}$ pairs of edges with y as a common endpoint. It follows that

$$\binom{n}{2} \leq |E| + \sum_{j=1}^n \binom{d_j}{2} \quad (1)$$

where $d_1 \geq d_2 \geq \dots \geq d_n$ are the degrees of the vertices. (Note that this is an inequality instead of an identity since some pairs of vertices w, x may correspond to more than one pair of edges.) As each d_j is at least 3 and the sum of degrees is $2|E|$, the sequence $(d_1, d_2, \dots, d_n)$ is majorized by $(2|E| - 3(n - 1), 3, 3, \dots, 3)$. By the majorization inequality, we have

$$\begin{aligned} \sum_{j=1}^n \binom{d_j}{2} &\leq \binom{2|E| - 3(n - 1)}{2} + (n - 1) \binom{3}{2} \\ &= \frac{(2|E| - 3n + 3)(2|E| - 3n + 2)}{2} + 3(n - 1). \end{aligned} \quad (2)$$

It follows from (1) and (2) that

$$n(n - 1) \leq 2|E| + (2|E| - 3n + 3)(2|E| - 3n + 2) + 6(n - 1).$$

Suppose on the contrary that $|E| \leq 2n - 6$. Then we obtain

$$n(n - 1) \leq 2(2n - 6) + (n - 9)(n - 10) + 6(n - 1).$$

(Note that $2|E| - 3n + 2 > 0$ as $2|E| \geq 3n$.) This implies $8n \leq 72$, and hence $n \leq 9$. However, we also have

$$2n - 6 \geq |E| = \frac{1}{2} \sum_{j=1}^n d_j \geq \frac{3n}{2}.$$

This yields $n \geq 12$, which contradicts $n \leq 9$.

In any case, the number of edges is at least $2n-5$ when $n \geq 5$. So this is the minimum number of rails needed. $\square$

Remarks. In graph theory, it is common to count the number of pairs of vertices since this is related to the degrees of the vertices. As such a relation involves the binomial coefficients, it is natural to use the majorization inequality to derive some useful bounds.

Definition 4.8. An *acyclic graph*(無環圖) is a simple graph with no cycles. In particular, a *tree*(樹) is a connected acyclic graph.

An acyclic graph is also called a *forest*(森林), because it must be the union of disjoint trees.

Tree is an important concept since we often come across graphs without cycles. The following result gives some characterizations of trees.

Proposition 4.3. Let $G = (V, E)$ be a simple graph. Then the following are equivalent.

- (i) G is a tree
- (ii) G is a minimal connected graph (i.e. G is connected, but is not connected if we remove any edge)
- (iii) G is a maximal acyclic graph (i.e. G is acyclic, but is not acyclic if we add any edge)

Proof. We prove four implications one by one.

- Suppose (i) holds. By definition, G is connected. If we remove an edge vw , we show that v and w are no longer connected. Indeed, if there exists a path $v, x_1, x_2, \dots, x_k, w$ in the new subgraph, then $v, x_1, x_2, \dots, x_k, w$ is a cycle in G as v and w are adjacent. This contradicts the assumption that G is a tree. Therefore, (ii) holds.
- Suppose (ii) holds. Then G is connected. If G contains a cycle $v, x_1, x_2, \dots, x_k, w$, we can remove the edge vw so that the remaining subgraph is still connected (since if we need to use the edge vw in a path, we could simply use the longer path $v, x_1, x_2, \dots, x_k, w$, possibly with some edges removed if we visit the same vertex more than once). This violates the condition. So (i) holds.
- Suppose (i) holds. By definition, G is acyclic. Since G is connected, for any vertices v and w , there exists a path $v, x_1, x_2, \dots, x_k, w$. If vw is not an edge and we add this new edge, we must have $k \geq 1$, and a cycle $v, x_1, x_2, \dots, x_k, w$ is formed. Since this holds for all pairs of vertices, (iii) holds.

- Suppose (iii) holds. Then G is acyclic. To show that it is connected, we assume on the contrary that there is no path between some vertices v and w . By adding the edge vw , there should not be any cycle in the new graph. This is because any such cycle must be of the form $v, x_1, x_2, \dots, x_k, w$, which means there is a path between v and w before vw is added. Therefore, the new graph is still acyclic, violating the condition. So (i) holds.

□

In view of this result, we see that every connected graph contains a spanning tree, i.e. a spanning subgraph which is a tree. This is because we can remove more and more edges while keeping the graph connected, until this cannot be done. As there are finitely many edges, the process must come to an end. This gives a minimal connected graph, which must be a tree. In general, every simple graph contains a spanning forest.

Another important property of a tree is the following.

Proposition 4.4. Let $G = (V, E)$ be a connected graph with $|V| = n$. Then G is a tree iff $|E| = n - 1$.

Proof. We first show that G is a tree implies $|E| = n - 1$ by induction on n . When $n = 2$, we must have one edge joining the two vertices. For the inductive step, we are done if there exists a vertex v with degree 1, since we can remove v and the edge incident with v and apply the inductive hypothesis (the remaining subgraph is clearly a tree). As G is connected, we may now assume $\deg(w) \geq 2$ for all $w \in V$. Using a similar argument as that in [problem 4.2](#), we know that G must contain a cycle, contradiction. Therefore, we must have $|E| = n - 1$ by induction.

Conversely, suppose $|E| = n - 1$. As G is connected, it contains a spanning tree. From above, this spanning tree consists of $n - 1$ edges. Thus, this tree is exactly G itself. □

Using this result, we easily see that every tree of $n \geq 2$ vertices has at least two *leaves* (葉), i.e. vertices of degree 1. Another general result is that an acyclic graph with n vertices and k components has exactly $n - k$ edges, since each component is a tree.

Problem 4.5. Let $G = (V, E)$ be a connected graph of $n \geq 2$ vertices. Prove that we can colour $\left\lfloor \frac{n}{2} \right\rfloor$ vertices red such that each vertex in G is either red or adjacent to a red vertex.

Analysis. An optimal example is a path of length n . In that case, the only natural idea is to colour the even vertices red by counting from one end of the path. This idea can be generalized to an arbitrary graph by using its spanning tree. We shall demonstrate this trick in the solution below.

Solution. Let H be a spanning tree of G . Choose any vertex v . Let S be the set of all vertices which have an odd distance from v in H , and let T be the set of all vertices which have an even distance from v in H (including v). Note that one of S and T has size at most $\left\lfloor \frac{n}{2} \right\rfloor$. So it suffices to show that we can colour all vertices in S red or colour all vertices in T red. (If there are fewer than $\left\lfloor \frac{n}{2} \right\rfloor$ vertices, we can colour more vertices red arbitrarily.)

We only need to show that each vertex in S is adjacent to a vertex in T , and vice versa. Indeed, for any $w \in S$, the distance between v to w is an odd integer d . By definition, there is a path $v, x_1, x_2, \dots, x_{d-1}, w$ in H . Clearly, the distance between v and x_{d-1} is $d-1$ (if the distance is smaller, the distance between v and w would also be smaller). Therefore, $x_{d-1} \in T$. This shows each $w \in S$ is adjacent to some vertex in T .

Similarly, for any $y \in T$ with $y \neq v$, y is adjacent to some vertex in S . In addition, there must be a vertex z adjacent to v in H . The distance between v and z is 1, and hence $z \in S$. Thus, $v \in T$ is adjacent to a vertex in S . This proves the result. $\square$

Problem 4.6. (IMO 1991 Problem 4) Suppose G is a connected graph with k edges. Prove that it is possible to label the edges $1, 2, \dots, k$ in such a way that at each vertex which belongs to two or more edges, the greatest common divisor of the integers labelling those edges is equal to 1.

Analysis. A simple idea is to use consecutive integers to label edges incident with the same vertex. This works well if the whole graph is a path or a cycle. More generally, we can use this idea if G is a tree, since a tree can be partitioned into many paths in an organized way (like the branches of a tree in reality). The remaining idea is to see how we can transform an arbitrary connected graph to a graph of these kinds by ‘moving’ the edges in some ways.

Solution. In the following, we say that a vertex is *good* if the gcd of the labels of the edges incident with this vertex is 1.

If G is a cycle, we can label the edges in this cycle from 1 to k in this order. Then for each vertex, the two edges incident with it are labelled by $j, j+1$ for some j or labelled by $1, k$. In both cases, the vertex is good.

If G is a tree, we prove a stronger result that such a labelling exists if we use the labels $m, m+1, \dots, m+k-1$ where m is an arbitrary positive integer. We can prove this by induction on k . The base case $k=1$ is trivial as the degree of each vertex is 1. For the inductive step, we first pick a path P between two leaves of G (which exist by the corollary of [proposition 4.4](#)). We label the edges in P by the smallest labels in order. Clearly, all vertices in P are good no matter how we label the remaining edges.

Consider the subgraph H of G obtained by deleting all edges in P . Then H is an acyclic graph with fewer edges. Each component of H is a tree. We can partition the remaining labels into groups of consecutive integers according to the number of vertices in each component of H . By the inductive hypothesis, we can use these integers to label the edges in each component of H such that all vertices are good in H . But this also means that all vertices are good in G . So we have proven the result by induction when G is a tree.

Finally, consider the case where G is neither a cycle nor a tree. Then we can find a cycle $v_1, v_2, \dots, v_n$ which is not the entire graph. As G is connected, we may assume $\deg(v_1) \geq 3$. We remove the edge v_1v_n and add a new vertex w together with a new edge v_nw . The number of edges remains unchanged. If there is a labelling in the new graph such that all vertices are good, then we can use the same labelling in G where v_1v_n has the same label as v_nw . Indeed, for v_1 , its degree in the new graph is at least 2, so the labels of edges incident with v_1 in the new graph have gcd 1. This remains true in G when we add back the edge v_1v_n . For all other vertices, the labels of edges incident with them remain unchanged, so they must be good as well.

Therefore, we can repeat the above process until we end up with a graph which is a cycle or a tree. To show that this process must terminate, note that initially we have $|V| \leq |E| + 1$ since G is connected (by [proposition 4.2](#)). Each time $|V|$ is increased by 1 while $|E|$ remains unchanged. Eventually, we must have $|V| = |E| + 1$, which means the graph becomes a tree by [proposition 4.4](#). Thus, we are done by combining the above results. $\square$

Definition 4.9. A *bipartite graph* (二分圖) is a graph $G = (V, E)$ for which V can be partitioned into two disjoint sets V_1, V_2 such that every edge must have one endpoint in V_1 and another endpoint in V_2 .

We may use the notation $G = (V_1, V_2, E)$ to emphasize the bipartition of a bipartite graph. A *complete bipartite graph* (完全二分圖) is a bipartite graph $G = (V_1, V_2, E)$ such that there is an edge between every vertex in V_1 and every vertex in V_2 . We denote such a graph by $K_{m,n}$ where $m = |V_1|$ and $n = |V_2|$. More generally, we can talk about an *r -partite graph* (多分圖) by partitioning V into r disjoint sets.

We first provide some well-known facts below.

Proposition 4.5. A simple graph $G = (V, E)$ is bipartite iff it does not contain any cycle of odd length.

Proof. Firstly, suppose $G = (V_1, V_2, E)$ is a bipartite graph. Consider any cycle $v_1, v_2, \dots, v_k$. WLOG assume $v_1 \in V_1$. Since G is bipartite, we have $v_2 \in V_2$. Inductively, $v_j \in V_1$ iff j is odd, and $v_j \in V_2$ iff j is even. Note that v_k must belong to V_2 since it is adjacent to v_1 . Therefore, the length k of the cycle must be even.

Conversely, suppose a graph does not contain any odd cycle. WLOG assume the graph is connected. Otherwise, we can simply consider each component to work out the partition. Firstly, we put a vertex v in V_1 . For any other vertex w , we consider any path from v to w . We put w in V_1 if the path has an even length, and put it in V_2 otherwise. This is well-defined. Indeed, if there exist paths $v, x_1, x_2, \dots, x_k, w$ and $v, y_1, y_2, \dots, y_\ell, w$ such that k and ℓ have different parities, then the circuit $v, x_1, x_2, \dots, x_k, w, y_\ell, y_{\ell-1}, \dots, y_1, v$ has an odd length. If $x_i = y_j$ for some i, j , we can always break this circuit into two or more cycles such that the total lengths of these cycles is the length of the circuit. Therefore, one of these cycles must have an odd length, which contradicts the assumption.

We have partitioned V into V_1, V_2 . Suppose there exists an edge joining some vertices a, b in the same set. Consider a path from v to a , followed by ab and another path from b to v . Since the two paths have the same parity, this gives a circuit of odd length. Again, this implies there is a cycle of odd length, contradiction. Therefore, $G = (V_1, V_2, E)$ is bipartite. $\square$

Theorem 4.3. ([Turán's theorem](#)(圖蘭定理)) Let $n \geq r$ be positive integers. Suppose $G = (V, E)$ is a simple graph which does not contain any K_{r+1} subgraph. Then $|E|$ is maximized when G is a complete r -partite graph with r -partition $(V_1, V_2, \dots, V_r)$ such that $|V_i|$ and $|V_j|$ differ by at most 1 for all $1 \leq i < j \leq r$.

Proof. Let $G = (V, E)$ be any simple graph without K_{r+1} such that $|E|$ is maximized. Consider the complement $\overline{G} = (V, \overline{E})$ which is the graph with the same vertex set and with the edge set $\overline{E}$ being the complement of E in $K_{|V|}$. Then $\overline{G}$ does not contain an *independent set*(獨立集) (a set of pairwise non-adjacent vertices) of $r+1$ vertices, and it is such a graph for which $|\overline{E}|$ is minimized.

Claim. $\overline{G}$ is a union of at most r complete graphs.

Proof. The base case $r = 1$ is trivial since the graph must be a complete graph. Consider the case r . If $\overline{G}$ is not connected, let $G_1, G_2, \dots, G_k$ be the components of $\overline{G}$. Clearly, each independent set of $\overline{G}$ corresponds to the union of independent sets of $G_1, G_2, \dots, G_k$. Suppose each G_i does not contain an independent set of $a_i + 1$ vertices for each i , where $a_1 + a_2 + \dots + a_k \leq r$. As $k \geq 2$ and $a_i \geq 1$, we have $a_i < r$. By the inductive hypothesis, G_i must be a union of complete graphs. Since it is connected, G_i is a complete graph. This shows $\overline{G}$ is a union of complete graphs if $\overline{G}$ is not connected.

If $\overline{G}$ is connected, we pick a vertex v with the smallest degree. Let $w_1, w_2, \dots, w_m$ be the neighbours of v . We first claim that $v, w_1, \dots, w_m$ is a complete subgraph. Suppose on the contrary that $m \geq 2$ and w_1, w_2 are not adjacent. We remove all edges incident with w_1 or w_2 , and join all the edges $vw_1, vw_2, w_1w_2, w_1w_j, w_2w_j$ for $3 \leq j \leq m$. In other words, v, w_1, w_2 are pairwise adjacent, and they have exactly the same neighbours other than themselves. In the new graph G' formed, it does not

contain any independent set of size $r + 1$. Indeed, if such a set S exists, it should not exist in $\overline{G}$. Therefore, S must contain w_1 or w_2 . Note that S contains at most one of v, w_1, w_2 (otherwise it is not independent). WLOG assume $w_1 \in S$. If we replace w_1 by v , the set is still independent as w_1 and v have the same neighbours. However, such a set exists in $\overline{G}$ as it does not involve w_1 or w_2 . This is a contradiction.

Therefore, G' does not contain any independent set of size $r + 1$. But we shall show that the number of edges is decreased. Indeed, let b_1, b_2 be the number of edges of the form $w_1 w_i, w_2 w_i$ (where $i \geq 3$) in the original graph $\overline{G}$, and let c_1, c_2 be the number of edges of the form $w_1 x, w_2 x$ where $x \neq w_i$ (possibly $x = v$). Then we have removed $b_1 + b_2 + c_1 + c_2$ edges and added $2m - 1$ edges. Recall that v has the smallest degree. Therefore,

$$(b_1 + c_1) + (b_2 + c_2) \geq m + m > 2m - 1.$$

This shows there is a graph with a smaller number of edges which does not contain any independent set of size $r + 1$. This contradicts the choice of G and $\overline{G}$. Therefore, any two of $v, w_1, \dots, w_m$ are adjacent. If this is not the whole graph $\overline{G}$, then WLOG assume w_1 is adjacent to another vertex y due to the connectedness. We can simply remove the edge $w_1 y$ without affecting the key property. This is due to the same reason as before. Indeed, if an independent set contains w_1, y and some other vertices, then the set cannot contain $v, w_2, \dots, w_m$, and the same holds if we replace w_1 by v . Thus, the size of this independent set cannot exceed r . So $\overline{G}$ is a complete graph in this case.

The **claim** is proved by induction. □

By the **claim**, let $\overline{G} = K_{t_1} \cup K_{t_2} \cup \dots \cup K_{t_\ell}$. We claim that any two of the t_i 's differ by at most 1. Suppose on the contrary that $t_1 - t_2 \geq 2$. Note that

$$(t_1, t_2) > \left(\left\lceil \frac{t_1 + t_2}{2} \right\rceil, \left\lfloor \frac{t_1 + t_2}{2} \right\rfloor \right).$$

A complete graph of size s has $\binom{s}{2}$ edges. By the majorization inequality, we have

$$\binom{t_1}{2} + \binom{t_2}{2} \geq \binom{\left\lceil \frac{t_1 + t_2}{2} \right\rceil}{2} + \binom{\left\lfloor \frac{t_1 + t_2}{2} \right\rfloor}{2}.$$

Equality holds when $t_1 = \left\lceil \frac{t_1 + t_2}{2} \right\rceil$, which is not the case. This contradicts the assumption that the number of edges is minimized.

Finally, we prove that $\ell = r$. If $\ell \geq r + 1$, we can pick a vertex from each component to form an independent set of size $\ell \geq r + 1$. If $\ell \leq r - 1$, we may assume $t_1 \geq 2$ since $t_1 + t_2 + \dots + t_\ell = n \geq r$. Then we can replace K_{t_1} by $K_{t_1-1} \cup K_1$. In the new graph, we still cannot find an independent set of size $r + 1$ as there are only $\ell + 1 \leq r$ components, but the number of edges is decreased since

$$\binom{t_1}{2} > \binom{t_1 - 1}{2}.$$

Therefore, we must have $\ell = r$.

Combining all these, we see that the complement of $\overline{G}$, which is G , has exactly the form as described in this problem. $\square$

The extremal case in this theorem is called the *Turán graph* (圖蘭圖).

Theorem 4.4. (Hall's theorem (霍爾定理)) Let $G = (V_1, V_2, E)$ be a bipartite graph. For any subset W of V_1 , let d_W be the number of vertices in V_2 which are neighbours of at least one vertex in W . Then the following are equivalent.

- (i) Let $V_1 = \{v_1, v_2, \dots, v_m\}$. There exists a set of vertices $\{w_1, w_2, \dots, w_m\}$ in V_2 such that $v_i w_i \in E$ for each $i = 1, 2, \dots, m$.
- (ii) For each subset W of V_1 , we have $d_W \geq |W|$.

Proof. Firstly, suppose (i) holds. For any subset W of V_1 , each element $v_i \in W$ is adjacent to a distinct element w_i in V_2 by (i), which is counted in d_W by definition. Therefore, $d_W \geq |W|$.

Conversely, suppose (ii) holds. We show that such a pairing exists by induction on m , the number of vertices in V_1 . For the base case $m = 1$, we simply take $W = \{v_1\}$, so that v_1 must be adjacent to a vertex w_1 in V_2 . Now, assume $m \geq 2$.

Let $V_1 = \{v_1, v_2, \dots, v_m\}$. Consider the subset $\{v_1, v_2, \dots, v_{m-1}\}$. By the inductive hypothesis, we can find $w_1, w_2, \dots, w_{m-1} \in V_2$ such that $v_i w_i \in E$ for $i = 1, 2, \dots, m-1$. It remains to find a neighbour for v_m following the steps given below.

- If v_m is adjacent to some other vertex $w_m \in V_2$, then we are done. Otherwise, WLOG we may assume v_m is only adjacent to $w_1, \dots, w_{i_1}$ for some $1 \leq i_1 \leq m-1$ (again i_1 must be at least 1).
- Consider $W = \{v_1, \dots, v_{i_1}, v_m\}$. Since $|W| = i_1 + 1$, condition (ii) shows the number of vertices adjacent to at least one of the vertices in W is at least $i_1 + 1$. So there should be some vertices other than $w_1, \dots, w_{i_1}$. If there exists such a vertex $w_m \in V_2$ not involved yet, say $v_{i_1} w_m \in E$, then we simply change the pair (v_{i_1}, w_{i_1}) to (v_{i_1}, w_m) and (v_m, w_{i_1}) . This gives a desired pairing. If not, WLOG the only vertices adjacent to at least one of $v_1, \dots, v_{i_1}, v_m$ are $w_1, \dots, w_{i_2}$ for some $i_1 < i_2 \leq m-1$.
- Consider $W = \{v_1, \dots, v_{i_2}, v_m\}$. Arguing in the same way, there are two possibilities. One possibility is that there exists a vertex $w_m \in V_2$ such that $v_{i_2} w_m \in E$ and $v_{i_1} w_{i_2} \in E$ (up to symmetry). Then we change the pairs (v_{i_1}, w_{i_1}) and (v_{i_2}, w_{i_2}) to (v_{i_1}, w_{i_2}) , (v_{i_2}, w_m) and (v_m, w_{i_1}) . The other possibility is that the only vertices adjacent to at least one of $v_1, \dots, v_{i_2}, v_m$ are $w_1, \dots, w_{i_3}$ for some $i_2 < i_3 \leq m-1$.

Repeating the process, we must come to an end since the indices $i_1, i_2, \dots$ are strictly increasing and bounded above. This means we must be able to find a vertex w_m not involved in previous steps such that we can change the pairs to form a new pairing including v_m . This completes the proof by induction. $\square$

Problem 4.7. Let $G = (V_1, V_2, E)$ be a bipartite graph such that $|V_1| = |V_2| = n$ and $|E| \geq n^2 - n + 1$. Prove that the vertices in V_1 can be paired up with the vertices in V_2 such that the vertices in each pair are adjacent.

Analysis. This is a simple application of Hall's theorem. According to the theorem, we only need to show that condition (ii) in the theorem is satisfied. This can be proved by using the fact that $|E|$ is large.

Solution. Consider any $W \subset V_1$. Let d be the number of vertices in V_2 which are adjacent to at least one vertex in W . We shall prove that $d \geq |W|$. Suppose on the contrary that $d \leq |W| - 1$. There are $n - d$ vertices in V_2 which are not adjacent to all vertices in W . Therefore, at least $(n - d)|W|$ edges are missing. It follows that

$$n^2 - n + 1 \leq |E| \leq n^2 - (n - d)|W| \leq n^2 - (n - d)(d + 1).$$

This implies $d(d - n + 1) \geq 1$. However, as $0 \leq d \leq |W| - 1 \leq n - 1$, we should have $d(d - n + 1) \leq 0$. This is a contradiction. It follows that condition (ii) in Hall's theorem holds. By Hall's theorem, the assertion follows readily. $\square$

Problem 4.8. (Korea MO 2021 Problem 4) For a positive integer n , there are two countries A and B with n airports each and $n^2 - 2n + 2$ airlines operating between the two countries. Each airline operates at least one flight. Exactly one flight by one of the airlines operates between each airport in A and each airport in B , and that flight operates in both directions. Also, there are no flights between two airports in the same country. For two different airports P and Q , denote by (P, Q) -travel route the list of airports $T_0, T_1, \dots, T_s$ satisfying the following conditions:

- $T_0 = P, T_s = Q$;
- $T_0, T_1, \dots, T_s$ are all distinct;
- there exists an airline that operates between the airports T_i and T_{i+1} for all $i = 0, 1, \dots, s - 1$.

Prove that there exist two airports P, Q such that there is no or exactly one (P, Q) -travel route.

Analysis. If we use a simple double counting to count the number of pairs of distinct airports, the bound obtained is not strong enough to justify the result. One reason is that it is not easy to reject the 'worst' case where there are exactly two (P, Q) -travel routes for many pairs of P, Q .

The main idea that leads to the following solution is the fact that there are many airlines compared to the number of flights. Therefore, most of the airlines only operate exactly one flight. We can first ignore these airlines since they only contribute at most 1 to the number of (P, Q) -travel route for each pair of P, Q . In other words, we only need to show that there is no (P, Q) -travel route for some P, Q after removing those airlines.

Solution. Consider the complete bipartite graph $G = (A, B, E)$ where A and B contain those airports in A and B respectively. We use $n^2 - 2n + 2$ colours to colour the edges in G such that each colour corresponds to an airline. Therefore, a (P, Q) -travel route is simply a monochromatic path between P and Q .

We say that a colour is *good* if it is used at least twice, and is *bad* if it is used once only. Suppose there are k good colours in total. The case $k = 0$ is trivial since all edges are of different colours in that case, and we can pick any P, Q . Suppose $k \geq 1$. For the i th good colour C_i , let $H_i = (V_i, E_i)$ be the subgraph of G that contains only edges in colour C_i and vertices which are incident with at least one edge in colour C_i .

By adding more colours if necessary, we may assume each H_i is connected (otherwise we use a new colour to colour all edges in a component), and there are $c \geq n^2 - 2n + 2$ colours in total. Trivially, there are at most $\binom{|V_i|}{2}$ pairs of vertices P, Q such that there is a path between P and Q in colour C_i . As H_i is connected, we have $|V_i| \leq |E_i| + 1$ by [proposition 4.2](#). Therefore, the number N of pairs of vertices P, Q such that there is a path between P and Q in a good colour is at most

$$\sum_{i=1}^k \binom{|V_i|}{2} \leq \sum_{i=1}^k \binom{|E_i| + 1}{2}.$$

Since $|E_i| \geq 2$, the sequence $(|E_1| + 1, |E_2| + 1, \dots, |E_k| + 1)$ (in a suitable order) is majorized by $(a - 2(k - 1) + 1, 3, 3, \dots, 3)$, where

$$a = |E_1| + |E_2| + \dots + |E_k| = n^2 - (c - k)$$

as each of the $c - k$ bad colours corresponds to exactly one edge. By the majorization inequality, we deduce

$$\begin{aligned} N &\leq \binom{n^2 - (c - k) - 2(k - 1) + 1}{2} + (k - 1) \binom{3}{2} \\ &= \frac{k^2 - (2n^2 - 2c - 1)k + (n^2 - c + 2)(n^2 - c + 3) - 6}{2}. \end{aligned} \quad (1)$$

As there are at least two edges of a good colour and one edge of a bad colour, we have $n^2 \geq 2k + (c - k)$. Therefore, $1 \leq k \leq n^2 - c$. It follows that (1) is maximized when $k = 1$ (or $k = n^2 - c$ when $n^2 - c = 2$), which means

$$N \leq \binom{n^2 - c + 2}{2} \leq \binom{2n}{2}$$

as $c \geq n^2 - 2n + 2$.

- If $N < \binom{2n}{2}$, then there exists a pair of vertices P, Q such that there is no path between P and Q in a good colour. Obviously, there is at most one path (which must be an edge) between P and Q in a bad colour. So this pair P, Q satisfies the requirement.
- If $N = \binom{2n}{2}$, all equalities hold. There are two possibilities.
 - In the first case, we have $k = 1$ and $|V_1| = |E_1| + 1$. By [proposition 4.4](#), we know that H_1 is a tree. Therefore, if PQ is an edge in H_1 , then there is only one path between P and Q in colour C_1 . As all other colours are bad, there is only one monochromatic path between P and Q , which satisfies the requirement.
 - In the second case, we have $k = 2$, $c = n^2 - 2$, $|E_1| = |E_2| = 2$ and $|V_1| = |V_2| = 3$. Thus, both H_1 and H_2 is a path of length 2. We can choose P and Q to be two adjacent vertices in H_1 . It is not hard to see that the only monochromatic path between P and Q is the one in H_1 , which satisfies the requirement.

We are done in all cases. □

Definition 4.10. A [Hamiltonian path](#)(哈密頓路徑) of a graph $G = (V, E)$ is a path that passes through all vertices in V . A [Hamiltonian cycle](#)(哈密頓圈) is a cycle that passes through all vertices in V . A [Hamiltonian graph](#)(哈密頓圖) is a graph that contains a Hamiltonian cycle.

Intuitively, if there are many edges, then the graph should be Hamiltonian. The following provides some simple results based on this intuition.

Theorem 4.5. ([Ore's theorem](#)(奧爾定理)) Suppose $G = (V, E)$ is a simple graph with $|V| = n \geq 3$ such that $\deg(v) + \deg(w) \geq n$ for all non-adjacent vertices v and w . Then G is Hamiltonian.

Proof. Suppose on the contrary that G is not Hamiltonian. WLOG assume G is a maximal counterexample, i.e. it becomes Hamiltonian if we add any new edge (otherwise we add more edges to G without affecting the condition). Clearly, G is not a complete graph. Let v, w be two non-adjacent vertices. As G becomes Hamiltonian by adding the edge vw , G must contain a Hamiltonian path $v, x_1, x_2, \dots, x_{n-2}, w$.

Note that v is adjacent to $\deg(v) - 1$ of $x_2, x_3, \dots, x_{n-2}$. In addition, w is adjacent to $\deg(w) - 1$ of $x_1, x_2, \dots, x_{n-3}$. As

$$(\deg(v) - 1) + (\deg(w) - 1) = (\deg(v) + \deg(w)) - 2 \geq n - 2,$$

there exists an index $j \in \{1, 2, \dots, n-3\}$ such that v is adjacent to x_{j+1} while w is adjacent to x_j by the pigeonhole principle. It follows that

$$v, x_{j+1}, x_{j+2}, \dots, x_{n-2}, w, x_j, x_{j-1}, \dots, x_1$$

is a Hamiltonian cycle, contradiction.

Therefore, G must be Hamiltonian. □

As a special case, we have the following result.

Theorem 4.6. (**Dirac's theorem**(狄拉克定理)) Suppose $G = (V, E)$ is a simple graph with $|V| = n \geq 3$ such that $\deg(v) \geq \frac{n}{2}$ for all $v \in V$. Then G is Hamiltonian.

Proof. This follows from Ore's theorem since $\deg(v) + \deg(w) \geq \frac{n}{2} + \frac{n}{2} = n$ for any $v, w \in V$, no matter they are adjacent or not. □

Next, we have another result for directed graphs.

Proposition 4.6. Let $G = (V, E)$ be a directed graph such that for any distinct $v, w \in V$, exactly one of vw and wv belongs to E . Then G is strongly connected iff it is Hamiltonian.

Proof. Using induction on $|V|$ is not that simple because the condition is completely different when a new vertex is added. This is because if there exists a path from v to w , the same need not hold when one of the vertices of the graph is removed. Thus, a better way is to use the extremal principle.

Consider any distinct $v, w \in V$. Since there exists a path from v to w and a path from w to v , G contains a circuit. If the same vertex is passed through more than once, we can remove the vertices in between. This shows G contains a cycle.

Consider a cycle $v_1, v_2, \dots, v_k$ with the greatest number of vertices. We are done if $k = |V|$. Otherwise, there exists $w \in V$ which is not in this cycle. If both $v_i w$ and $w v_{i+1}$ are edges in G for some i (where $v_{k+1} = v_1$), then we can form a larger cycle

$$v_1, v_2, \dots, v_i, w, v_{i+1}, \dots, v_k.$$

This contradicts the maximality of k . Therefore, if $v_i w \in E$ for some i , then $v_j w \in E$ for all j by induction. Similarly, if $w v_i \in E$ for some i , then $w v_j \in E$ for all j .

Since there exists a path from any vertex not in the cycle to v_1 , there must be a vertex w such that $w v_i \in E$ for all i . Next, let $v_1, x_1, x_2, \dots, x_m, w$ be a path. We may assume $x_1, x_2, \dots, x_m$ are different from the v_i 's as we can simply remove all vertices

in between two v_i 's (and replace v_1 by another vertex v_j , which does not affect the argument). But then we have found a larger cycle

$$v_1, x_1, x_2, \dots, x_m, w, v_2, v_3, \dots, v_k.$$

This is a contradiction. Therefore, we must have $k = |V|$, which means G is Hamiltonian. $\square$

Instead of finding a path or a cycle that passes through all vertices, it may be useful to find a path or a cycle that uses all edges. This leads to the following definition.

Definition 4.11. An *Eulerian trail*(歐拉軌跡) of a graph $G = (V, E)$ is a trail that uses all edges in E . An *Eulerian circuit*(歐拉迴路) is a circuit that uses every edge in E exactly once. An *Eulerian graph*(歐拉圖) is a graph that contains an Eulerian circuit.

Proposition 4.7. Let $G = (V, E)$ be a simple connected graph. Then G is Eulerian iff $\deg(v)$ is even for all $v \in V$.

Proof. If G is Eulerian, let $e_1, e_2, \dots, e_n$ be an Eulerian circuit where $e_i \in E$. For each $v \in V$, if v is the terminal vertex of exactly k edges in the circuit, then it is also the starting vertex of exactly k edges in the circuit. Therefore, we have $\deg(v) = 2k$, which is even.

Conversely, suppose $\deg(v)$ is even for all $v \in V$. We prove that G is Eulerian by induction on $n = |E|$. It is impossible that $n = 1, 2$. When $n = 3$, the only such graph is a cycle of length 3. The result is obvious. Suppose the result holds for all smaller cases. For the inductive step, note that $\deg(v) \geq 2$ for all $v \in V$ since G is connected and $\deg(v)$ is even. Using a similar argument as that in [problem 4.2](#), we know that G must contain a circuit. Let $e_1, e_2, \dots, e_m$ be a circuit in G where $e_i \in E$. Consider the subgraph H of G obtained by removing the edges $e_1, e_2, \dots, e_m$.

Note that the degree of each vertex in H is still even since each degree is decreased by an even number when we remove the edges. By the inductive hypothesis, each component of H is Eulerian. Therefore, all edges in H , and hence G , can be partitioned into a finite number of disjoint circuits (with distinct edges, not vertices).

If $v_1v_2, v_2v_3, \dots, v_{s-1}v_s, v_sv_1$ and $w_1w_2, w_2w_3, \dots, w_{t-1}w_t, w_tw_1$ are disjoint circuits, where $v_i, w_j \in V$, such that $v_s = w_t$, then we can combine them to form a larger circuit

$$v_1v_2, v_2v_3, \dots, v_{s-1}v_s, w_tw_1, w_1w_2, w_2w_3, \dots, w_{t-1}w_t, v_sv_1.$$

Since G is connected, we can combine the disjoint circuits found in the previous step to form a single circuit. This means G is Eulerian by induction. $\square$

Similarly, a simple connected graph has an Eulerian trail iff $\deg(v)$ is odd for exactly 0 or 2 vertices.

4.2 Applications of Graph Theory

Knowledge in graph theory is not only useful in solving problems in graph theory. In fact, for most applications of graph theory in Olympiad problems, it is not obvious how the problem can be rephrased in terms of graph theory. The hardest step is to construct a graph based on the setting of the problem. We shall go through several examples of this kind in this subsection.

Problem 4.9. (*De Bruijn sequence*(德布萊英序列)) Let n be a positive integer. Prove that there exists a binary string $a_1a_2\cdots a_{2^n}$ of length 2^n (where $a_j = 0, 1$ for each j) such that for every binary string $b_1b_2\cdots b_n$ of length n , there exists k such that $a_{k+j} = b_j$ for $j = 1, 2, \dots, n$ (where the indices of a_i are taken modulo 2^n).

Analysis. Since there are 2^n binary strings of length n in total, each such binary string must occur exactly once in $a_1a_2\cdots a_{2^n}$. Therefore, the problem is the same as arranging these 2^n binary strings in some order such that a string $b_1b_2\cdots b_n$ can only be followed by the string $b_2b_3\cdots b_n0$ or $b_2b_3\cdots b_n1$.

There are two possible ways to interpret this constraint in terms of a directed graph. For example, we can regard each sequence $b_1b_2\cdots b_n$ as a vertex, and draw an edge from it to $b_2b_3\cdots b_n0$ and $b_2b_3\cdots b_n1$. Alternatively, we can regard each sequence $b_1b_2\cdots b_n$ as an edge that goes from the vertex $b_1b_2\cdots b_{n-1}$ to $b_2b_3\cdots b_n$. The problem is transformed to different problems in graph theory under these interpretations. We shall work on the latter interpretation since it trivializes the problem.

Solution. The problem is trivial if $n = 1$, as we can take $a_1a_2 = 10$. For $n \geq 2$, consider a directed graph $G = (V, E)$ where the vertices are all binary strings of length $n - 1$. We draw an edge from each vertex $b_1b_2\cdots b_{n-1}$ to the vertex $b_2b_3\cdots b_{n-1}b_n$ where $b_n = 0, 1$. Also, we label this edge by $b_1b_2\cdots b_n$.

Note that there are 2^n edges in G in total, which correspond to the 2^n binary strings of length n . Our goal is to find a circuit that contains all edges in E . If this can be done, such a circuit must be of the form

$$a_1a_2\cdots a_n, a_2a_3\cdots a_{n+1}, \dots, a_{2^n}a_1a_2\cdots a_{n-1} \text{ (which are edges)}$$

in view of the definition of G . Therefore, $a_1a_2\cdots a_{2^n}$ is a desired de Bruijn sequence.

Thus, it remains to show that G is Eulerian (with a more general definition for directed graphs). Analogous to [proposition 4.7](#), we only need to prove that G is strongly connected and $\deg^-(v) = \deg^+(v)$ for all $v \in V$.

Firstly, to go from any vertex $b_1b_2\cdots b_{n-1}$ to $c_1c_2\cdots c_{n-1}$, we can follow the walk

$$b_1b_2\cdots b_{n-1}, b_2b_3\cdots b_{n-1}c_1, b_3b_4\cdots b_{n-1}c_1c_2, \dots, c_1c_2\cdots c_{n-1}.$$

This shows G is strongly connected. Secondly, the only edges that end at $b_1b_2\cdots b_{n-1}$ are $0b_1b_2\cdots b_{n-1}$ and $1b_1b_2\cdots b_{n-1}$. The only edges that start from this vertex are $b_1b_2\cdots b_{n-1}0$ and $b_1b_2\cdots b_{n-1}1$. This shows $\deg^-(v) = 2$ and $\deg^+(v) = 2$ for all $v \in V$. This completes the proof. $\square$

Problem 4.10. (MEMO 2021 Individual Problem 2) Let m and n be positive integers. Some squares of an $m \times n$ board are coloured red. A sequence $a_1, a_2, \dots, a_{2r}$ of $2r \geq 4$ pairwise distinct red squares is called a *bishop circuit* if for every $k \in \{1, \dots, 2r\}$, the squares a_k and a_{k+1} lie on a diagonal, but the squares a_k and a_{k+2} do not lie on a diagonal (here $a_{2r+1} = a_1$ and $a_{2r+2} = a_2$).

In terms of m and n , determine the maximum possible number of red squares on an $m \times n$ board without a bishop circuit.

(Remark: Two squares lie on a diagonal if the line passing through their centres intersects the sides of the board at an angle of 45° .)

Analysis. The word ‘circuit’ suggests we may consider a graph and some circuits in this graph. The main question is how to set up such a graph. One may attempt to use vertices to denote the squares and draw edges between squares on the same diagonal. However, this is hard to distinguish between red squares and other squares, and hard to know whether we have changed the diagonal when we go from one edge to another edge.

In view of these, it may be better to use vertices to denote the diagonals and draw edges between diagonals that intersect at a red square. This setting solves the above two problems, and also greatly simplifies the problem.

Solution. The cases $m = 1$ or $n = 1$ are trivial since we can colour all squares red. Suppose $m, n \geq 2$. Consider a graph $G = (V, E)$ where the vertices are the diagonals of the board. An edge is drawn between every two diagonals that intersect at a red square. Note that a bishop circuit corresponds to a cycle in G , while the number of red squares is the number of edges in G . Therefore, we need to maximize $|E|$ given that G is acyclic.

One may intend to apply [proposition 4.4](#) directly. However, it is not true that G is connected. Indeed, by considering the chessboard colouring, we see that each diagonal only contains squares of the same colour. Therefore, we can never go from a black diagonal to a white diagonal. This shows G has at least two components.

Let $a_1, a_2, \dots, a_k$ be the number of vertices of the $k \geq 2$ components of G respectively. As each component is a tree, the number of edges in the i th component is $a_i - 1$ by

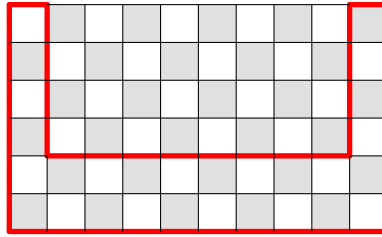
proposition 4.4. This implies

$$|E| = \sum_{i=1}^k (a_i - 1) = \left(\sum_{i=1}^k a_i \right) - k = |V| - k \leq |V| - 2.$$

Using a simple counting, we see that there are $2m + 2n - 2$ diagonals in the board (for example, each diagonal going from north-west to south-east must start at a different square in the first column or the first row, so there are $m + n - 1$ such diagonals). This implies $|V| = 2m + 2n - 2$, and hence

$$|E| \leq 2m + 2n - 4.$$

Equality can be attained. For example, we can colour every square in the first column, the last column, and the last two rows in red.



To show that this construction works, consider the chessboard colouring. Clearly, every black diagonal contains at least one red square. Therefore, to show that we can go from any black diagonal to another black diagonal, it suffices to show that we can go from any black square in red to another black square in red. Indeed, every black square in the first column (or the last column) can move to a black square in the bottom two rows in one step, and it is clear that all black squares in the bottom two rows can move to each other along a zigzag path. This shows all vertices in G corresponding to the black diagonals form a component. Similarly, all vertices corresponding to the white diagonals form a component. Thus, G only contains two components. As the number of edges is two less than the number of vertices, both components are trees, which means G is acyclic.

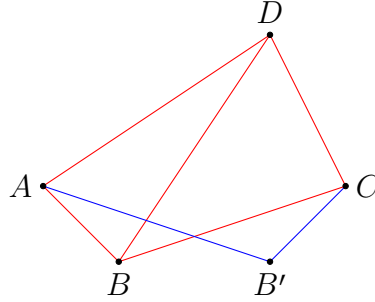
In conclusion, the maximum number of red squares is $\begin{cases} 2m + 2n - 4 & \text{if } m, n \geq 2, \\ n & \text{if } m = 1, \\ m & \text{if } n = 1. \end{cases} \quad \square$

Problem 4.11. (Kharkiv Masters Tournament 2019 Grade 10–11 Problem 6) Given some triangulation T of a convex 100-gon, we construct another polygon $F(T)$ in the following way: we draw a copy of this 100-gon and draw a diagonal AC in this copy if there exist two more vertices B and D in the 100-gon such that $ABCD$ is convex, and all the segments AB, BC, CD, DA, BD are edges in T . Let $i(T)$ denote the number of pairs of diagonals in $F(T)$ that have a common point (inside). What is the maximum value of $i(T)$?

Analysis. Each diagonal AC in $F(T)$ clearly corresponds to a pair of triangles in the triangulation that share a common side. This gives a better way to interpret $F(T)$. This also leads to a natural construction of the graph used below, with vertices representing the triangles and edges represent such pairs of triangles. It remains to look for some special properties of this graph and use language in graph theory to interpret the problem.

Solution. Consider a graph $G = (V, E)$ where the vertices are the triangles in T . An edge is drawn between two triangles that share a common side. For convenience, we can use the centroids of the triangles to represent the vertices. In this way, a cycle in G can be visualized as a closed polygon line in the 100-gon. However, such a polygon line must contain an interior point which is a vertex of the 100-gon. This is impossible as the polygon is convex. Therefore, G is acyclic. Indeed, it must be a tree as it is clearly connected in view of the triangulation. Note that $|V| = 98$ since an n -gon can be partitioned into $n - 2$ triangles. It follows that $|E| = 97$ by [proposition 4.4](#). (Alternatively, one can also count $|E|$ directly based on its meaning in the triangulation.)

For any diagonal AC in $F(T)$, the vertices B and D given in the question are uniquely determined. Indeed, suppose both B, D and B', D' satisfy the conditions where $ABCD$ and $AB'CD'$ have the same orientation, where $(B, D) \neq (B', D')$. WLOG assume $B \neq B'$, and assume $ABB'CD$ is convex. Then the diagonals AB' and BC must cross each other, which is impossible as T is a triangulation. Thus, there is a bijection between diagonals in $F(T)$ and edges in G (mapping AC to the edge joining $\triangle ABD$ and $\triangle CBD$).

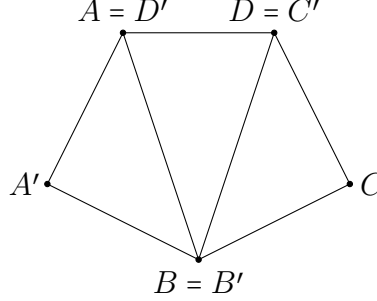


Next, we study the meaning of a pair of diagonals AC and $A'C'$ in $F(T)$ which have a common interior point. Suppose $ABCD$ and $A'B'C'D'$ are the corresponding quadrilaterals in anticlockwise orientation. We want to show $(B', C', D') = (B, D, A)$ up to symmetry. We use interval notations $(X, Y), (X, Y], [X, Y), [X, Y]$ to stand for the set of all vertices between X and Y in the anticlockwise direction.

WLOG assume A' and B lie on the same side of the line AC , so that C' and D lie on the other side.

- If $A' \in (A, B)$, in order that $A'B'$ and $A'D'$ do not cross AB , we must have $B' \in (A', B]$ and $D' \in [A, A')$. If $B' \neq B$, then $B'C'$ must cross AB . Therefore,

we must have $B' = B$. Similarly, we must have $D' = A$. It is easy to see that $C' = D$ is the only possibility.



- Each of the cases $A' \in (B, C)$, $C' \in (C, D)$ and $C' \in (D, A)$ is analogous.
- If $A' = B$ and $C' = D$, then BD and $B'D'$ must cross each other. This is impossible.

It follows that $ABCD$ and $A'B'C'D'$ share a common triangle. Equivalently, the edge joining $\triangle BCD$ and $\triangle ABD$ shares a common endpoint with the edge joining $\triangle A'B'D'$ and $\triangle B'C'D'$. Conversely, it is clear that any pair of edges in G sharing a common endpoint corresponds to such a pair of intersecting diagonals in $F(T)$.

Thus, it remains to maximize the number of pairs of edges in G sharing a common endpoint. Let $d_1, d_2, \dots, d_{98}$ be the degrees of the vertices in G . Note that $d_j = 1, 2, 3$ because of the triangulation. Also, we have

$$d_1 + d_2 + \dots + d_{98} = 2|E| = 2(97) = 194$$

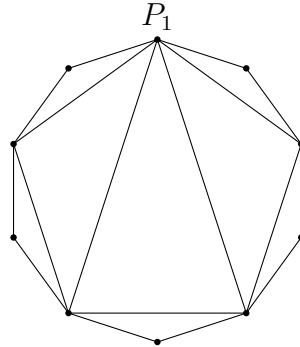
by the handshaking theorem. We need to maximize $\sum_{j=1}^{98} \binom{d_j}{2}$. As

$$\underbrace{(3, 3, \dots, 3)}_{48 \text{ copies}} \underbrace{(1, 1, \dots, 1)}_{50 \text{ copies}} > (d_1, d_2, \dots, d_{98}),$$

we have

$$\binom{d_1}{2} + \binom{d_2}{2} + \dots + \binom{d_{98}}{2} \leq 48 \binom{3}{2} + 50 \binom{1}{2} = 144$$

by the majorization inequality.



The maximum value 144 can be attained. For example, consider the triangulation of a polygon $P_1P_2\cdots P_{100}$ given by

$$\triangle P_1P_2P_3, \triangle P_3P_4P_5, \dots, \triangle P_{99}P_{100}P_1, \triangle P_1P_3P_5, \triangle P_1P_5P_7, \dots, \triangle P_1P_{97}P_{99}.$$

It is not hard to see that each vertex in G has degree 1 or 3 (meaning that each triangle is adjacent to 1 or 3 triangles), so the maximum value is attained. $\square$

Problem 4.12. (Serbia TST 2021 Problem 4) Given that $a_1, a_2, \dots, a_{2020}$ are integers, find the maximal number of subsequences $a_i, a_{i+1}, \dots, a_j$ ($0 < i \leq j < 2021$) with sum 2021.

Analysis. Using a similar idea as [problem 2.5](#), we may consider a new sequence defined by $s_j = a_1 + a_2 + \cdots + a_j$. This sequence is useful if we need to work on the sum of consecutive terms of $\{a_i\}$. We now need to maximize the number of pairs of s_i, s_j such that $i < j$ and $s_j - s_i = 2021$. Therefore, it may be helpful to consider the graph formed by the vertices s_i (or the indices i) and edges which join such pairs of terms. The crucial observation is that such a graph cannot contain any odd cycle.

Solution. Let $s_0 = 0$ and $s_j = a_1 + a_2 + \cdots + a_j$. Consider a graph $G = (V, E)$ with vertices $0, 1, \dots, 2020$ such that vertices i and j are adjacent if $s_j - s_i = \pm 2021$. (We choose to weaken the condition by including $s_j - s_i = -2021$, so that we do not need to check the order of i and j . This only works if we are finding an upper bound only.) Note that if $a_{i+1} + a_{i+2} + \cdots + a_j = 2021$, then $s_j - s_i = 2021$, and so there is an edge joining vertices i and j . Thus, the number of such subsequences cannot exceed the number of edges in G .

If G has an odd cycle $i_1, i_2, \dots, i_k$, then we have

$$0 = (s_{i_1} - s_{i_2}) + (s_{i_2} - s_{i_3}) + \cdots + (s_{i_k} - s_{i_1}) = \pm 2021 \pm 2021 \pm \cdots \pm 2021.$$

This is impossible since k is odd. Therefore, G has no odd cycle, which means it is a bipartite graph by [proposition 4.5](#). Let (V_1, V_2) be the bipartition, and let $|V_1| = m$. Then $|V_2| = 2021 - m$, and hence

$$|E| \leq m(2021 - m) \leq 1010 \cdot 1011 = 1021110.$$

Now, consider the sequence $a_j = 0$ for $j \neq 1010$ and $a_{1010} = 2021$. Then all subsequences $a_i, a_{i+1}, \dots, a_j$ with $1 \leq i \leq 1010 \leq j \leq 2020$ have sum 2021. There are $1010 \cdot 1011 = 1021110$ such subsequences. So the maximum value is 1021110. $\square$

Problem 4.13. (IMO Shortlist 2020 C4) The Fibonacci numbers $F_0, F_1, F_2, \dots$ are defined inductively by $F_0 = 0$, $F_1 = 1$, and $F_{n+1} = F_n + F_{n-1}$ for $n \geq 1$. Given an integer $n \geq 2$, determine the smallest size of a set S of integers such that for every $k = 2, 3, \dots, n$ there exist some $x, y \in S$ such that $x - y = F_k$.

Analysis. WLOG we may assume the smallest element in S is 0. It is easy to work out some small examples. When $n = 2$, the minimum value of $|S|$ is 2, with example $S = \{0, 1\}$. When $n = 3, 4$, the minimum value of $|S|$ is 3, with example $S = \{0, 1, 3\}$ or $S = \{0, 2, 3\}$. When $n = 5, 6$, the minimum value of $|S|$ is 4. One of the examples is $S = \{0, 1, 3, 8\}$. From this, it is easy to work out an optimal example for the general case, which is $\{F_0, F_2, F_4, \dots, F_{2m}\}$ where $m = \left\lfloor \frac{n}{2} \right\rfloor$.

It is a lot harder to prove the lower bound. If we wish to construct a graph based on the setting of the question, one natural way is to use vertices to denote the elements in S , and join two vertices x, y satisfying $x - y = F_k$ for some $k = 2, 3, \dots, n$. But then we cannot say much about the properties of this graph, and could hardly work out a lower bound for the number of vertices. The crucial step in the solution below is to focus on the subgraph which contains some edges joining two vertices x, y with $x - y = F_k$ for some odd k . It turns out that we can already deduce the correct lower bound by working on this subgraph.

Solution. We prove that the smallest size of S is $m + 1$ where $m = \left\lfloor \frac{n}{2} \right\rfloor$.

Firstly, consider $S = \{F_0, F_2, F_4, \dots, F_{2m}\}$. For $j = 1, 2, \dots, m$, we have

$$\begin{aligned} F_{2j} - F_{2(j-1)} &= F_{2j-1}, \\ F_{2j} - F_0 &= F_{2j}. \end{aligned}$$

Thus, there are elements $x, y \in S$ with difference $F_2, F_3, \dots, F_{2m}$ respectively. Since $2m \geq n$, this set meets the condition, and we have $|S| = m + 1$ in this case.

Secondly, we show that $|S| \geq m + 1$. Construct a simple graph $G = (V, E)$ as follows. The vertices are elements in S . For each $j = 1, 2, \dots, m$, we choose one pair (not all pairs) of $x, y \in S$ such that $x - y = F_{2j-1}$, and draw an edge between x and y . Thus, we have $|E| = m$, and we need to prove $|V| \geq m + 1$. Because of this linkage, we may think of proving that G is acyclic. This is also a reason for the construction of this mysterious graph.

Now, suppose G contains a cycle $v_1, v_2, \dots, v_k$ where each $v_j \in S$ and $k \geq 3$. Let $|v_{j+1} - v_j| = F_{a_j}$ for $j = 1, 2, \dots, k-1$, and let $|v_k - v_1| = F_b$, where a_j and b are distinct odd numbers. WLOG assume $b > a_j$ for all j . By the triangle inequality, we have

$$\begin{aligned} F_b &= |v_k - v_1| \\ &= |(v_k - v_{k-1}) + (v_{k-1} - v_{k-2}) + \dots + (v_2 - v_1)| \\ &\leq |v_k - v_{k-1}| + |v_{k-1} - v_{k-2}| + \dots + |v_2 - v_1| \\ &= F_{a_{k-1}} + F_{a_{k-2}} + \dots + F_{a_1} \\ &\leq F_1 + F_3 + \dots + F_{b-2} \\ &= F_{b-1}. \end{aligned}$$

The last equality is a well-known property of the Fibonacci numbers, which can be proved easily by induction. Now, as $b \geq 5$, we must have $F_{b-1} < F_b$, contradiction.

Therefore, G is acyclic. By [proposition 4.4](#), for each component of G , the number of vertices is one more than the number of edges. Since there is at least one component, we must have $|V| \geq |E| + 1 = m + 1$. This means $|S| \geq m + 1$. So we are done. $\square$

Problem 4.14. (IMO Shortlist 2020 C6/Problem 3) There are $4n$ pebbles of weights $1, 2, 3, \dots, 4n$. Each pebble is coloured in one of n colours and there are four pebbles of each colour. Show that we can arrange the pebbles into two piles so that the following two conditions are both satisfied:

- The total weights of both piles are the same.
- Each pile contains two pebbles of each colour.

Analysis. It is easy to distribute four pebbles of the same colour into two piles such that each pile contains two pebbles. However, it is not that easy to distribute the pebbles such that the total weights of the two piles are the same. Therefore, the first question to ask is how to put $1, 2, 3, \dots, 4n$ into two groups of $2n$ numbers with the same sum.

After trying out some examples, the only natural way to distribute the numbers is to put every pair of numbers $(k, 4n + 1 - k)$ in the same group, so that the sum of each pair is fixed (or with a modification of this method). Other methods could hardly be generalized as there is no obvious pattern of the numbers in each group. After pairing up these numbers, each time we need to put two pebbles in the same group. This forms some linkage between the colours of these pebbles (which may be the same or not). This motivates the construction of a graph where the edges join these colours (as vertices) together.

Solution. We pair up the numbers $(1, 4n), (2, 4n - 1), \dots, (2n, 2n + 1)$ such that the sum of each pair is $4n + 1$. Therefore, the first condition is satisfied if we put n pairs in one pile and n pairs in another pile. It remains to show that there exists a grouping such that each pile contains two pebbles of each colour.

Construct a multigraph $G = (V, E)$ as follows. The vertices are the n colours. For each pair $(k, 4n + 1 - k)$ of numbers, we draw an edge between the two vertices that represent the colours of k and $4n + 1 - k$. Note that each edge can be a loop, and there can be more than one edge between two vertices.

Note that $\deg(v) = 4$ for each $v \in V$ as there are four pebbles of each colour. In view of this condition, we may think of using [proposition 4.7](#) (or a generalized result for multigraph). Indeed, this shows every component of G is Eulerian. Consider a particular component H of G with m vertices. Suppose H is the Eulerian circuit $e_1, e_2, \dots, e_k$ where $e_j \in E$. Since the sum of degrees of vertices in H is $4m$, we have $k = 2m$ by the handshaking theorem. We put the pairs of numbers corresponding to $e_1, e_3, \dots, e_{2m-1}$ in one pile, and put the pairs of numbers corresponding to $e_2, e_4, \dots, e_{2m}$ in another pile. As the terminal vertex of e_j coincides with the initial vertex of e_{j+1} (where $e_{2m+1} = e_1$), the two piles have an equal number of pebbles of a

certain colour. This means for each colour in H , there are two pebbles of this colour in one pile and two pebbles in another pile. By doing this for every component of G , we end up with two piles that satisfy the two conditions. $\square$

The following summarizes some important concepts and results in graph theory.

- complete graph ([problem 4.1](#))
- path and cycle ([problems 4.2, 4.5, 4.6, 4.8, 4.10, 4.12, 4.13](#))
- degree ([problems 4.2, 4.3, 4.4, 4.11](#))
- connected graph and components ([problems 4.4, 4.8, 4.9, 4.10, 4.13, 4.14](#))
- tree ([problems 4.5, 4.6, 4.8, 4.10, 4.11, 4.13](#))
- bipartite graph ([problems 4.8, 4.12](#))
- handshaking theorem ([problems 4.4, 4.11, 4.14](#))
- $|E| \geq |V| - 1$ for a connected graph ([problems 4.4, 4.6, 4.8](#))
- a connected graph is a tree iff $|E| = |V| - 1$ ([problems 4.6, 4.8, 4.10, 4.11, 4.13](#))
- a simple graph is bipartite iff it does not contain any odd cycle ([problem 4.12](#))
- Hall's theorem ([problem 4.7](#))
- a connected graph is Eulerian iff $\deg(v)$ is even for all v ([problems 4.9, 4.14](#))

5 Combinatorial Geometry

Combinatorial geometry mainly refers to combinatorial problems with geometric settings. Those problems do not require deep knowledge in geometry, but rather emphasize more on combinatorial thinking. Many techniques introduced, such as induction, double counting and colouring, are useful in this kind of problems.

5.1 Combinatorial Ideas

We first go through some problems in combinatorial geometry that can be solved using the standard ideas in combinatorics.

Problem 5.1. (JBMO Shortlist 2015 C2) 2015 points are given in a plane such that from any five points we can choose two points with distance less than 1 unit. Prove that 504 of the given points lie on a unit disc.

Analysis. Since 504 is about $\frac{1}{4}$ of 2015 and we know some properties about any 5 given points, it is natural to use the pigeonhole principle. For example, we can use 4 given points as centre to draw some unit discs, and show that many points must fall into the same disc. But first we need to choose the 4 points carefully. The argument does not work if some of these points have a distance less than 1 unit.

Solution. Let k be the largest positive integer such that there exist k given points with pairwise distance larger than or equal to 1 unit. Note that k exists and must be less than 5 by the given condition.

Now, let $A_1, A_2, \dots, A_k$ be given points such that $A_i A_j \geq 1$ for all $i \neq j$. Consider the unit discs with centres $A_1, A_2, \dots, A_k$ respectively. All the 2015 points must lie in at least one of these unit discs due to the maximality of k . By the pigeonhole principle, one of the unit discs contains at least

$$\left\lceil \frac{2015}{k} \right\rceil \geq \left\lceil \frac{2015}{4} \right\rceil = 504$$

points as desired. □

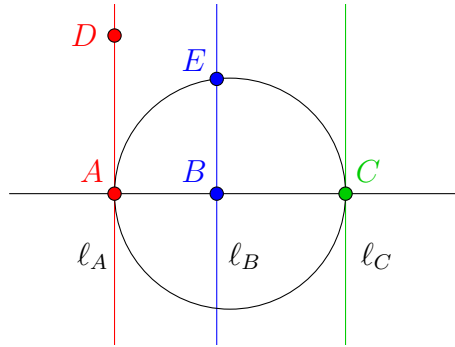
Problem 5.2. (All-Russian MO 2004 Grade 11 Problem 1) Each grid point of the Cartesian plane is coloured with one of three colours, whereby all three colours are used. Show that one can always find a right-angled triangle, whose three vertices have pairwise different colours.

Analysis. One idea is to consider a class of right-angled triangles that share common vertices. For example, by fixing two vertices A and B , we can look for a third vertex

which lies on the line through A perpendicular to AB . Then we can find some relationship between the colours of the points on this line. The rest is just to consider all cases one by one.

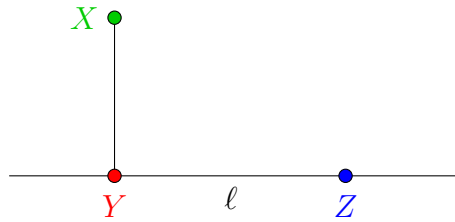
Solution. Suppose the colours are red, blue and green. We say that a right-angled triangle is *good* if its three vertices are of different colours. There are two cases to consider.

Case 1. There exists a line containing three points A, B, C with pairwise different colours.



WLOG assume A is red, B is blue and C is green, and assume B lies between A and C . Let ℓ_A, ℓ_B, ℓ_C be the lines perpendicular to AB and passing through A, B, C respectively. For any D on ℓ_A , if D is blue, then $\triangle DAC$ is good. If D is green, then $\triangle DAB$ is good. Therefore, we may assume every point on ℓ_A is red. Similarly, assume every point on ℓ_B is blue and every point on ℓ_C is green. Let E be an intersection point of ℓ_B and the circle with AC as diameter. Then $\triangle ACE$ is good as $\angle AEC = 90^\circ$.

Case 2. Every line consists of points in at most two different colours.



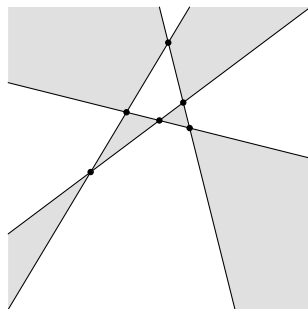
Pick a line ℓ passing through a red point and a blue point. Then every point on ℓ is red or blue. Let X be a green point, which cannot lie on ℓ . Let Y be the projection of X onto ℓ . WLOG assume Y is red. Pick a blue point Z on ℓ . Then $\triangle XYZ$ is good.

In any case, we have found a good triangle. □

Problem 5.3. (EGMO 2017 Problem 3) There are 2017 lines in the plane such that no three of them go through the same point. Turbo the snail sits on a point on exactly one of the lines and starts sliding along the lines in the following fashion: she moves on a given line until she reaches an intersection of two lines. At the intersection, she follows her journey on the other line turning left or right, alternating her choice at each intersection point she reaches. She can only change direction at an intersection point. Can there exist a line segment through which she passes in both directions during her journey?

Analysis. With some trial and errors, one should know that the answer is negative. Since the directions chosen at each intersection point alternate, the parity should be the key thing to consider. In combinatorial geometry, one usual way to consider the parity is to use colouring. The following way of colouring the regions in two colours is a standard trick in solving this kind of problems.

Solution. The 2017 lines divide the plane into a finite number of regions, some of which are finite and some are not. We claim that we can colour these regions in black and white such that every two adjacent regions (which share a common edge) have distinct colours.



This claim can be proved by induction on the number of lines. When there is one line, it divides the plane into two halves. So we can colour one half black and the other half white. Suppose a colouring for n lines exists. We add one more line to the plane. This line divides the plane into two halves. We keep the colours on one side of the line and change the colours on the other side of the line. The adjacent regions which share a common edge formed by this line have distinct colours. Also, any two regions which were adjacent before this line is added still have distinct colours. By induction, a colouring exists for the case $n = 2017$.

Initially, suppose the region to the right of the snail (region I) is black and the region to the left (region II) is white. When the snail reaches an intersection point, there are two possibilities. If the snail turns to the right, then the region to the right of her is still region I, which is black in colour. If the snail turns to the left, then the region to the left of her is still region II, which is white in colour. The same argument applies whenever the snail reaches an intersection point. Therefore, no matter how far she goes, the region to her right must be black and the region to her left must be

white. So it is obvious that the direction travelled by the snail on every line segment is uniquely determined. The answer is negative. $\square$

Problem 5.4. (The University of Melbourne School Mathematics Competition 2017) A finite number of points in the plane are chosen, and each is coloured either red or blue. Suppose that for any line ℓ in the plane, the difference between the number of red points on ℓ and the number of blue points on ℓ is at most 1. Prove that the points all lie on a single line.

Analysis. This time we shall use the method of double counting. We count the number of pairs of points according to their colours. This sets up some equations relating the total number of red/blue points and the number of red/blue points on those lines passing through at least two points. Then we use algebraic ways to show that there must be a line with many points.

Solution. Let r and b be the number of red points and the number of blue points respectively. Let $n = r + b$ be the total number of points. It suffices to consider those lines which pass through at least two of the points. There are only finitely many of them. These lines can be partitioned into 3 groups depending on the differences of the numbers of points of the two colours on them.

- For a line in group I, the number of red points and the number of blue points are the same. Suppose there are x_i red points and x_i blue points.
- For a line in group II, the number of red points is greater than the number of blue points. Suppose there are y_i red points and $y_i - 1$ blue points.
- For a line in group III, the number of red points is smaller than the number of blue points. Suppose there are $z_i - 1$ red points and z_i blue points.

The number of pairs of red points is $\binom{r}{2}$. This is equal to the sum of pairs of red points on all the lines. Hence, we obtain

$$\sum \binom{x_i}{2} + \sum \binom{y_i}{2} + \sum \binom{z_i - 1}{2} = \binom{r}{2}. \quad (1)$$

Each sum runs through all lines in the respective groups. The sum could be empty. Similarly, we have

$$\sum \binom{x_i}{2} + \sum \binom{y_i - 1}{2} + \sum \binom{z_i}{2} = \binom{b}{2}. \quad (2)$$

By counting the number of pairs of points with distinct colours, we obtain

$$\sum x_i^2 + \sum y_i(y_i - 1) + \sum (z_i - 1)z_i = rb. \quad (3)$$

We first rewrite (1) and (2) by expanding the binomial coefficients. This gives

$$\sum x_i(x_i - 1) + \sum y_i(y_i - 1) + \sum (z_i - 1)(z_i - 2) = r(r - 1) \quad (4)$$

and

$$\sum x_i(x_i - 1) + \sum (y_i - 1)(y_i - 2) + \sum z_i(z_i - 1) = b(b - 1). \quad (5)$$

We observe that the left-hand side of (3) is larger than those of (4) and (5), but the right-hand side is not in general. Thus, there should be some tight inequalities that have to be satisfied. We combine these equations by considering $2 \times (3) - (4) - (5)$. This gives

$$\sum 2x_i + \sum 2(y_i - 1) + \sum 2(z_i - 1) = r + b - (r - b)^2 \leq n. \quad (6)$$

Note that each of $2x_i$, $2(y_i - 1)$ and $2(z_i - 1)$ is approximately the number of points on a line. Hence, (6) tells us that it is almost certain that there can only be one line, since otherwise some points contribute more than 1 to the left-hand side and thus the sum is greater than n . We can make this precise algebraically. Adding (4) and (5), we obtain

$$\sum 2x_i(x_i - 1) + \sum 2(y_i - 1)^2 + \sum 2(z_i - 1)^2 = r(r - 1) + b(b - 1) \geq \frac{n(n - 2)}{2} \quad (7)$$

as $r^2 + b^2 \geq \frac{(r + b)^2}{2} = \frac{n^2}{2}$. Suppose the line with the greatest number of points contains M points. Then $x_i \leq \frac{M}{2}$, $y_i \leq \frac{M + 1}{2}$ and $z_i \leq \frac{M + 1}{2}$. Applying these bounds to the left-hand side of (7), we get

$$\frac{M - 1}{2} (\sum 2x_i + \sum 2(y_i - 1) + \sum 2(z_i - 1)) \geq \frac{n(n - 2)}{2}.$$

Using (6), we have $M \geq n - 1$, meaning that there is a line containing almost all points.

If $M = n - 1$, suppose ℓ contains $n - 1$ points and we may assume there is a red point P not on ℓ . Since ℓ contains at least 2 points, it must contain a red point Q . Then the line PQ does not meet the condition. This is a contradiction. Hence, $M = n$, and we are done. $\square$

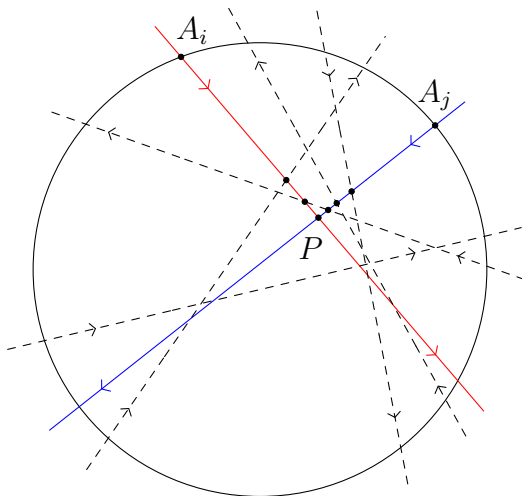
Problem 5.5. (IMO Shortlist 2016 C7/Problem 6) There are $n \geq 2$ line segments in the plane such that every two segments cross, and no three segments meet at a point. Geoff has to choose an endpoint of each segment and place a frog on it, facing the other endpoint. Then he will clap his hands $n - 1$ times. Every time he claps, each frog will immediately jump forward to the next intersection point on its segment. Frogs never change the direction of their jumps. Geoff wishes to place the frogs in such a way that no two of them will ever occupy the same intersection point at the same time.

- (a) Prove that Geoff can always fulfill his wish if n is odd.
- (b) Prove that Geoff can never fulfill his wish if n is even.

Analysis. For a setting involving many pairwise intersecting segments/lines, a useful technique is to draw a sufficiently large circle that includes all intersection points inside. In this way, every line intersects the circle at two points A_i, B_i such that the points on the circle are $A_1, A_2, \dots, A_n, B_1, B_2, \dots, B_n$ in this order. This gives us a way to order these arbitrary lines. The setting is similar to the one in [problem 5.3](#). The same invariance trick is useful as parity is involved.

Solution. Firstly, consider a sufficiently large circle that includes all intersection points inside. We extend each segment to form a straight line. Each such line ℓ_i intersects the circle at two points A_i, B_i . For distinct i, j , since $A_i B_i$ and $A_j B_j$ meet inside the circle, the points A_j and B_j must lie on different sides of the line $A_i B_i$. Thus, it is easy to see that the intersection points on the circle are $A_1, A_2, \dots, A_n, B_1, B_2, \dots, B_n$ in the clockwise order (by swapping A_i and B_i if necessary). WLOG we may assume the endpoints of each segment are exactly A_i, B_i .

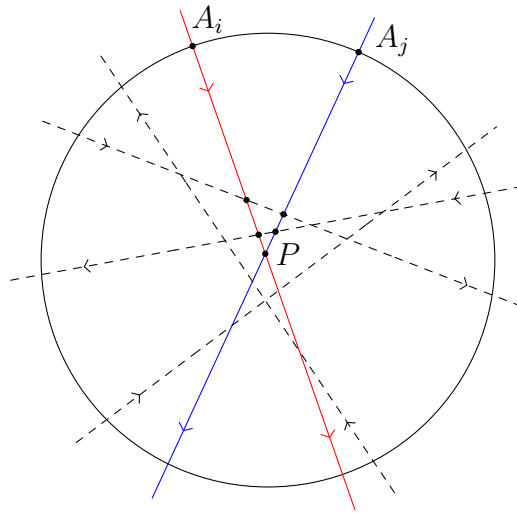
- (a) For odd n , we travel along the circle and mark each of the points A_i or B_i ‘in’ and ‘out’ alternately. For each $i = 1, 2, \dots, n$, there are exactly $n-1$ points between A_i and B_i , which is even. Therefore, one of A_i and B_i is marked ‘in’ and the other is marked ‘out’. Thus, Geoff can put a frog on the endpoint of each segment which is marked ‘in’.



Consider any distinct indices i, j . We want to show that the frogs on ℓ_i and ℓ_j do not meet. WLOG assume the frogs start at A_i and A_j respectively. Let ℓ_i intersect ℓ_j at P . Note that there is an odd number of points on $\overline{A_i A_j}$. Each of these points belongs to a line ℓ_k . Such a line ℓ_k must intersect exactly one of the segments $A_i P$ and $A_j P$, making an odd number of intersections. For the other lines, they may intersect both segments $A_i P$ and $A_j P$, or meet none of them. Therefore, the total number of intersection points on segments $A_i P$ and $A_j P$ (excluding P) is odd. However, if the frogs arrive at P at the same time, then there should be an equal number of intersections on $A_i P$ and $A_j P$, which gives

an even number of intersections. This is a contradiction. This means the frogs do not meet each other. As this holds for all i, j , Geoff can fulfill his wish. $\square$

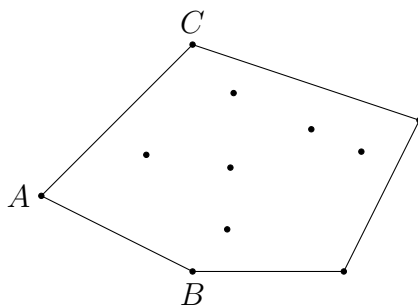
- (b) For even n , we consider any way that Geoff places the frogs and mark each of the points A_i or B_i ‘in’ and ‘out’ according to the directions travelled by the frogs. By working on some examples, it is not hard to see that two frogs starting at two adjacent intersection points would meet each other.



Indeed, since n is even, there must be two adjacent points which are marked ‘in’, say A_i and A_j (otherwise, we have to mark the intersection points ‘in’ and ‘out’ alternately, but then A_i and B_i are both marked ‘in’ or both marked ‘out’). Let P be the intersection of the segments A_iB_i and A_jB_j . Then any other segment meeting segments A_iP or A_jP must also meet another, and so the number of intersections on A_iP and A_jP are the same. This shows the frogs starting at A_i and A_j will meet at P . $\square$

5.2 Point Set Problems

Point set problems refer to those problems involving many discrete points on the plane. They form a common kind of problems in combinatorial geometry. A useful trick in solving such problems is the following fact. If there is a finite set S containing points on the plane which do not lie on the same line, then there exists a unique convex polygon containing all these points with the smallest area. Such a polygon is called the *convex hull*(凸包) of the point set S . Clearly, the vertices of the convex hull must belong to S .

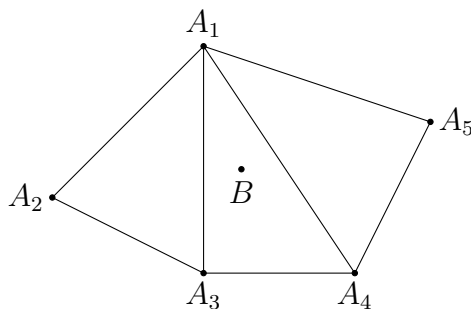


The convex hull is useful because we have a simpler configuration to work on, compared to a random set of points. For example, if AB is a side of the convex hull, then all points in S lie on the same side of the line AB . In particular, if C is another neighbouring vertex of A , then all points in S lie in $\angle BAC$. For every other point $D \in S$, we can measure $\angle BAD$, and order the points in S according to the size of all these angles.

Problem 5.6. Let $\mathcal{P}$ be a polygon with $n \geq 4$ vertices. Suppose every four vertices of $\mathcal{P}$ form a convex quadrilateral. Prove that $\mathcal{P}$ is a convex polygon.

Analysis. We use this simple problem to demonstrate how to use the convex hull to simplify the argument.

Solution. Let $A_1A_2\cdots A_m$ be the convex hull of the set of vertices of $\mathcal{P}$, where $m \leq n$. If $m < n$, then there exists a point B inside $\mathcal{P}$. Consider a triangulation of the convex hull. The point B must lie inside one of the triangles $A_iA_jA_k$. Then $A_iA_jA_kB$ cannot be a convex quadrilateral, contradiction. Thus, we must have $m = n$, which means $\mathcal{P}$ is convex. $\square$

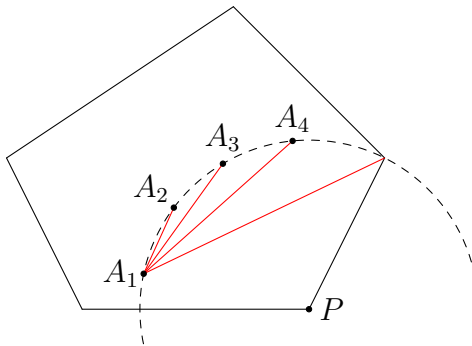


Problem 5.7. (IMO HK TST 2020 Test 2 Problem 2) Suppose there are 2019 distinct points in a plane and the distances between pairs of them attain k different values. Prove that k is at least 44.

Analysis. We assume the contrary. If we fix one of the points P , then we can show by the pigeonhole principle that there are at least 47 points lying on the same circle with P as centre. However, we cannot derive any contradiction by focusing on these 47 points. For example, if they form a regular polygon, then there are only 23

different distances between some pairs of them. We can get rid of this if these points lie on the same semicircle only. This can be done if P is a vertex of the convex hull of all the points.

Solution. Suppose on the contrary that $k < 44$. Clearly, the points cannot belong to the same line. Let P be a vertex of the convex hull of the set of these 2019 points.



Since there are at most 43 different distances from P , all other 2018 points lie on at most 43 circles with centre P . By the pigeonhole principle, at least $\left\lceil \frac{2018}{43} \right\rceil = 47$ points lie on the same circle. They must lie on the same semicircle since P is on the boundary of the convex hull. Let $A_1, A_2, \dots, A_{45}$ be 45 of these points in clockwise order. Then the distances

$$A_1A_2 < A_1A_3 < \dots < A_1A_{45}$$

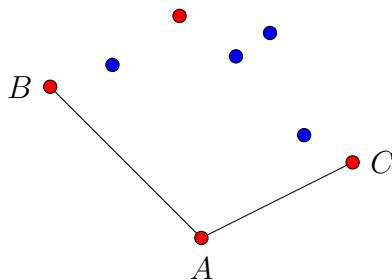
are distinct, which contradicts the assumption that there are fewer than 44 distances. Thus, $k \geq 44$. $\square$

Problem 5.8. Let $n \geq 2$ be an integer. There are n red points and n blue points on the plane such that no three of them are collinear. Prove that there exists a line joining two of the points such that there is an equal number of red points and blue points (possibly none) on each side determined by the line.

Analysis. Clearly, the line must be formed by a red point and a blue point. In that case, it suffices to show that one side of the line contains an equal number of red points and blue points. The trick of rotating a line with a fixed pivot and making use of the discreteness of some function with range $\mathbb{Z}$ is a useful tool in combinatorial geometry.

Solution. Consider the convex hull of the points. Let B, A, C be three consecutive vertices of the convex hull. WLOG assume A is red. If B is blue, then the line AB satisfies the condition since all the other $n - 1$ red points and $n - 1$ blue points lie on the same side of AB , and there is no point on the other side. Thus, we may assume B is red. Similarly, assume C is red.

Consider a ray ℓ with endpoint A . Initially, ℓ goes in the direction AB . We rotate ℓ with A as the centre so that it sweeps across the convex hull until it goes in the direction AC . Let r and b be the number of red points and blue points respectively on the same side of ℓ as B . We consider the value $r - b$.



When ℓ meets the first point, we have $r - b = 1 - 0 = 1$ since only B lies on one side of ℓ . When ℓ reaches AC , we have $r - b = (n - 2) - n = -2$ since all points except A and C lie on one side of ℓ . Each time when ℓ sweeps across a point, the value of $r - b$ is increased or decreased by 1, depending on whether that point is red or blue. Hence, there must be some situations for which $r - b$ equals 0. We consider the last moment for which $r - b = 0$.

At this moment, ℓ must pass through a point D other than A . If D is blue, then there are r red points and $b = r$ blue points on one side of ℓ , and there are $n - 1 - r$ red points and $n - 1 - b = n - 1 - r$ blue points on the other side. So this is the line we need. If D is red, then $r - b$ becomes 1 immediately after ℓ sweeps across D . Again due to the discreteness, there must be another moment afterwards for which $r - b = 0$. This is a contradiction. Therefore, we can always get such a line. $\square$

Problem 5.9. (IMO Shortlist 2011 C3/Problem 2) Let $\mathcal{S}$ be a finite set of at least two points in the plane. Assume that no three points of $\mathcal{S}$ are collinear. A *windmill* is a process that starts with a line ℓ going through a single point $P \in \mathcal{S}$. The line rotates clockwise about the *pivot* P until the first time that the line meets some other point belonging to $\mathcal{S}$. This point, Q , takes over as the new pivot, and the line now rotates clockwise about Q , until it next meets a point of $\mathcal{S}$. This process continues indefinitely.

Show that we can choose a point P in $\mathcal{S}$ and a line ℓ going through P such that the resulting windmill uses each point of $\mathcal{S}$ as a pivot infinitely many times.

Analysis. One may attempt to start with a vertex of the convex hull and see how ℓ moves by using this vertex as a pivot. However, by trying some more examples, one would find that it is extremely hard to work in this direction since there can be many points inside the convex hull, making it hard to trace the path of ℓ .

For a correct approach, the first observation is that the number of points of $\mathcal{S}$ on each side of ℓ is fixed throughout the process (except when ℓ passes through two

points in $\mathcal{S}$). The second observation is that once we know the number of points on each side and we know the direction of ℓ , we can determine the position of ℓ and the current pivot uniquely. Naturally, we can choose ℓ such that the points in $\mathcal{S}$ are divided into two (almost) equal halves. This choice is admissible as justified by some small examples.

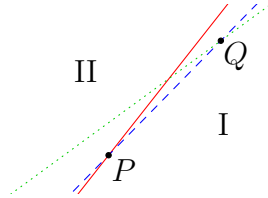
Solution. Let $|\mathcal{S}| = n$. In the following, we assign a positive direction to each line, so that if the line rotates by 180° , it will point in the opposite direction, and is considered as a different line as the original one. Note that every line divides the plane into two regions. We label them as region I and region II such that region I is on the left of the line when the upward direction is the positive direction. The main result we need is the following.

Claim. For any $P \in \mathcal{S}$, there exists a line ℓ_P passing through P such that region I contains $\left\lceil \frac{n-1}{2} \right\rceil$ points and region II contains $\left\lfloor \frac{n-1}{2} \right\rfloor$ points. (Equivalently, region I has 0 or 1 point more than region II.)

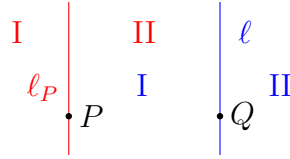
Proof. Consider any line ℓ_P passing through P but not any other point in $\mathcal{S}$. Let a and b be the number of points of $\mathcal{S}$ in region I and region II respectively. Consider the quantity $c = a - b$.

When we rotate ℓ_P about P , every time when ℓ_P sweeps over a point in $\mathcal{S}$, this point goes from region I to region II or vice versa. Therefore, the value of c is increased or decreased by 2. If c_0 is the initial value of c , then its value becomes $-c_0$ after ℓ_P has rotated by 180° , because regions I and II are interchanged. By the discreteness, there must be a moment for which $c = 0$ or $c = 1$. At this moment, the line ℓ_P is the one we need. $\square$

We return to the original problem. We choose any $P \in \mathcal{S}$ and choose $\ell = \ell_P$ as defined in the **claim**. After some time, suppose ℓ meets a new point $Q \in \mathcal{S}$. WLOG assume Q lies in region I. Afterwards, Q becomes the pivot, and P goes to region I. Therefore, the numbers of points in region I and region II remain unchanged throughout the process, except when ℓ passes through two points in $\mathcal{S}$.



Since there are finitely many points in $\mathcal{S}$, there exists a positive constant θ such that ℓ needs to rotate by at least θ radians every time before meeting a new point. Therefore, ℓ will rotate for an infinite number of cycles (about different pivots during the process). In particular, for each $P \in \mathcal{S}$ (not necessarily the starting point), there are infinitely many times when ℓ is parallel to ℓ_P .



We want to show that $\ell = \ell_P$. If not, suppose ℓ passes through $Q \in \mathcal{S}$ different from P . WLOG label the regions as shown above. Note that region I of ℓ contains P , while region I of ℓ_P does not. As the former region contains the latter region, the former region contains more points. This contradicts the fact that region I always contains $\left\lceil \frac{n-1}{2} \right\rceil$ points. Therefore, we must have $\ell = \ell_P$. This means P is the pivot infinitely many times, which is our final goal. $\square$

5.3 Geometric Ideas

In this subsection, we go through more problems in combinatorial geometry which require some basic knowledge in geometry. Everyone is supposed to have this knowledge, so there is no need to introduce the related concepts and results. Our main purpose is to highlight some geometric facts which are commonly used when solving combinatorial geometry problems.

Problem 5.10. (APMO 2011 Problem 2) Five points A_1, A_2, A_3, A_4, A_5 lie on a plane in such a way that no three among them lie on a same straight line. Determine the maximum possible value that the minimum value for the angles $\angle A_i A_j A_k$ can take where i, j, k are distinct integers between 1 and 5.

Analysis. Since this problem is about angles, it is obvious that we need some basic knowledge about angles, such as the angle sum of polygons.

Considering the convex hull of the points is a useful step. This is because all points lie in an interior angle of this convex hull. For example, if we start with the smallest interior angle, we are able to bound some angles. It is not hard to guess that the optimal case is a regular pentagon.

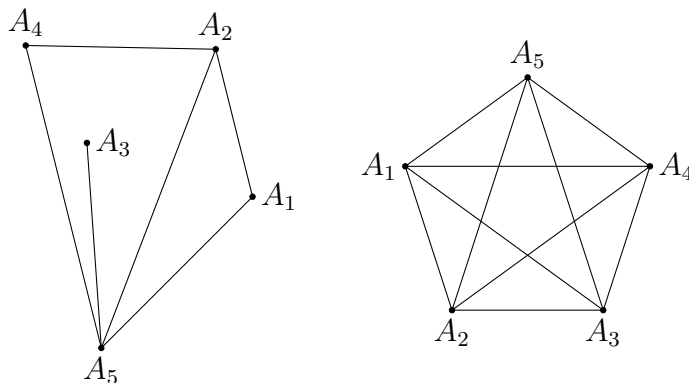
Solution. WLOG assume $\angle A_1 A_5 A_4$ is one of the smallest interior angles of the convex hull formed by the 5 points. If the convex hull has n sides, where $n = 3, 4, 5$, then

$$\angle A_1 A_5 A_4 \leq \frac{(n-2) \times 180^\circ}{n} \leq \frac{(5-2) \times 180^\circ}{5} = 108^\circ.$$

Next, note that A_2 and A_3 lie in $\angle A_1 A_5 A_4$. We may assume the lines $A_5 A_1, A_5 A_2, A_5 A_3, A_5 A_4$ lie in that order. By the pigeonhole principle, one of

$$\angle A_1 A_5 A_2, \angle A_2 A_5 A_3, \angle A_3 A_5 A_4$$

is at most $\frac{1}{3} \angle A_1 A_5 A_4$, which is at most $\frac{1}{3} \times 108^\circ = 36^\circ$. Thus, the answer cannot exceed 36° .



On the other hand, if $A_1 A_2 A_3 A_4 A_5$ is a regular pentagon, then each angle is equal to one of

$$\angle A_1 A_5 A_2, \angle A_2 A_5 A_3, \angle A_1 A_5 A_3, \angle A_1 A_5 A_4$$

due to symmetry. These are equal to $36^\circ, 36^\circ, 72^\circ, 108^\circ$ respectively. In this case the smallest angle is 36° .

We conclude that the maximum possible value of the smallest angle is 36° . $\square$

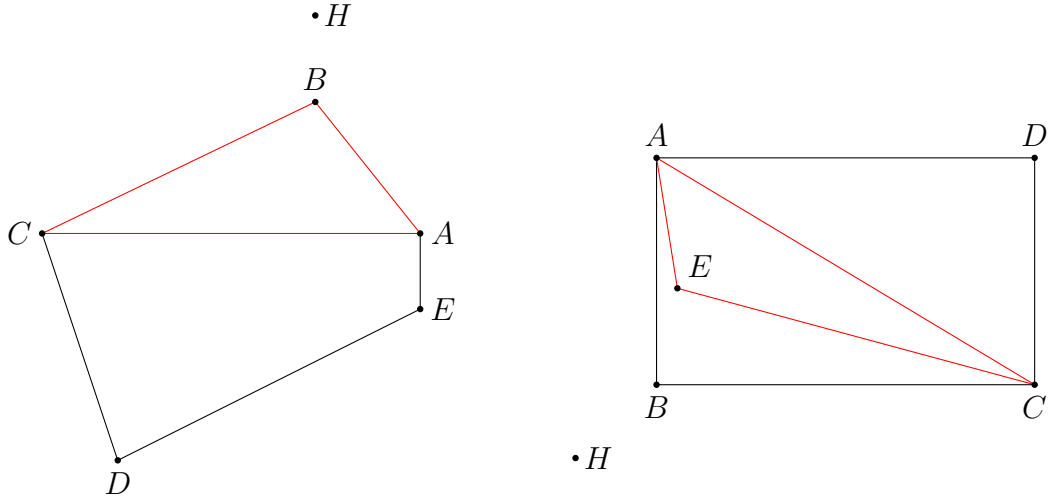
Problem 5.11. Find all integers $n \geq 1$ and all sets $\mathcal{S}$ of n points on the plane such that no three points in $\mathcal{S}$ are collinear and the orthocentre of the triangle formed by any three distinct points in $\mathcal{S}$ also belongs to $\mathcal{S}$.

Analysis. This problem has more geometric ingredient. The key property that one should know is that if H is the orthocentre of $\triangle ABC$, then C is the orthocentre of $\triangle ABH$, etc. Also, the position of the orthocentre is important. If $\triangle ABC$ is acute, then H lies inside the triangle. If $\triangle ABC$ is obtuse, then H lies outside the triangle. If $\triangle ABC$ is a right-angled triangle, then H coincides with one of the vertices.

Again, we may first consider those triangles formed by three vertices of the convex hull.

Solution. For $n = 1, 2$, $\mathcal{S}$ can be any set as the condition is trivially satisfied.

For $n \geq 3$, consider the convex hull of $\mathcal{S}$. If the convex hull is a polygon of $m \geq 5$ vertices, then one of the interior angles $\angle ABC$ of this convex polygon is at least $\frac{m-2}{m} \cdot 180^\circ > 90^\circ$. Then the orthocentre H of $\triangle ABC$ lies outside $\triangle ABC$ and lies on the same side of AC as B . This implies H does not lie inside the convex hull, and hence $H \notin \mathcal{S}$.



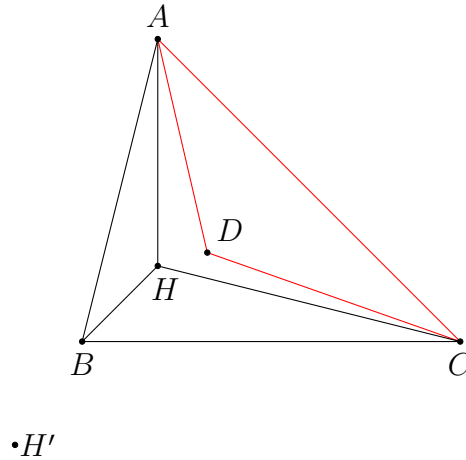
Therefore, the convex hull can only be a quadrilateral or a triangle. For the former case, it has to be a rectangle or otherwise we can use the same argument to an obtuse interior angle.

Suppose the convex hull of $\mathcal{S}$ is a rectangle. If $n = 4$, then $\mathcal{S}$ satisfies the condition since the orthocentre of a right-angled triangle is the vertex with the right angle. If $n \geq 5$, let E be a point in $\mathcal{S}$ inside $ABCD$. WLOG assume E lies inside $\triangle ABC$. Then $\angle CEA > \angle CBA = 90^\circ$. The orthocentre H of $\triangle ACE$ satisfies

$$\angle AHC = 180^\circ - \angle AEC < 90^\circ = \angle ABC.$$

Hence, H lies outside the circumcircle of $\triangle ABC$. This implies H lies outside $ABCD$, and so it cannot belong to $\mathcal{S}$.

Suppose the convex hull of $\mathcal{S}$ is $\triangle ABC$. Again, $\triangle ABC$ cannot be obtuse. If $\triangle ABC$ is right-angled and $n = 3$, then the condition is satisfied. If $\angle ABC = 90^\circ$ and $n \geq 4$, let D be a point in $\mathcal{S}$ inside $\triangle ABC$. Analogous to the argument of the rectangular case, we deduce that the orthocentre of $\triangle ACD$ cannot belong to $\mathcal{S}$.



If $\triangle ABC$ is acute, its orthocentre H must lie inside the triangle. If $n = 4$, then the condition is satisfied since the orthocentre of the triangle formed by any three of A, B, C, H is the remaining point. If $n \geq 5$, let D be a point in $\mathcal{S}$. WLOG, assume D lies inside $\triangle ACH$. Let H' be the orthocentre of $\triangle ACD$. Then

$$\angle AH'C = 180^\circ - \angle ADC < 180^\circ - \angle AHC = \angle ABC.$$

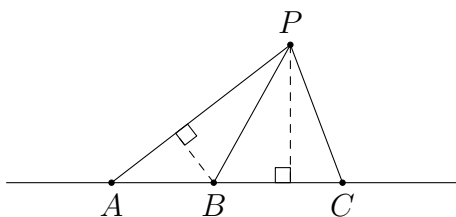
Again, this implies H' lies outside $\triangle ABC$, and hence H' does not belong to $\mathcal{S}$.

We conclude that n can only be 1, 2, 3, 4. For $n = 1, 2$, $\mathcal{S}$ can be anything. For $n = 3$, the three points in $\mathcal{S}$ form a right-angled triangle. For $n = 4$, the four points in $\mathcal{S}$ either form a rectangle or a triangle together with its orthocentre. $\square$

Problem 5.12. ([Sylvester-Gallai theorem](#)(西爾維斯特-加萊定理)) Suppose there are $n \geq 3$ points on the plane which do not lie on the same straight line. Prove that there exists a line passing through exactly two of these points.

Analysis. This solution is a standard proof of this theorem. The key idea is to consider the distances from the points to the lines formed by joining some of the points. We choose a line ℓ and a point P not on ℓ such that the distance from P to ℓ is the smallest. If ℓ contains three distinct points A, B, C , then we derive a contradiction by finding a further smaller distance.

Solution. Consider any pair (ℓ, P) where ℓ is a line formed by two of the points and P is a point not lying on ℓ . Such a pair exists by the condition. Among all these pairs, we choose the one such that the distance from P to ℓ is the smallest. We claim that ℓ contains exactly two points.



Suppose on the contrary that there are three distinct points A, B, C on ℓ in that order. As $\angle PBA + \angle PBC = 180^\circ$, we may assume $\angle PBA \geq 90^\circ$. Consider $\triangle PAB$. As $\angle PBA$ is the largest interior angle, PA is the longest side of this triangle. By considering the area of $\triangle PAB$, we find that the distance from B to PA is smaller than the distance from P to AB . This contradicts the minimal choice of the pair (ℓ, P) . Therefore, ℓ only contains two points as desired. $\square$

Remarks. In the above solution, we use the following simple fact about triangles: In $\triangle ABC$, $\angle A$ is the largest angle iff its opposite side BC has the longest length. This fact is commonly used in combinatorial geometry as we often need some loose bounds to compare the lengths of some segments.

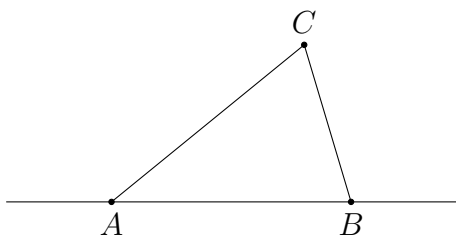
Problem 5.13. Let $\mathcal{S}$ be an infinite set of points on the plane. Suppose for any $A, B, C \in \mathcal{S}$, the distance from A to the line BC is an integer. Prove that all the points in $\mathcal{S}$ are collinear.

Analysis. Note that we cannot use Sylvester-Gallai theorem since the theorem only applies to a finite set of points, but $\mathcal{S}$ is infinite in this problem.

The major condition in this problem is the integer length. If we are able to work out an upper bound for some lengths, then there are only finitely many choices for the lengths. This may contradict the assumption that $\mathcal{S}$ is an infinite set.

The only geometric ingredient that we need is the computation of lengths using trigonometric functions.

Solution. Suppose on the contrary that not all points in $\mathcal{S}$ are collinear. We fix two points A and B in $\mathcal{S}$. We first prove that there is only a finite number of points not on AB .



For each point $C \in \mathcal{S}$ not lying on AB , the distance from B to AC is $AB \sin \angle BAC$. As $\sin \theta \leq 1$, this distance is an integer lying between 1 and AB . Therefore, it has finitely many different possibilities. Each possible distance corresponds to a fixed value of $\sin \angle BAC$, which corresponds to at most two possible values of $\angle BAC$. For each fixed $\angle BAC$, there are two possible directions of the line AC . It follows that C must lie on a finite number of lines $\ell_1, \ell_2, \dots, \ell_k$ (different from AB) passing through A . Similarly, C lies on a finite number of lines $L_1, L_2, \dots, L_K$ passing through B . As each pair of ℓ_i and L_j has at most one intersection point, there are only finitely many such points C which can lie on some ℓ_i and some L_j simultaneously. This proves our claim.

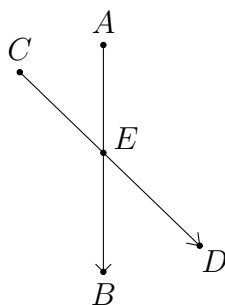
Now, as $\mathcal{S}$ is infinite, there are infinitely many points on AB . Let C be any point not on AB . Applying the claim to the line AC , we see that there are finitely many points not on AC . But then every point on AB other than A does not lie on AC . This is a contradiction. Therefore, all points in $\mathcal{S}$ are collinear. $\square$

Problem 5.14. (239 Open MO 2018 Senior Problem 1) Prove that in any convex polygon where all pairwise distances between vertices are distinct, there exists a vertex such that the closest vertex of the polygon is adjacent to it.

Analysis. Intuitively, if B is the closest vertex to a vertex A , and D is the closest vertex to a vertex C , then C and D should lie on the same side of the line AB . If this is true, then we can argue repeatedly to obtain closer and closer vertices until they are adjacent. Alternatively, we can use the extremal principle.

Indeed, we can establish the above intuitive fact by using the triangle inequality. This is another commonly used geometric fact in combinatorial geometry problems.

Solution. For each vertex P , we let $f(P)$ be the vertex which is closest to P . Suppose on the contrary that there is no P such that P and $f(P)$ are adjacent.



Choose a vertex A such that A and $B = f(A)$ are separated by the smallest number of vertices. Let C be the vertex adjacent to A that lies on the side of the line AB with fewer vertices. By the assumption, C is distinct from A and B , and $D = f(C)$ should not lie on the same side of the line AB as C by the choice of A . Therefore, the segments AB and CD intersect at some point E . Note that $AD > AB$ and $BC > CD$ since $B = f(A)$ and $D = f(C)$. By the triangle inequality, we have

$$AB + CD = (AE + DE) + (BE + CE) > AD + BC > AB + CD.$$

This is a contradiction. Therefore, there must be a vertex P such that P and $f(P)$ are adjacent. $\square$

Problem 5.15. (Canada MO 2017 Problem 5) There are 100 circles of radius one in the plane. A triangle formed by the centres of any three given circles has area at most 2017. Prove that there is a line intersecting at least three of the circles.

Analysis. In this problem, we introduce one more useful idea in solving combinatorial geometry problems. The idea is to project some points and/or objects onto a line, such as the coordinate axes. This collapses a two-dimensional setting to a one-dimensional system for consideration, which simplifies the structure. For example, if we project the centres of the circles to the x -axis, we are able to show that these centres cannot be too close to each other. Thus, the two furthest centres should be far away from each other, so we may obtain a contradiction by considering a triangle with these centres as two of the vertices.

Solution. Suppose on the contrary that every line intersects at most two circles. Let A and B be the centres of two circles such that AB is maximized. We set up a

coordinate system such that AB is the x -axis. Let $P_j(x_j, y_j)$ be the centres of the circles where $j = 1, 2, \dots, 100$ and $x_1 \leq x_2 \leq \dots \leq x_{100}$. Since $P_1 P_{100} \geq x_{100} - x_1 \geq AB$, equality must hold by the choice of A, B . Thus, we have $P_1 = A$ and $P_{100} = B$ up to symmetry.

We claim that $AB \geq 98$. Indeed, if $AB < 98$, by partitioning $[x_1, x_{100}]$ into 49 intervals of equal length, the length of each interval is less than 2. By the pigeonhole principle, there exist x_j, x_{j+1}, x_{j+2} lying in the same interval. Then the straight line $x = \frac{x_j + x_{j+2}}{2}$ intersects the three circles with centres P_j, P_{j+1}, P_{j+2} , since $\left| \frac{x_j + x_{j+2}}{2} - x_i \right| < 1$ for $i = j, j+1, j+2$. This is a contradiction. So $AB \geq 98$.

Similarly, if we arrange the y -coordinates of the points in the order $y_{i_1} \leq y_{i_2} \leq \dots \leq y_{i_{100}}$, we must have $y_{i_{100}} - y_{i_1} \geq 98$. Since A, B have the same y -coordinate, there exists a circle with centre C (with y -coordinate y_{i_1} or $y_{i_{100}}$) which has a distance ≥ 49 from AB . This gives

$$[ABC] \geq \frac{98 \times 49}{2} = 2401 > 2017,$$

contradicting the assumption. Thus, there is always a line intersecting at least three circles. $\square$

We summarize the useful ideas in combinatorial geometry below.

- combinatorial ideas:
 - pigeonhole principle ([problems 5.1, 5.7, 5.10, 5.15](#))
 - extremal principle ([problems 5.1, 5.10, 5.12, 5.14, 5.15](#))
 - exhaustion argument ([problems 5.2, 5.11](#))
 - colouring, invariance ([problems 5.3, 5.5, 5.9](#))
 - double counting ([problem 5.4](#))
 - discreteness ([problems 5.8, 5.9, 5.13](#))
- convex hull ([problems 5.6, 5.7, 5.8, 5.10, 5.11](#))
- geometric ideas:
 - larger angle, longer side ([problem 5.12](#))
 - triangle inequality ([problem 5.14](#))
 - rotating lines ([problems 5.8, 5.9](#))
 - drawing a large circle that includes all intersection points ([problem 5.5](#))
 - projecting objects onto a line ([problem 5.15](#))

6 Generating Functions

Generating function can be used to solve a certain kind of counting problems. Suppose the number of ways to do a task corresponding to a certain quantity n (where n is a nonnegative integer) is a_n . Then we can try to form a function using the numbers $a_0, a_1, \dots$ and deduce some useful properties of the function formed. This sometimes allows us to find the values of a_n in an algebraic way.

The most common type of generating function is

$$f(x) = a_0 + a_1x + a_2x^2 + \dots$$

This means we use $a_0, a_1, \dots$ as the coefficients to form a formal power series. We call it a formal power series but not a polynomial because it has infinitely many terms in most cases. For a formal power series, it means we do not need to worry whether the infinite sum is convergent or not. Indeed, as long as x remains as a variable, this does not cause any trouble. Yet we can still do the arithmetic manipulation as usual. For example, we have

$$(a_0 + a_1x + a_2x^2 + \dots)(b_0 + b_1x + b_2x^2 + \dots) = c_0 + c_1x + c_2x^2 + \dots$$

where $c_n = a_0b_n + a_1b_{n-1} + \dots + a_nb_0$ for each $n = 0, 1, 2, \dots$

One major fact which makes generating function useful is the following generalized binomial theorem. To state the theorem, we need to use the generalized binomial coefficients. For any real number a and any positive integer r , we define

$$\binom{a}{r} = \frac{a(a-1)\dots(a-r+1)}{r!}.$$

We also define $\binom{a}{0} = 1$. These definitions coincide with the definition for integer $a \geq r$. Also, it can be proved easily that $\binom{a}{r} = 0$ if a, r are positive integers with $a < r$. For positive integers n and r , we find that

$$\begin{aligned} \binom{-n}{r} &= \frac{(-n)(-n-1)\dots(-n-r+1)}{r!} \\ &= (-1)^r \frac{(n)(n+1)\dots(n+r-1)}{r!} \\ &= (-1)^r \binom{n+r-1}{r}. \end{aligned}$$

We have the following theorem.

Theorem 6.1. (Generalized binomial theorem(廣義二項式定理)) Let a be a real number. Then

$$(x+y)^a = \sum_{r=0}^{\infty} \binom{a}{r} x^{a-r} y^r.$$

The proof of this generalized version is omitted. The result is indeed the same as the ordinary binomial theorem. The main difference is that the expansion may be an infinite series, since for non-integers a , we do not have $\binom{a}{r} = 0$ for large positive integers r .

For example, we often come across a geometric sequence such as

$$(1 + x + x^2 + \cdots + x^k)^m.$$

This is equal to $\left(\frac{x^{k+1} - 1}{x - 1}\right)^m = (x^{k+1} - 1)^m (x - 1)^{-m}$. Then we can use the generalized binomial theorem to expand the terms $(x^{k+1} - 1)^m$ and $(x - 1)^{-m}$.

We now demonstrate the use of generating functions by going through a few problems.

Problem 6.1. (IMO HK TST 2014 Test 1 Problem 4) A number is *good* if the sum of its digits is 18 (for example 130518 is a good number). Find the number of good 6-digit numbers.

Analysis. We give an alternative solution to this problem using generating function. It demonstrates the fundamental usage of generating functions.

Solution. For each positive integer n , let a_n be the number of 6-digit numbers whose sums of digits are n . Consider the generating function $f(x) = a_1x + a_2x^2 + a_3x^3 + \cdots$. Note that the exponent corresponds to the sum of digits. If we only consider the leading digit, then the corresponding generating function should be $x + x^2 + \cdots + x^9$. This is because the leading digit can be one of $1, 2, \dots, 9$, and there is one possibility for each digit. Similarly, each of the remaining digits corresponds to $1 + x + \cdots + x^9$. To form a 6-digit number, we can choose the 6 digits one by one. The choice for each digit is independent of each other. Therefore, we have

$$f(x) = (x + x^2 + \cdots + x^9)(1 + x + \cdots + x^9)^5.$$

(For example, the number 130518 corresponds to the product $(x)(x^3)(1)(x^5)(x)(x^8)$ in the expansion.) Each of the geometric sequences can be expanded in the way as discussed. We have

$$f(x) = \frac{x^{10} - x}{x - 1} \cdot \left(\frac{x^{10} - 1}{x - 1}\right)^5 = (x^{10} - x)(x^{10} - 1)^5(x - 1)^{-6}.$$

Using the generalized binomial theorem, we find that

$$f(x) = (-x + x^{10}) \left(\sum_{r=0}^{\infty} \binom{5}{r} (-1)^{5-r} x^{10r} \right) \left(\sum_{s=0}^{\infty} \binom{-6}{s} (-1)^{-6-s} x^s \right).$$

We need to find the coefficient of x^{18} . Thus, we can restrict the sum in r to $r = 0$ and $r = 1$ (as all subsequent terms exceed x^{18}). Then we obtain

$$f(x) = (x - x^{10} - 5x^{11} + \cdots) \left(\sum_{s=0}^{\infty} \binom{-6}{s} (-1)^{-6-s} x^s \right)$$

where the part not shown consists of terms x^k with $k > 18$. Hence, the answer is

$$\begin{aligned} a_{18} &= \binom{-6}{17}(-1)^{-6-17} - \binom{-6}{8}(-1)^{-6-8} - 5\binom{-6}{7}(-1)^{-6-7} \\ &= \binom{22}{17} - \binom{13}{8} - 5\binom{12}{7} \\ &= 21087. \end{aligned}$$

□

Remarks. The last formula $\binom{22}{17} - \binom{13}{8} - 5\binom{12}{7}$ is actually the same as the one we found using the inclusion-exclusion principle in [problem 1.4](#). One may see the linkage between the two methods.

Problem 6.2. Let n be a positive integer. In how many ways can we tile a $2 \times n$ board with 1×2 and 2×1 dominoes?

Analysis. Recall that such problems can be solved using recurrence relations. In this problem, we demonstrate how to use generating functions to find the general term of a recurrence relation.

Solution. Let a_n be the number of ways to tile a $2 \times n$ board, and let $a_0 = 1$. Note that $a_1 = 1$ as we can only use a 2×1 domino.

Consider any $n \geq 2$. If we use a 2×1 domino to fill up the first column, there are a_{n-1} ways to tile the remaining $2 \times (n-1)$ board. If we use a 1×2 domino to cover the cell in the top left corner, we must use another 1×2 domino to cover the cell in the bottom left corner. Then there are a_{n-2} ways to tile the remaining $2 \times (n-2)$ board (which also holds if $n = 2$). This establishes the recurrence relation

$$a_n = a_{n-1} + a_{n-2}.$$



Define $g(x) = a_0 + a_1x + a_2x^2 + \dots$. Using the recurrence relation, we have

$$\begin{aligned} g(x) &= 1 + x + a_2x^2 + a_3x^3 + a_4x^4 + \dots \\ &= 1 + x + (a_0 + a_1)x^2 + (a_1 + a_2)x^3 + (a_2 + a_3)x^4 + \dots \\ &= 1 + x + x^2(a_0 + a_1x + a_2x^2 + \dots) + x(a_1x + a_2x^2 + a_3x^3 + \dots) \\ &= 1 + x + x^2g(x) + x(g(x) - 1) \\ &= 1 + x^2g(x) + xg(x). \end{aligned}$$

Solving for $g(x)$, we obtain $g(x) = -\frac{1}{x^2 + x - 1}$. To express this using an infinite series, since we only know how to do so for a linear denominator, we shall use partial fraction. As $x^2 + x - 1 = (x - \alpha)(x - \beta)$ where $\alpha = \frac{-1 + \sqrt{5}}{2}$ and $\beta = \frac{-1 - \sqrt{5}}{2}$, we equate $g(x) = \frac{A}{x - \alpha} + \frac{B}{x - \beta}$ to obtain

$$\begin{aligned} g(x) &= \frac{1}{\alpha - \beta} \left(-\frac{1}{x - \alpha} + \frac{1}{x - \beta} \right) \\ &= \frac{1}{\alpha - \beta} \left(-\sum_{r=0}^{\infty} \binom{-1}{r} (-\alpha)^{-1-r} x^r + \sum_{r=0}^{\infty} \binom{-1}{r} (-\beta)^{-1-r} x^r \right) \\ &= \frac{1}{\alpha - \beta} \left(-\sum_{r=0}^{\infty} (-1)^r (-\alpha)^{-1-r} x^r + \sum_{r=0}^{\infty} (-1)^r (-\beta)^{-1-r} x^r \right) \\ &= \frac{1}{\alpha - \beta} \left(\sum_{r=0}^{\infty} \alpha^{-1-r} x^r - \sum_{r=0}^{\infty} \beta^{-1-r} x^r \right). \end{aligned}$$

Note that $\alpha^{-1} = -\beta$ and $\beta^{-1} = -\alpha$. Hence, this becomes

$$g(x) = \frac{1}{\alpha - \beta} \sum_{r=0}^{\infty} ((-\beta)^{r+1} - (-\alpha)^{r+1}) x^r.$$

From this, it is obvious that $a_n = \frac{(-\beta)^{n+1} - (-\alpha)^{n+1}}{\alpha - \beta}$. □

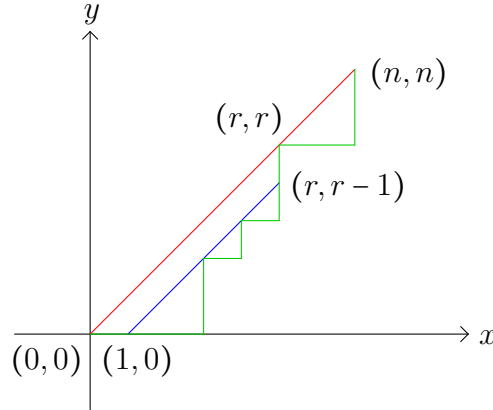
Remarks. Of course $a_n = F_{n+1}$, which is the Fibonacci sequence.

Problem 6.3. Let n be a positive integer. An object is moved from $(0, 0)$ to (n, n) in the coordinate plane in the following way. In each step, the object can be moved to one step rightward or one step upward. Suppose the object never goes to a point (x, y) for which $y > x$. Find the number of ways to move the object.

Analysis. Let C_n be the number of ways to move the object. With some trials, one can obtain $C_1 = 1$, $C_2 = 2$, $C_3 = 5$ and $C_4 = 14$, etc. There seems to be no obvious pattern. We shall try to find a recurrence relation for the sequence $\{C_n\}$ and use generating function to solve for the general term. In this way, there is no need to guess the answer.

Solution. Let C_n be the number of ways to move the object. We further define $C_0 = 1$ for convenience. For $n \geq 1$, suppose the first time when the object reaches the line $y = x$ (other than $(0, 0)$) is at point (r, r) where $r = 1, 2, \dots, n$.

For the path from $(0, 0)$ to (r, r) , the first step must go to $(1, 0)$ and the step before (r, r) must go to $(r, r - 1)$. This means we count the number of paths from $(1, 0)$ to $(r, r - 1)$ such that the object never goes to a point (x, y) with $y + 1 > x$ (we cannot have $y = x$ since this contradicts the smallest choice of r). By translating the configuration one step to the left, this number is exactly C_{r-1} .



For the path from (r, r) to (n, n) , the object is allowed to reach the line $y = x$ in the intermediate steps. Therefore, it corresponds to the original problem moving from $(0, 0)$ to $(n - r, n - r)$. There are C_{n-r} ways to do so. Note that the same holds when $r = n$ as $C_0 = 1$.

By the multiplication principle and summing over all possible r , we obtain the recurrence relation

$$C_n = C_0 C_{n-1} + C_1 C_{n-2} + \cdots + C_{n-1} C_0.$$

We use generating function to solve for C_n . Note that the right-hand side of the recurrence relation looks like the coefficient of the square of a power series. Indeed, if we define

$$f(x) = C_0 + C_1 x + C_2 x^2 + \cdots$$

as usual, we have

$$\begin{aligned} f(x)^2 &= (C_0 + C_1 x + C_2 x^2 + \cdots)^2 \\ &= C_0 C_0 + (C_0 C_1 + C_1 C_0) x + (C_0 C_2 + C_1 C_1 + C_2 C_0) x^2 + \cdots \\ &= C_1 + C_2 x + C_3 x^2 + \cdots \\ &= \frac{f(x) - 1}{x}. \end{aligned}$$

We can solve the quadratic equation $xf(x)^2 - f(x) + 1 = 0$ as if $f(x)$ is a real number.

This gives $f(x) = \frac{1 \pm \sqrt{1 - 4x}}{2x}$. The square root is just $(1 - 4x)^{\frac{1}{2}}$, and so it can be expanded using the generalized binomial theorem. We have

$$f(x) = \frac{1}{2x} (1 \pm (1 - 4x)^{\frac{1}{2}}) = \frac{1}{2x} \left(1 \pm \sum_{r=0}^{\infty} \binom{\frac{1}{2}}{r} (-4x)^r \right).$$

Comparing with $f(x) = C_0 + C_1 x + C_2 x^2 + \cdots$, we easily see that the negative sign should be chosen (otherwise there is a term $\frac{1}{x}$). Therefore, we find that

$$f(x) = -\frac{1}{2} \sum_{r=1}^{\infty} \binom{\frac{1}{2}}{r} (-4)^r x^{r-1}.$$

Hence,

$$C_n = -\frac{1}{2} \binom{\frac{1}{2}}{n+1} (-4)^{n+1} = \binom{\frac{1}{2}}{n+1} (-1)^n 2^{2n+1}.$$

This can be further simplified. Indeed, using the definition, we have

$$\begin{aligned} \binom{\frac{1}{2}}{n+1} &= \frac{\left(\frac{1}{2}\right)\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)\cdots\left(-\frac{2n-1}{2}\right)}{(n+1)!} \\ &= (-1)^n 2^{-n-1} \frac{1 \times 3 \times 5 \times \cdots \times (2n-1)}{(n+1)!} \\ &= (-1)^n 2^{-n-1} \frac{(2n)!}{(2 \times 4 \times 6 \times \cdots \times (2n))(n+1)!} \\ &= (-1)^n 2^{-2n-1} \frac{(2n)!}{n!(n+1)!} \\ &= (-1)^n 2^{-2n-1} \frac{1}{n+1} \binom{2n}{n}. \end{aligned}$$

This implies $C_n = \frac{1}{n+1} \binom{2n}{n}$. □

Remarks. The terms in the sequence $\{C_n\}$ are called *Catalan numbers*. There are various combinatorial meanings of this sequence.

Before proceeding to the next problem, we recall some facts about complex numbers. A complex number can be represented in the form $z = a + bi$ (where $a, b \in \mathbb{R}$ and i is the number satisfying $i^2 = -1$), $z = r(\cos \theta + i \sin \theta)$ (where $r \geq 0$ and $0 \leq \theta < 2\pi$) or $z = re^{i\theta}$. For each positive integer n , the solutions to the equation $z^n = 1$ are

$$z = 1, e^{i\frac{2\pi}{n}}, e^{i\frac{4\pi}{n}}, \dots, e^{i\frac{2(n-1)\pi}{n}}.$$

Let $\omega = e^{i\frac{2\pi}{n}}$. Then the solutions can be represented as $1, \omega, \omega^2, \dots, \omega^{n-1}$. These solutions are called *roots of unity*. If $z \neq 1$ is an n th root of unity, then

$$z^{n-1} + z^{n-2} + \cdots + 1 = 0$$

since $(z-1)(z^{n-1} + z^{n-2} + \cdots + 1) = z^n - 1 = 0$ and $z \neq 1$.

Problem 6.4. For any set A of integers, let $s(A)$ denote the sum of elements of A . Let $S = \{1, 2, \dots, 1999\}$. Find the number of elements in

$$\mathcal{F}_0 = \{T : T \subset S, s(T) \equiv 0 \pmod{7}\}.$$

Analysis. This means we need to count the number of subsets of S whose sums of elements are congruent to 0 modulo 7. This kind of problems can be solved via generating functions with the aid of roots of unity, as they can extract an arithmetic sequence easily.

Solution. When we form a subset of S , each element can be included or excluded. If an element k is included, then its contribution to the sum is k . Otherwise, the contribution is 0. Therefore, each element corresponds to a generating function $1+x^k$. Hence, the generating function corresponding to this problem is

$$f(x) = (1+x)(1+x^2)\cdots(1+x^{1999}). \quad (1)$$

If we let $f(x) = a_0 + a_1x + \cdots + a_dx^d$ (where $d = 1 + 2 + \cdots + 1999$), then each a_k counts the number of subsets of S whose sums of elements are k . This gives

$$|\mathcal{F}_0| = a_0 + a_7 + a_{14} + \cdots.$$

Let $\omega = e^{i\frac{2\pi}{7}}$ be a 7th root of unity. We sum up

$$\begin{aligned} f(1) &= a_0 + a_1 + a_2 + \cdots + a_d, \\ f(\omega) &= a_0 + a_1\omega + a_2\omega^2 + \cdots + a_d\omega^d, \\ f(\omega^2) &= a_0 + a_1\omega^2 + a_2\omega^4 + \cdots + a_d\omega^{2d}, \\ &\vdots \\ f(\omega^6) &= a_0 + a_1\omega^6 + a_2\omega^{12} + \cdots + a_d\omega^{6d} \end{aligned}$$

using the properties of roots of unity. Note that a_1 is multiplied to the term

$$1 + \omega + \omega^2 + \cdots + \omega^6 = 0.$$

This holds for many other terms. Indeed, for each $k = 0, 1, 2, \dots, d$, the coefficient a_k is multiplied to the term

$$1 + \omega^k + \omega^{2k} + \cdots + \omega^{6k}.$$

If $k \not\equiv 0 \pmod{7}$, then ω^k is a 7th root of unity different from 1. As explained, the expression $1 + (\omega^k) + (\omega^k)^2 + \cdots + (\omega^k)^6$ is equal to 0. If $k \equiv 0 \pmod{7}$, then $\omega^k = 1$ (as $\omega^7 = 1$) and so the expression is equal to 7. Therefore, we obtain

$$f(1) + f(\omega) + f(\omega^2) + \cdots + f(\omega^6) = 7(a_0 + a_7 + a_{14} + \cdots) = 7|\mathcal{F}_0|.$$

It remains to find the particular values on the left using formula (1). Note that the exponents in the formula can be taken modulo 7 since x is now a root of unity. Thus, the main task is to compute

$$g(x) = (1+x)(1+x^2)\cdots(1+x^7).$$

As 7 is a prime number, when we substitute $x = \omega^k$ in $g(x)$ for $k = 1, 2, \dots, 6$, the exponents form a complete set of residues modulo 7. This shows

$$g(\omega^k) = g(\omega) = (1+\omega)(1+\omega^2)\cdots(1+\omega^7).$$

Note that $h(x) = (x+\omega)(x+\omega^2)\cdots(x+\omega^7)$ is the monic polynomial of degree 7 whose roots are $-\omega, -\omega^2, \dots, -\omega^7$. Hence, we have $h(x) = x^7 + 1$. It follows that $g(\omega^k) = h(1) = 1 + 1 = 2$. Also, it is obvious that $g(1) = 2^7$.

We have

$$f(\omega) = g(\omega)^{285}(1 + \omega)(1 + \omega^2)(1 + \omega^3)(1 + \omega^4) = 2^{285}(1 + \omega^3)$$

since $(1 + \omega)(1 + \omega^2)(1 + \omega^4) = 1 + \omega + \omega^2 + \dots + \omega^7 = \omega^7 = 1$. Similarly, we find that

$$f(\omega^k) = 2^{285}(1 + \omega^{3k})$$

for $k = 1, 2, \dots, 6$. For $k = 0$, we have $f(1) = 2^{1999}$. Therefore, we finally obtain

$$|\mathcal{F}_0| = \frac{1}{7} \left(2^{1999} + 2^{285}((1 + \omega^3) + (1 + \omega^6) + \dots + (1 + \omega^{18})) \right) = \frac{2^{1999} + 5 \times 2^{285}}{7}.$$

□

Remarks. In a similar manner, we can obtain the number of subsets of S whose sums of elements are congruent to r modulo 7 for some $r = 1, 2, \dots, 6$. For example, for $r = 1$, we can consider the sum

$$f(1) + \omega^6 f(\omega) + \omega^{12} f(\omega^2) + \dots + \omega^{36} f(\omega^6).$$

The trick is to have each coefficient a_{7k+1} multiplied to a 7ℓ th power of ω (which is 1).

Although we usually cannot substitute a number in a generating function, we can do so in this problem because the generating function here is a finite sum, which is simply a polynomial.

Problem 6.5. (USA TSTST 2018 Problem 6) Let $S = \{1, \dots, 100\}$, and for every positive integer n define

$$T_n = \{(a_1, \dots, a_n) \in S^n : a_1 + \dots + a_n \equiv 0 \pmod{100}\}.$$

Determine which n have the following property: if we colour any 75 elements of S red, then at least half of the n -tuples in T_n have an even number of coordinates with red elements.

Analysis. The congruence condition looks similar to the one in the previous problem. We can construct a similar generating function and use roots of unity to work out some useful identities. In order to distinguish between the red elements and the non-red elements, we may construct two polynomials instead, one corresponding to each kind of elements.

Solution. Let S' be the set of all red elements in S . Define

$$f(x) = \sum_{a \in S'} x^a \quad \text{and} \quad g(x) = \sum_{a \in (S - S')} x^a.$$

As usual,

$$(x + x^2 + \dots + x^{100})^n = (f(x) + g(x))^n = \sum_{k=0}^n \binom{n}{k} f(x)^k g(x)^{n-k}$$

is the generating function that corresponds to all possible n -tuples of elements in S . In particular, the generating function that corresponds to those n -tuples with exactly k red elements is

$$h_k(x) = \binom{n}{k} f(x)^k g(x)^{n-k}.$$

The number of elements N_k in T_n with exactly k red coordinates is the sum of coefficients of $1, x^{100}, x^{200}, \dots$ in $h_k(x)$. Similar to [problem 6.4](#), we can single out these coefficients by using the 100th root of unity $\omega = e^{i\frac{2\pi}{100}}$. Indeed, we have

$$N_k = \frac{1}{100} \sum_{j=0}^{99} h_k(\omega^j) = \frac{1}{100} \binom{n}{k} \sum_{j=0}^{99} f(\omega^j)^k g(\omega^j)^{n-k}.$$

Define

$$A = \sum_{\substack{k=0 \\ 2|k}}^n N_k \quad \text{and} \quad B = \sum_{\substack{k=0 \\ 2 \nmid k}}^n N_k.$$

The requirement that at least half of the n -tuples in T_n have an even number of coordinates with red elements is equivalent to $A \geq B$. Therefore, we consider

$$\begin{aligned} A - B &= \sum_{\substack{k=0 \\ 2|k}}^n N_k - \sum_{\substack{k=0 \\ 2 \nmid k}}^n N_k \\ &= \sum_{k=0}^n (-1)^k N_k \\ &= \sum_{k=0}^n (-1)^k \frac{1}{100} \binom{n}{k} \sum_{j=0}^{99} f(\omega^j)^k g(\omega^j)^{n-k} \\ &= \frac{1}{100} \sum_{j=0}^{99} \sum_{k=0}^n \binom{n}{k} (-f(\omega^j))^k g(\omega^j)^{n-k} \\ &= \frac{1}{100} \sum_{j=0}^{99} (g(\omega^j) - f(\omega^j))^n. \end{aligned}$$

Recall that $f(x) + g(x) = x + x^2 + \dots + x^{100}$. This is equal to 100 when $x = 1$, and is equal to 0 when $x = \omega^j$ where $1 \leq j \leq 99$. This implies $g(\omega^j) - f(\omega^j) = 2g(\omega^j)$ for $1 \leq j \leq 99$, and hence

$$\begin{aligned} A - B &= \frac{1}{100} \left((g(1) - f(1))^n + 2^n \sum_{j=1}^{99} g(\omega^j)^n \right) \\ &= \frac{1}{100} \left((25 - 75)^n + 2^n \sum_{j=1}^{99} g(\omega^j)^n \right) \\ &= \frac{1}{100} \left((-50)^n + 2^n \sum_{j=1}^{99} g(\omega^j)^n \right). \end{aligned}$$

The sum can be expressed in terms of its combinatorial meaning by using exactly the same method. Indeed,

$$\frac{1}{100} \sum_{j=0}^{99} g(\omega^j)^n$$

counts the number C of n -tuples in T_n with all coordinates non-red. Obviously, $C \geq 0$. It follows that

$$A - B = \frac{1}{100}((-50)^n + 2^n(100C - g(1)^n)) = \frac{1}{100}((-50)^n - 50^n) + 2^n C.$$

In view of this formula, it is natural to consider the parity of n .

- If n is even, we have $A - B = 2^n C \geq 0$. The condition is satisfied.
- If n is odd, we have $A - B = -50^{n-1} + 2^n C$. It is not hard to work out an example for which $A - B < 0$. This happens when C is small. For example, we can colour all elements in S not congruent to 2 (mod 4) red. There are exactly 75 red elements. Since all non-red elements are of the form $4t + 2$, the sum of any odd number of these elements cannot be a multiple of 4, hence cannot be a multiple of 100. This shows $C = 0$ by definition. So $A - B = -50^{n-1} < 0$.

To conclude, the answer is all even n . □

Problem 6.6. Let a be a positive integer. Find all $m \times n$ rectangles which can be tiled by $1 \times a$ (and $a \times 1$) rectangles completely.

Analysis. We revisit the same problem as [problem 2.8](#). This time we shall use a multivariable generating function to solve the problem. Instead of using different colours to colour the cells, we may associate each cell with a monomial $x^i y^j$ in two variables, depending on its position. By doing so, we can work out an identity analogous to the double counting technique.

Solution. Such a tiling clearly exists when $a \mid m$ or $a \mid n$. In the following, we show that these are the only possible cases.

In the (i, j) th cell of the rectangle, where $1 \leq i \leq m$ and $1 \leq j \leq n$, we write down a monomial $x^i y^j$. The sum of all monomials in the rectangle is

$$f(x, y) = \sum_{i=1}^m \sum_{j=1}^n x^i y^j = \left(\sum_{i=1}^m x^i \right) \left(\sum_{j=1}^n y^j \right) = \frac{x(x^m - 1)}{x - 1} \cdot \frac{y(y^n - 1)}{y - 1}.$$

Consider each $1 \times a$ rectangle. If the leftmost cell covered by this rectangle is the (i, j) th cell, then the sum of monomials covered by this rectangle is

$$x^i y^j + x^i y^{j+1} + \dots + x^i y^{j+a-1} = \frac{x^i y^j (y^a - 1)}{y - 1}.$$

Similarly, the sum of monomials covered by each $a \times 1$ rectangle is

$$x^i y^j + x^{i+1} y^j + \dots + x^{i+a-1} y^j = \frac{x^i y^j (x^a - 1)}{x - 1}.$$

If a tiling exists, the small rectangles together must cover each monomial once. Thus, there exist indice sets S and T such that

$$f(x, y) = \sum_{(i,j) \in S} \frac{x^i y^j (y^a - 1)}{y - 1} + \sum_{(i,j) \in T} \frac{x^i y^j (x^a - 1)}{x - 1}.$$

Note that this is an identity. We can substitute some values of x and y to simplify the expression. For example, by setting $x = y = \omega$, where $\omega = e^{i\frac{2\pi}{a}}$ is a root of unity, we deduce $f(\omega, \omega) = 0$ since $\omega^a - 1 = 0$ (and $\omega - 1 \neq 0$). This implies

$$\frac{\omega(\omega^m - 1)}{\omega - 1} \cdot \frac{\omega(\omega^n - 1)}{\omega - 1} = 0,$$

or simply $(\omega^m - 1)(\omega^n - 1) = 0$. It follows that $\omega^m = 1$ or $\omega^n = 1$. This only holds when $a \mid m$ or $a \mid n$ (as $(e^{i\frac{2\pi}{a}})^k = e^{i\frac{2k\pi}{a}}$ is equal to 1 iff $a \mid k$). This completes the proof $\square$

Remarks. In fact, this solution is closely related to the solution given in [problem 2.8](#) using colouring. For example, the (i, j) th cell and the $(i + a, j)$ th cell have the same colour. The monomials written down in these cells are $x^i y^j$ and $x^{i+a} y^j$ respectively, which are equal when $x = y = \omega$. In general, the values of two monomials are the same iff the two cells are of the same colour. In some sense, the two solutions are different ways to present the same idea.

Problem 6.7. (IMO Shortlist 2022 C4) Let $n > 3$ be a positive integer. Suppose that n children are arranged in a circle, and n coins are distributed between them (some children may have no coins). At every step, a child with at least 2 coins may give 1 coin to each of their immediate neighbours on the right and left. Determine all initial distributions of coins from which it is possible that, after a finite number of steps, each child has exactly one coin.

Analysis. If we use a_j to denote the number of coins of the j th child, we can form a generating function $\sum_{j=1}^n a_j x^j$. In general, performing an operation to the j th child means to add $x^{j+1} - 2x^j + x^{j-1}$ to the function. However, there is some trouble if $j = 1$ or $j = n$. Indeed, the coefficients of x^k and x^{n+k} should be combined together because the children are arranged in a circle. To get rid of this algebraically, we can consider everything modulo $x^n - 1$. Given two polynomials $f(x)$ and $g(x)$ with integer coefficients, we write $f(x) \equiv g(x) \pmod{x^n - 1}$ if there exists a polynomial $h(x)$ with integer coefficients such that $f(x) - g(x) = (x^n - 1)h(x)$.

Solution. Let a_j be the number of coins of the j th child ($j = 1, 2, \dots, n$), where the children are labelled in clockwise order. Define

$$f(x) = \sum_{j=1}^n a_j x^j.$$

We shall consider all polynomials modulo $x^n - 1$. In that case, performing an operation to the j th child means to add

$$x^{j+1} - 2x^j + x^{j-1} = x^{j-1}(x-1)^2$$

to f . Therefore, we can obtain all possible configurations described by the function

$$f(x) + g(x)(x-1)^2 \pmod{x^n - 1},$$

where $g(x)$ is a polynomial with nonnegative integer coefficients. The nonnegative condition can actually be removed. Indeed, since

$$(x^{n-1} + x^{n-2} + \dots + 1)(x-1)^2 = (x^n - 1)(x-1) \equiv 0 \pmod{x^n - 1},$$

we can add some copies of $x^{n-1} + x^{n-2} + \dots + 1$ to $g(x)$ until all its coefficients are nonnegative. As the final configuration we want is $h(x) = x^n + x^{n-1} + \dots + x$, our remaining task is to determine all $f(x)$ such that

$$f(x) + g(x)(x-1)^2 \equiv h(x) \pmod{x^n - 1}$$

for some $g(x) \in \mathbb{Z}[x]$. Equivalently, we need to find all $f(x)$ such that

$$f(x) - h(x) = p(x)(x^n - 1) - g(x)(x-1)^2 \tag{1}$$

for some $p(x), g(x) \in \mathbb{Z}[x]$.

Firstly, note that the right-hand side of (1) is divisible by $x-1$. Thus, we must have $f(1) = h(1)$. This means $a_1 + a_2 + \dots + a_n = n$, i.e. the total number of coins is n (which is obvious anyway). Assuming this, (1) is the same as

$$q(x) = \frac{f(x) - h(x)}{x-1} = p(x)(x^{n-1} + x^{n-2} + \dots + 1) - g(x)(x-1). \tag{2}$$

Putting $x = 1$ in (2), we must have $q(1) = p(1)n \equiv 0 \pmod{n}$. We claim that this is also a sufficient condition. Indeed, if $n \mid q(1)$, then we can set $p(x) = \frac{q(1)}{n} \in \mathbb{Z}[x]$ (which is a constant polynomial). Then

$$q(x) - p(x)(x^{n-1} + x^{n-2} + \dots + 1)$$

is divisible by $x-1$ by the factor theorem as $q(1) - p(1)(1^{n-1} + 1^{n-2} + \dots + 1) = 0$. This shows (2) holds for some $g(x) \in \mathbb{Z}[x]$.

It remains to study the function $q(x)$. We have

$$q(x) = \frac{1}{x-1} \sum_{j=1}^n (a_j - 1)x^j = \frac{1}{x-1} \sum_{j=1}^n (a_j - 1)(x^j - 1)$$

since $\sum_{j=1}^n (a_j - 1) = \sum_{j=1}^n a_j - n = n - n = 0$. It follows that

$$q(x) = \sum_{j=1}^n (a_j - 1)(x^{j-1} + x^{j-2} + \cdots + 1).$$

Therefore, the second condition $n \mid q(1)$ means $\sum_{j=1}^n j(a_j - 1) = \sum_{j=1}^n ja_j - \frac{n(n+1)}{2}$ is divisible by n . Thus, the answer is all configurations $(a_1, a_2, \dots, a_n)$ with

$$\sum_{j=1}^n a_j = n \quad \text{and} \quad \sum_{j=1}^n ja_j \equiv \frac{n(n+1)}{2} \pmod{n}.$$

□

The following lists out some tricks and concepts we have used in this section about generating functions.

- generalized binomial theorem ([problems 6.1, 6.2, 6.3](#))
- roots of unity ([problems 6.4, 6.5, 6.6](#))
- multivariate function ([problem 6.6](#))
- polynomial congruence ([problem 6.7](#))

7 Probabilistic Method

Finally, we introduce the probabilistic method, which is used to solve some existence problems conveniently. One is assumed to have some basic knowledge about probability. We briefly go through the necessary knowledge below.

A *random variable*(隨機變量) is a quantity that depends on some random events. It has no fixed value, but we can tell the probability that certain events may occur depending on this random variable. For example, let X be a random variable which takes the values 1, 2, 3 at probabilities $\frac{1}{2}, \frac{1}{3}, \frac{1}{6}$ respectively. Then we may write

$$\mathbb{P}(X = 1) = \frac{1}{2}, \quad \mathbb{P}(X = 2) = \frac{1}{3}, \quad \mathbb{P}(X = 3) = \frac{1}{6}.$$

Apart from specifying the probability that it takes a certain value, we can express the probability for a certain event to occur in many different ways. For example, we can write

$$\mathbb{P}(X \geq 2) = \frac{1}{3} + \frac{1}{6} = \frac{1}{2}$$

to mean the probability that $X \geq 2$ holds. Note that if the only possible outcomes are $x_1, x_2, \dots, x_n$, we must have

$$\sum_{j=1}^n \mathbb{P}(X = x_j) = 1$$

as a property of probability.

Let X be a random variable that has finitely many possible outcomes $x_1, x_2, \dots, x_n$. Then the *expected value*(期望值) or *expectation* of X is defined as

$$\mathbb{E}[X] = \sum_{j=1}^n x_j \mathbb{P}(X = x_j).$$

For example, if we roll a fair dice, then the expected value of the outcome is

$$\frac{1}{6} \cdot 1 + \frac{1}{6} \cdot 2 + \frac{1}{6} \cdot 3 + \frac{1}{6} \cdot 4 + \frac{1}{6} \cdot 5 + \frac{1}{6} \cdot 6 = \frac{7}{2}.$$

This tells us the average value of the outcome.

Given two random variables X and Y , their sum $X + Y$ is also a random variable. Its definition is simple. For example, if X and Y satisfy

$$\begin{aligned} \mathbb{P}(X = 1) &= \frac{1}{3}, & \mathbb{P}(X = 2) &= \frac{2}{3}, \\ \mathbb{P}(Y = 1) &= \frac{1}{4}, & \mathbb{P}(Y = 2) &= \frac{1}{4}, & \mathbb{P}(Y = 3) &= \frac{1}{2}, \end{aligned}$$

then $X + Y$ can take the values 2, 3, 4, 5, and

$$\mathbb{P}(X + Y = 3) = \mathbb{P}(X = 1)\mathbb{P}(Y = 2) + \mathbb{P}(X = 2)\mathbb{P}(Y = 1) = \frac{1}{4},$$

for example.

A useful result that we need is the following.

Proposition 7.1. Let X and Y be two random variables. Then

$$\mathbb{E}[X + Y] = \mathbb{E}[X] + \mathbb{E}[Y].$$

Proof. We only provide a proof when both X and Y have finitely many possible outcomes. By definition,

$$\begin{aligned} \mathbb{E}[X + Y] &= \sum_c c \mathbb{P}(X + Y = c) \\ &= \sum_c c \sum_{x_i + y_j = c} \mathbb{P}(X = x_i) \mathbb{P}(Y = y_j) \\ &= \sum_{x_i} \sum_{y_j} (x_i + y_j) \mathbb{P}(X = x_i) \mathbb{P}(Y = y_j) \\ &= \sum_{x_i} x_i \mathbb{P}(X = x_i) \sum_{y_j} \mathbb{P}(Y = y_j) + \sum_{y_j} y_j \mathbb{P}(Y = y_j) \sum_{x_i} \mathbb{P}(X = x_i) \\ &= \sum_{x_i} x_i \mathbb{P}(X = x_i) + \sum_{y_j} y_j \mathbb{P}(Y = y_j) \\ &= \mathbb{E}[X] + \mathbb{E}[Y]. \end{aligned}$$

□

The result is known as the linearity of expectation. In general, we have

$$\mathbb{E}\left[\sum_{j=1}^n X_j\right] = \sum_{j=1}^n \mathbb{E}[X_j]$$

for any random variables $X_1, X_2, \dots, X_n$. The result is obvious if the random variables are independent. However, the same holds even if they are dependent, which indicates the usefulness of this result. For example, if we roll 10 fair dices at the same time, then the expected value of the sum of outcomes is simply $10 \cdot \frac{7}{2} = 35$. However, if we use the definition to compute this, we need to compute

$$\sum_{k=10}^{60} k \mathbb{P}(X_1 + X_2 + \dots + X_{10} = k),$$

which is extremely tedious.

To use expected values to show the existence of something, a simple strategy is as follows. Recall that the expected value is somehow the average value. If the average

value of some numbers is k , there must be a number which is at least k , and there must be a number which is at most k . In this way, we are able to tell the existence of some objects with certain properties. This is analogous to the use of the pigeonhole principle.

Let us go through some problems below to demonstrate this.

Problem 7.1. (IMO Shortlist 1987) Prove that there exists a four-colouring of the set $M = \{1, 2, \dots, 1987\}$ such that any arithmetic progression with 10 terms in the set M is not monochromatic.

Analysis. For many applications of expectation, the process is as follows. Suppose we need to show the existence of some objects such that a certain quantity X (depending on the chosen objects) is at most k . Then we can choose these objects randomly and show that $\mathbb{E}[X] = k$. In addition, if we know that X must be an integer, we only need a weaker bound for that. For example, if $\mathbb{E}[X] = r$, then we know that there must be a choice such that $X = \lfloor r \rfloor$. Therefore, it is good enough if $k = \lfloor r \rfloor$.

Solution. We colour each element of M randomly. Then we define a random variable X , which is the number of monochromatic arithmetic progression with 10 terms in M . In addition, for each arithmetic progression with 10 terms, we define a random variable X_i , such that $X_i = 1$ if this arithmetic progression is monochromatic and $X_i = 0$ otherwise. By the linearity of expectation, we have

$$\mathbb{E}[X] = \sum_{i=1}^n \mathbb{E}[X_i],$$

where n is the number of arithmetic progressions. (The indicator functions X_i are useful in counting like this.)

Each $\mathbb{E}[X_i]$ is easy to compute. Indeed, by fixing the colour of the first term of the arithmetic progression, all other 9 terms need to have the same colour in order that the arithmetic progression is monochromatic. This implies

$$\mathbb{E}[X_i] = \frac{1}{4^9}.$$

It is also easy to find n . The idea is exactly the same as that in [problem 1.3](#), so we shall skip some of the details. Since there are 220 numbers in M congruent to each of $0, 8 \pmod{9}$ and 221 numbers in M congruent to each of $1, 2, \dots, 7 \pmod{9}$, one can find that

$$n = 2 \binom{220}{2} + 7 \binom{221}{2} = 2 \times 110 \times 219 + 7 \times 221 \times 110 = 218350.$$

It follows that

$$\mathbb{E}[X] = \sum_{i=1}^n \mathbb{E}[X_i] = \frac{n}{4^9} = \frac{218350}{262144}.$$

This shows there is a colouring for which $X \leq \left\lfloor \frac{218350}{262144} \right\rfloor = 0$, meaning that there is no monochromatic arithmetic progression with 10 terms in M . So we are done. $\square$

Problem 7.2. (IMC 2002 Problem 8) 200 students participated in a math contest. They had 6 problems to solve. Each problem was correctly solved by at least 120 participants. Prove that there must be 2 participants such that every problem was solved by at least one of these two students.

Analysis. This kind of problems can be solved by double counting. Indeed, a solution using probabilistic method is essentially the same as a solution using double counting, but we simply rephrase everything in terms of probability and expectation. The double counting process is somehow contained in the linearity of expectation, which makes the solution neat. The nature of this problem is very similar to the previous problem.

Solution. We choose two students randomly. Let X be the random variable that counts the number of problems solved by at least one of them, and let X_i be a random variable such that $X_i = 1$ if at least one of them solved the i th problem, and is 0 otherwise. By the linearity of expectation, we have

$$\mathbb{E}[X] = \mathbb{E}[X_1] + \mathbb{E}[X_2] + \cdots + \mathbb{E}[X_6].$$

For each problem, the probability that a participant did not solve it is $1 - \frac{120}{200} = \frac{2}{5}$. Therefore,

$$\mathbb{E}[X_i] = 1 - \mathbb{P}(\text{both students did not solve the } i\text{th problem}) = 1 - \left(\frac{2}{5}\right)^2 = \frac{21}{25}.$$

This implies

$$\mathbb{E}[X] = 6 \times \frac{21}{25} = \frac{126}{25}.$$

Thus, we are able to find two students such that $X \geq \left\lceil \frac{126}{25} \right\rceil = 6$, which is exactly the result we are looking for. $\square$

Remarks. To compute the expected value, sometimes we need some simple facts about probabilities.

Problem 7.3. (CHKMO 2020 Problem 1) There is a table with n rows and 18 columns. Each of its cells contains a 0 or a 1. The table satisfies the following properties:

- (i) Every two rows are different.
- (ii) Each row contains exactly 6 cells that contain 1.

- (iii) For every three rows, there exists a column so that the intersection of the column with the three rows (the three cells) all contain 0.

What is the greatest possible value of n ?

Analysis. One obvious construction of a table that satisfies all properties is to set a zero column. This already provides a bound $n \geq \binom{17}{6}$. Since this bound is large and there does not seem to be any other nice construction, this is likely to be the optimal value.

This time we demonstrate how to use expectation to work out an upper bound. To do so, we may assume $n > \binom{17}{6}$ and assume properties (i) and (ii) are satisfied, and try to show that there exist three rows violating property (iii). This is essentially an existence problem, and that's why expectation can be useful.

Solution. We can identify each row as a 6-element subset of $S_{18} = \{1, 2, \dots, 18\}$, so that the elements correspond to the numbers of columns which contain 1. Then we have to maximize the number of such subsets so that no three of them form a partition of S_{18} , since forming a partition is the only case for their complement to have an empty intersection.

First we construct an example of $\binom{17}{6} = 12376$ subsets. We simply need to consider all 6-element subsets of $\{1, 2, \dots, 17\}$. Since in any three subsets, there are $6 \times 3 = 18$ elements, there must be an element belonging to at least two of the subsets by the pigeonhole principle. Therefore, the condition is satisfied, and there are $\binom{17}{6}$ subsets.

Next we show that $n \leq 12376$. Let $A_1, A_2, \dots, A_n$ be the 6-element subsets of S_{18} which correspond to the rows. If we follow the previous strategy, we may choose three of these subsets randomly, and show that they form a partition of S_{18} on average. However, one may find it hard to rephrase this property in terms of something that can be counted in a simple way. Rather than that, we may try to do it in another way.

We choose a partition B of S_{18} into three 6-element subsets randomly, and let X be a random variable that counts the number of sets in B which are some of the A_i 's. Let X_i be a random variable such that $X_i = 1$ if A_i is one of the sets in B , and is 0 otherwise. By the linearity of expectation, we have

$$\mathbb{E}[X] = \sum_{i=1}^n \mathbb{E}[X_i].$$

Note that the total number of partitions is

$$\frac{1}{3!} \binom{18}{6} \binom{12}{6}$$

since there are $\binom{18}{6}$ ways to choose the first set and $\binom{12}{6}$ ways to choose the second set, and we need to divide by $3!$ because the order does not matter. Similarly, for each A_i , there are

$$\frac{1}{2!} \binom{12}{6}$$

partitions containing A_i . Therefore, we have

$$\mathbb{E}[X_i] = \frac{\frac{1}{2!} \binom{12}{6}}{\frac{1}{3!} \binom{18}{6} \binom{12}{6}} = \frac{3}{\binom{18}{6}} = \frac{2}{\binom{17}{6}} = \frac{2}{12376}.$$

This implies

$$\mathbb{E}[X] = \frac{2n}{12376}.$$

If $n > 12376$, this yields $\mathbb{E}[X] > 2$, and so there is a partition B such that $X = 3$. This means there are three of the A_i 's which form a partition of S_{18} , contradiction. Therefore, we must have $n \leq 12376$. This shows the greatest possible value of n is 12376. $\square$

Problem 7.4. (International Zhautykov Olympiad 2016 Problem 3) There are 60 towns in Graphland every two countries of which are connected by only a directed way. Prove that we can colour four towns to red and four towns to green such that every way between green and red towns are directed from red to green.

Analysis. There are two ways to work on this problem. We can colour four towns red randomly and show that it is possible (in the average sense) to colour four more towns green such that the condition is satisfied, or colour some towns green and then red. These are essentially the same up to symmetry. This time we show the use of the majorization inequality to obtain a bound for the expectation. This is often needed because many expressions in counting involve the binomial coefficients.

Solution. Consider a directed graph $G = (V, E)$ where each vertex corresponds to a town and each edge corresponds to a directed way. There are 60 vertices in total.

We colour four vertices v_1, v_2, v_3, v_4 red randomly. Let X be the number of remaining vertices w such that each $v_i w$ is an edge. For each $w \in V$, there are $\deg^-(w)$ edges with w as the terminal vertex. Therefore, w contributes 1 to X for exactly $\binom{\deg^-(w)}{4}$ quadruples of vertices. This implies

$$\mathbb{E}[X] = \frac{1}{\binom{60}{4}} \sum_{w \in V} \binom{\deg^-(w)}{4}.$$

Since

$$\frac{1}{60} \sum_{w \in V} \deg^-(w) = \frac{1}{60} \binom{60}{2} = \frac{59}{2},$$

by the majorization inequality, we have

$$\mathbb{E}[X] \geq \frac{1}{\binom{60}{4}} \cdot 60 \binom{\frac{59}{2}}{4} = \frac{60 \cdot \frac{59}{2} \cdot \frac{57}{2} \cdot \frac{55}{2} \cdot \frac{53}{2}}{60 \cdot 59 \cdot 58 \cdot 57} = \frac{2915}{928}.$$

It follows that there exist v_1, v_2, v_3, v_4 such that $X \geq \left\lceil \frac{2915}{928} \right\rceil = 4$. We can colour any four of the vertices counted in X green. Then the condition is satisfied. $\square$

Problem 7.5. (USAMO 1995 Problem 5) Suppose that in a certain society, each pair of persons can be classified as either *amicable* or *hostile*. We shall say that each member of an amicable pair is a *friend* of the other, and each member of a hostile pair is a *foe* of the other. Suppose that the society has n persons and q amicable pairs, and that for every set of three persons, at least one pair is hostile. Prove that there is at least one member of the society whose foes include $q \left(1 - \frac{4q}{n^2}\right)$ or fewer amicable pairs.

Analysis. In terms of graph theory, if we use an edge to denote an amicable pair, then we need to count the number of triples v_1, v_2, v_3 such that v_1 is not adjacent to v_2 or v_3 , while v_2v_3 is an edge. It is not easy to count such triples directly. On the other hand, it is easier to count the number of triples v_1, v_2, v_3 such that v_1v_2 and v_1v_3 are edges, while v_2 and v_3 are not adjacent. We can use the degrees of the vertices to count this number. The use of expectation is straightforward.

Solution. Consider a graph $G = (V, E)$ where each vertex represents a person and each edge represents an amicable pair. Then $n = |V|$ and $q = |E|$. It is given that there is no triangle. We wish to show that there exists a vertex v such that there are at most $q \left(1 - \frac{4q}{n^2}\right)$ edges incident with two vertices which are non-adjacent to v .

Naturally, we may pick a vertex v randomly, and let X be the number of edges incident with two vertices which are non-adjacent to v . For each edge wx , let $f(wx)$ be the number of vertices which are non-adjacent to both w and x . Then the edge wx is counted in the random variable X for exactly $f(wx)$ times. This implies

$$\mathbb{E}[X] = \frac{1}{n} \sum_{wx \in E} f(wx).$$

Let $g(wx)$ be the number of vertices which are adjacent to exactly one of w and x . Since there is no triangle in G , we have $f(wx) + g(wx) = n - 2$. It follows that

$$\mathbb{E}[X] = \frac{1}{n} \sum_{wx \in E} (n - 2 - g(wx)) = \frac{q(n - 2)}{n} - \frac{1}{n} \sum_{wx \in E} g(wx) = \frac{q(n - 2)}{n} - \frac{2N}{n},$$

where N is the number of unordered triples of vertices w, x, y such that there are exactly two edges among w, x, y . Clearly, we have

$$N = \sum_{v \in V} \binom{\deg(v)}{2} \geq n \binom{\frac{2q}{n}}{2} = \frac{q(2q - n)}{n}$$

by the majorization inequality. This yields

$$\mathbb{E}[X] \leq \frac{q(n-2)}{n} - \frac{2q(2q-n)}{n^2} = \frac{q(n^2-4q)}{n^2} = q \left(1 - \frac{4q}{n^2}\right).$$

Thus, we can always pick a vertex v such that $X \leq q \left(1 - \frac{4q}{n^2}\right)$, which is our final goal. $\square$

Problem 7.6. (MEMO 2011 Team Problem 4) Let $n \geq 3$ be an integer. At a MEMO-like competition, there are $3n$ participants, there are n languages spoken, and each participant speaks exactly three different languages. Prove that at least $\left\lceil \frac{2n}{9} \right\rceil$ of the spoken languages can be chosen in such a way that no participant speaks more than two of the chosen languages.

Analysis. The first idea is to choose $\left\lceil \frac{2n}{9} \right\rceil$ languages randomly, and see if there is a choice that not all three languages spoken by each participant are chosen. However, it is not easy to count in this way.

This time we introduce a new idea in using expectation. We can work out a quantity in more than one step. For this problem, we may first select more languages than needed in a random way, and remove some languages so that the condition is satisfied. For example, if all three languages spoken by a participant are chosen in the first step, we can remove any one of these languages arbitrarily. Although the two processes are dependent, we can somehow regard them as independent events by using the linearity of expectation.

Solution. We first construct a set of languages S such that the probability that each language is included in S is p for some $p \in (0, 1)$. Afterwards, for each of the $3n$ participants, we check whether the three languages spoken by this participant are all in S or not. If they are all in S , we remove any one of these languages arbitrarily. After doing this for every participant, we obtain a set of languages T which satisfies the condition. Let $X = |T|$. It remains to show that $\mathbb{E}[X] \geq \frac{2n}{9}$ for some p .

Let $Y = |S|$ and let Z be the number of languages removed from S . Also, for each participant, we define a random variable Z_i such that $Z_i = 1$ if all three languages spoken by this participant are in S , and is 0 otherwise. By the linearity of expectation, we have

$$\mathbb{E}[X] = \mathbb{E}[Y] - \mathbb{E}[Z] \geq \mathbb{E}[Y] - \sum_{i=1}^{3n} \mathbb{E}[Z_i].$$

Note that this is an inequality instead of an identity since we may remove the same language from S which corresponds to different participants. Now, each term is easy to compute. Indeed, we have $\mathbb{E}[Y] = np$ as there are n languages and each language has a chance of p to be included in S . Also, we have $\mathbb{E}[Z_i] = p^3$ because we need all three languages to be included in S . It follows that

$$\mathbb{E}[X] \geq np - 3np^3.$$

As $\frac{d}{dp}(np - 3np^3) = n(1 - 3p)(1 + 3p)$, it is not hard to see that the optimal choice of p is $p = \frac{1}{3}$. For this choice of the probability, we have

$$\mathbb{E}[X] \geq \frac{2n}{9}$$

as desired. □

Remarks. If one fully understands the linearity of expectation, there is no need to define so many random variables in the proof. We can simply work on each step independently (even if the steps are actually dependent) to compute the average of the quantity. For example, we can simply deduce $|S| = np$ on average, and we need to remove at most $3np^3$ languages from S on average to obtain $\mathbb{E}[X] \geq np - 3np^3$.

Problem 7.7. (China TST 2023 Test 1 Problem 11) Let n be a positive integer, and let $a_{ijk} \in \{-1, 1\}$ ($1 \leq i, j, k \leq n$). Prove that there exist $x_1, \dots, x_n, y_1, \dots, y_n, z_1, \dots, z_n \in \{-1, 1\}$ such that

$$\left| \sum_{i=1}^n \sum_{j=1}^n \sum_{k=1}^n a_{ijk} x_i y_j z_k \right| > \frac{n^2}{3}.$$

Analysis. We go through a more difficult problem which can be solved using expectation. This problem is more about algebra, but still involves some combinatorial nature, as we need to show the existence of some coefficients such that a quantity is large.

A simple observation is that $\frac{(-1) + 1}{2} = 0$. This means the expected values of many quantities are 0 if we choose the coefficients randomly. This clearly does not provide any useful information to us. To get rid of this, we may generate the square of some variables. This can be done by squaring a sum. We shall illustrate this technique in this problem.

Solution. Firstly, note that

$$S = \sum_{i=1}^n \sum_{j=1}^n \sum_{k=1}^n a_{ijk} x_i y_j z_k = \sum_{i=1}^n x_i \left(\sum_{j=1}^n \sum_{k=1}^n a_{ijk} y_j z_k \right).$$

To maximize this, we should choose x_i 's such that all summands of the outermost sum have positive signs. In other words, we should choose the signs such that

$$S = \sum_{i=1}^n |S_i|,$$

where

$$S_i = \sum_{j=1}^n \sum_{k=1}^n a_{ijk} y_j z_k.$$

As mentioned, we consider $\mathbb{E}[S_i^2]$ where y_j 's and z_k 's are chosen randomly. We first deduce

$$S_i^2 = \sum_{j_1, j_2=1}^n \sum_{k_1, k_2=1}^n a_{ij_1 k_1} a_{ij_2 k_2} y_{j_1} y_{j_2} z_{k_1} z_{k_2}.$$

If $j_1 \neq j_2$, then $\mathbb{E}[y_{j_1} y_{j_2}] = \mathbb{E}[y_{j_1}] \mathbb{E}[y_{j_2}] = 0$ since $\mathbb{E}[y_{j_1}] = \mathbb{E}[y_{j_2}] = \frac{(-1)+1}{2} = 0$. Similarly, if $k_1 \neq k_2$, the contribution of this sum to the expected value is 0. When $j_1 = j_2$ and $k_1 = k_2$, we have $y_{j_1} y_{j_2} = 1$ and $z_{k_1} z_{k_2} = 1$. Therefore,

$$\mathbb{E}[S_i^2] = \sum_{j=1}^n \sum_{k=1}^n a_{ijk}^2 = \sum_{j=1}^n \sum_{k=1}^n 1 = n^2. \quad (1)$$

We would like to obtain a lower bound for $\mathbb{E}[|S_i|]$ instead. To do so, we need another expected value involving a higher power of $|S_i|$ (as we can use Hölder's inequality afterwards). Thus, we further consider $\mathbb{E}[S_i^4]$. As above, we have

$$S_i^4 = \sum_{j_1, j_2, j_3, j_4=1}^n \sum_{k_1, k_2, k_3, k_4=1}^n a_{ij_1 k_1} a_{ij_2 k_2} a_{ij_3 k_3} a_{ij_4 k_4} y_{j_1} y_{j_2} y_{j_3} y_{j_4} z_{k_1} z_{k_2} z_{k_3} z_{k_4}.$$

If there is an index which appears an odd number of times in j_1, j_2, j_3, j_4 , then the contribution of this term to $\mathbb{E}[S_i^4]$ is 0. Thus, we may assume either j_1, j_2, j_3, j_4 are all equal, or they consist of two pairs of equal indices. The same holds for k_1, k_2, k_3, k_4 . In that case, the expected value of each summand is 1. Therefore,

$$\mathbb{E}[S_i^4] = \left(\sum_{j=1}^n 1 + \binom{4}{2} \sum_{1 \leq j_1 < j_2 \leq n} 1 \right)^2 = n^2 (3n-2)^2 < 9n^4. \quad (2)$$

By Hölder's inequality, we have

$$\mathbb{E}[|S_i|] \mathbb{E}[|S_i|] \mathbb{E}[S_i^4] \geq \mathbb{E}[S_i^2]^3.$$

(One can express these expected values as a sum to see why it corresponds to Hölder's inequality.) Using (1) and (2), this yields $\mathbb{E}[|S_i|] > \frac{n}{3}$. Finally,

$$\mathbb{E}[S] = \sum_{i=1}^n \mathbb{E}[|S_i|] > \frac{n^2}{3}.$$

This means there is a choice of the coefficients such that $S > \frac{n^2}{3}$. □

This short section is about the use of expectation. The linearity of expectation is useful because it allows us to focus on the average case only. However, one should make sure that such an argument can be written rigorously in terms of some random variables even if one does not wish to write down all such details in the proof.

Links

Theorems

- [Addition principle](#)
- [Binomial theorem](#)
- [Bonferroni inequalities](#)
- [Dirac's theorem](#)
- [Generalized binomial theorem](#)
- [Hall's theorem](#)
- [Handshaking theorem](#)
- [Inclusion-exclusion principle](#)
- [Karamata's inequality](#)
- [Majorization inequality](#)
- [Mantel's theorem](#)
- [Multiplication principle](#)
- [Ore's theorem](#)
- [Pascal's identity](#)
- [Pigeonhole principle](#)
- [Product rule](#)
- [Sperner's theorem](#)
- [Sum rule](#)
- [Sylvester-Gallai theorem](#)
- [Turán's theorem](#)

Propositions

- [1.1: Properties of permutations](#)
- [1.2: Properties of combinations](#)
- [1.3: General terms of recurrence sequences](#)
- [4.1: Partitioning a graph into cycles](#)
- [4.2: Number of edges of a connected graph](#)
- [4.3: Criteria for a graph to be a tree](#)
- [4.4: Number of edges of a tree](#)

- 4.5: Criterion for a graph to be a bipartite graph
- 4.6: A criterion for a directed graph to be Hamiltonian
- 4.7: A criterion for a connected graph to be Eulerian
- 7.1: Linearity of expectation

Problems

- 1.1: Basic counting 1
- 1.2: Basic counting 2
- 1.3: IMO Prelim 2012 Problem 14
- 1.4: IMO HK TST 2014 Test 1 Problem 4 (inclusion-exclusion principle)
- 1.5: IMO Prelim 2016 Problem 17
- 1.6: CHKMO 2003 Problem 2
- 1.7: APMO 2017 Problem 3
- 1.8: Austria MO Final Round 2019 Problem 3
- 1.9: International Zhautykov Olympiad 2015 Problem 5
- 1.10: CMO 2001 Problem 5
- 1.11: IMO Shortlist 2002 C3
- 2.1: European Mathematical Cup 2021 Senior Problem 1
- 2.2: Extremal combinatorics and induction
- 2.3: Czech-Polish-Slovak Match 2015 Problem 2 modified
- 2.4: IMO 2009 Problem 6
- 2.5: Pigeonhole principle
- 2.6: CHKMO 2006 Problem 1
- 2.7: CHKMO 2012 Problem 2
- 2.8: Tiling with rectangles
- 2.9: IMO HK TST 2015 Test 2 Problem 4
- 2.10: IMO 2004 Problem 3
- 2.11: EMMO 2016
- 2.12: An identity for binomial coefficients
- 2.13: Sperner's theorem
- 2.14: Counting pairs
- 2.15: CHKMO 2007 Problem 3

- 2.16: IMO 2001 Problem 3
- 2.17: Extremal principle 1
- 2.18: Extremal principle 2
- 2.19: APMO 2008 Problem 2
- 2.20: CHKMO 2015 Problem 4
- 2.21: China TST 1992 Problem 1
- 2.22: CHKMO 2011 Problem 4
- 3.1: Two-person game
- 3.2: CMO 2010 Problem 5
- 3.3: IMO Shortlist 2009 C8
- 3.4: IMO 2010 Problem 5
- 3.5: IMO 2014 Problem 5
- 3.6: Invariance 1
- 3.7: Invariance 2
- 3.8: APMO 2009 Problem 1
- 3.9: IMO Shortlist 2005 C5
- 3.10: IMO 2000 Problem 3
- 4.1: APMO 2003 Problem 5
- 4.2: Path and cycle
- 4.3: CMO 1987 Problem 3
- 4.4: Connected graph
- 4.5: Spanning tree
- 4.6: IMO 1991 Problem 4
- 4.7: Hall's theorem
- 4.8: Korea MO 2021 Problem 4
- 4.9: De Bruijn sequence
- 4.10: MEMO 2021 Individual Problem 2
- 4.11: Kharkiv Masters Tournament 2019 Grade 10–11 Problem 6
- 4.12: Serbia TST 2021 Problem 4
- 4.13: IMO Shortlist 2020 C4
- 4.14: IMO 2020 Problem 3
- 5.1: JBMO Shortlist 2015 C2

- 5.2: All-Russian MO 2004 Grade 11 Problem 1
- 5.3: EGMO 2017 Problem 3
- 5.4: The University of Melbourne School Mathematics Competition 2017
- 5.5: IMO 2016 Problem 6
- 5.6: Convex hull
- 5.7: IMO HK TST 2020 Test 2 Problem 2
- 5.8: Discreteness
- 5.9: IMO 2011 Problem 2
- 5.10: APMO 2011 Problem 2
- 5.11: Orthocentre
- 5.12: Sylvester-Gallai theorem
- 5.13: Integer distances
- 5.14: 239 Open MO 2018 Senior Problem 1
- 5.15: Canada MO 2017 Problem 5
- 6.1: IMO HK TST 2014 Test 1 Problem 4 (generating function)
- 6.2: Generating function and recurrence sequence
- 6.3: Catalan numbers
- 6.4: Generating function and roots of unity
- 6.5: USA TSTST 2018 Problem 6
- 6.6: Multivariable generating function
- 6.7: IMO Shortlist 2022 C4
- 7.1: IMO Shortlist 1987
- 7.2: IMC 2002 Problem 8
- 7.3: CHKMO 2020 Problem 1
- 7.4: International Zhautykov Olympiad 2016 Problem 3
- 7.5: USAMO 1995 Problem 5
- 7.6: MEMO 2011 Team Problem 4
- 7.7: China TST 2023 Test 1 Problem 11

Terminologies

- acyclic graph
- adjacent

- binomial coefficient $\binom{n}{r}$
- binomial function $\binom{n}{r}$
- bipartite graph
- Catalan numbers
- characteristic equation
- circuit
- combination $\binom{n}{r}$, C_r^n
- complete bipartite graph $K_{m,n}$
- complete graph K_n
- component
- connected graph
- convex hull
- cycle
- de Bruijn sequence
- degree $\deg(v)$
- dictionary order
- directed graph
- distance
- edge
- Eulerian circuit
- Eulerian graph
- Eulerian trail
- expectation $\mathbb{E}[X]$
- expected value $\mathbb{E}[X]$
- forest
- Hamiltonian cycle
- Hamiltonian graph
- Hamiltonian path
- in-degree $\deg^-(v)$
- incident with
- independent set

- induced subgraph
- leaf
- lexicographic order
- loop
- majorize $>, <$
- method of exhaustion
- multigraph
- neighbouring
- out-degree $\deg^+(v)$
- path
- permutation P_r^n
- Petersen graph
- r -partite graph
- random variable
- roots of unity
- simple graph
- spanning subgraph
- strongly connected
- subgraph
- trail
- tree
- Turán graph
- vertex
- walk

References

- A Path to Combinatorics for Undergraduates: Counting Strategies
(by Titu Andreescu, Zuming Feng)
- Modern Graph Theory (by Béla Bollobás)
- Probabilistic Methods in Combinatorics (by Po-Shen Loh)

「遠看山有色，近聽水無聲。春去花還在，人來鳥不驚。」

《畫》